

Abhijit Singh Scientific Paper Series

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Abhijit Singh

Series contents

This bound volume is a reading convenience. Every paper is also supplied as an independent submission package with its own PDF, source archive, abstract, metadata, and checksums. Release 8.0.1 integrates the occurrence-sensitive prime relation, the nine-state thirty-six-relation closure theorem, its transparent-prime and finite spectral-label interpretation, and the all-prime predictive-closure correction.

ID	Paper	Pages	Volume pages
P00	Recurrence Exponents, Accessibility Growth, and Predictive-State Promotion	9	5–13
P01	Future-Complete Quotients and Recursive Distinction	43	14–56
P02	Sequential Quantum Instruments: Depletion, Regeneration, and Minimal Predictive State	12	57–68
P03	Provenance-Preserving Representation Genesis in Finite Deterministic Systems	9	69–77
P04	Stochastic Frontier Information, Finite-Memory Dissipation, and Cycle Holonomy	8	78–85
P05	Coherent Alternatives, Complex Amplitudes, and Marked Quantum Dynamics	9	86–94
P06	Arity-Indexed Refinement Posets, Forced Inverses, and Rotational Root Geometry	9	95–103
P07	Capacity Complexes, One-Bit Persistence, and Operator-Ideal Criticality	8	104–111
P08	Critical Capacity Completion and Equivalent Forms of the Riemann Hypothesis	8	112–119
P09	Twin-Prime Survivors, Square-Horizon Occupation, and Moving-Boundary Dispersion	8	120–127
P10	Future-Relevant Representation Disagreement and Chronological Feature Restoration	8	128–135
P11	Reenactable Closure and Conditional Quantum-Causal Representation	9	136–144
P12	Local Indistinguishability, Global Obstruction, and Minimal Predictive Memory	7	145–151
P13	Directional Information Along Alpha-Particle Tracks and a Gas-Ladder Test	8	152–159
P14	Past-Unbounded Cosmology and an Exact Dust-Vacuum Return Condition	8	160–167
P15	Reflexive Predictive Integration: A Testable Systems Model of Conscious Processing	7	168–174
P16	Adaptive Predictive Systems with Native Representation Repair	10	175–184

ID	Paper	Pages	Volume pages
P17	Prime-Indexed Return Sectors and the Thirty-Six-Relation Closure of a Nine-State Carrier	8	185–192
A00	Master Scientific Atlas and Evidence Map	8	193–200

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Recurrence Exponents

Accessibility Growth and Predictive-State Promotion

Abhijit Singh

Contents

Abstract	3
1 Residual dynamics	4
2 Finite-dimensional theorem	4
3 The zero-exponent band	4
4 Accessibility exponent	5
5 Primitive Markov realization	5
6 Quantum-channel realization	5
7 Arithmetic realizations	6
8 Finite empirical promotion	6
9 Selection theorem	6
10 Scientific claim	6
11 Detailed recurrence calculus	7
12 Exact finite realizations	7
13 Nine-state critical-return realization	8
14 Finite promotion evidence and inference boundary	9
15 Falsification programme	9
References	9

Abstract

A residual left by an insufficient representation is assigned a cocycle and a typed growth exponent. Exact finite-dimensional theorems show that negative exponent gives exponential assimilation and positive exponent gives exponential escape. The zero-exponent class is proved to be a critical band requiring a secondary polynomial or persistence index. An independent accessibility exponent measures growth of source multiplicity hidden by the available continuation algebra. The calculus is realized in primitive Markov laws and quantum channels, related through explicit bridges to arithmetic capacity exponents, and compared with two finite held-out representation-restoration studies. Stationary witness margin, operator-ideal regularity, and finite empirical gain are separated from the recurrence exponent rather than conflated with it. Positive escape becomes an exact trigger for predictive-state enlargement. A new exact finite realization is included: a nine-state carrier with a 36-cell pair-relation frontier in which side-forgetting produces a fixed report while the complete source relation has critical period-two recurrence.

Keywords: Lyapunov exponent; predictive state; representation repair; quantum channel; criticality; accessibility

Scope and status: The recurrence theorems are exact under displayed cocycle assumptions. Cross-domain identifications require explicit intertwiners. Empirical sections establish finite future relevance rather than asymptotic exponents.

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1 Residual dynamics

A reporter compresses source histories into a current state. When a newly admitted continuation separates two histories that the reporter merged, the difference is a residual of the present representation. Let B_0 be a residual and let G_n be the linearized residual propagator between representation updates. With intrinsic increments $\Delta\tau_n > 0$,

$$B_n = G_{0:n}B_0, \quad G_{0:n} = G_{n-1} \cdots G_0, \quad \tau_n - \tau_0 = \sum_{j<n} \Delta\tau_j. \quad (1)$$

Define

$$\chi(B_0) = \limsup_{n \rightarrow \infty} \frac{\log \|G_{0:n}B_0\| - \log \|B_0\|}{\tau_n - \tau_0}. \quad (2)$$

The exponent is attached to a declared residual, propagator, norm class, and clock. It is not a free-floating label for any quantity that happens to have a boundary at one half.

2 Finite-dimensional theorem

In finite dimension all norms are equivalent, so χ is norm independent for a fixed clock.

Theorem 2.1 (Constant-operator formula). *Let G be a linear operator, $\tau_n = n$, and V_B the cyclic invariant subspace generated by nonzero B . Then*

$$\chi(B) = \log r(G|_{V_B}), \quad (3)$$

where r is spectral radius.

Proof. Jordan form expresses $G^n B$ as a finite sum of terms $n^k \lambda^n$. Polynomial factors vanish after logarithm and division by n ; the largest modulus eigenvalue with nonzero component in B remains. $\square$

Corollary 2.2 (Exponential trichotomy). *If $\chi(B) < 0$, the residual is exponentially assimilated. If $\chi(B) > 0$, it grows exponentially along a subsequence and eventually exceeds every bounded residual budget. The case $\chi(B) = 0$ is the critical exponential band.*

3 The zero-exponent band

Zero exponential rate is not one dynamical behaviour. Define

$$\beta(B) = \limsup_{n \rightarrow \infty} \frac{\log(\|B_n\|/\|B_0\|)}{\log(1+n)}, \quad p_\infty(B) = \limsup_{n \rightarrow \infty} \|B_n\|. \quad (4)$$

Within $\chi = 0$:

- $\beta < 0$ gives subexponential assimilation;
- $\beta > 0$ gives subexponential escape;
- $p_\infty > 0$ gives retained contrast;
- bounded unitary or oscillatory modes have $\chi = 0$ and nonzero persistence.

The selection law is therefore: assimilate when $\chi < 0$; promote when $\chi > 0$; and run a secondary critical-band test when $\chi = 0$.

4 Accessibility exponent

Residual growth and inaccessible provenance are different. For a resource-bounded continuation family $\Lambda(B_\tau)$, define

$$\eta = \limsup_{\tau \rightarrow \infty} \frac{1}{\tau} \log \mu([F]_{\Lambda(B_\tau)}), \quad (5)$$

where the equivalence class contains source occurrences indistinguishable to the available continuation algebra. The pair (χ, η) distinguishes:

1. $\chi < 0, \eta \approx 0$: transparent assimilation;
2. $\chi < 0, \eta > 0$: local relaxation with growing hidden provenance;
3. $\chi \approx 0, \eta > 0$: critical recurrence with growing inaccessible correlation;
4. $\chi > 0$: state-space failure and representation genesis.

This two-coordinate system is required for horizon and thermodynamic interpretations in which geometry can be stable while accessibility changes.

5 Primitive Markov realization

For a primitive column-stochastic matrix C with stationary vector π , let

$$\Pi = \pi \mathbf{1}^\top, \quad R = C - \Pi. \quad (6)$$

Then $C^n = \Pi + R^n$ and every zero-sum contrast d obeys

$$\chi(d) = \log r(R|_{V_d}) < 0. \quad (7)$$

For the corrected four-state repeated instrument, the nontrivial eigenvalues are

$$0.35301358, \quad 0.27498875, \quad 0.19041902. \quad (8)$$

The slowest contrast exponent is

$$\chi_{\text{mark}} = \log(0.35301358) \approx -1.0412 \quad (9)$$

per use. This explains exponential loss of source memory while the stationary entropy rate remains positive.

6 Quantum-channel realization

For a primitive quantum channel $\mathcal{T}$, a traceless eigenoperator X with eigenvalue λ has

$$\chi(X) = \log |\lambda|. \quad (10)$$

A product of unread projective channels contracts toward the common commutant. Under irreducibility only the identity remains fixed, so every traceless state contrast has $\chi < 0$.

A contextual witness margin

$$\Delta_{\text{wit}} = q - \beta_{\text{NC}} \quad (11)$$

is a different coordinate. A channel can have $\chi_{\text{state}} < 0$ while sustaining a positive state-independent witness. The complete protocol class is therefore at least two-dimensional:

$$(\chi_{\text{state}}, \Delta_{\text{wit}}). \quad (12)$$

7 Arithmetic realizations

The orientation operator has a Schatten boundary at $\Re s = 1/2$, but this operator-regularity line is not itself a recurrence exponent. The recurrence coordinates are separately defined:

$$\chi_{\text{cap}} = \limsup_{N \rightarrow \infty} \frac{\log(A_N/N)}{\log N}, \quad (13)$$

for the weighted capacity excess, and

$$\chi_{\text{RH}} = \limsup_{N \rightarrow \infty} \frac{1}{N} \log(1 + \Theta_N) \quad (14)$$

for the causal-tail escape representation. Within the frozen arithmetic bridge, RH is equivalent to $\chi_{\text{cap}} = 0$ and to absence of positive transverse escape. The half-cost operator boundary and recurrence zero line coincide only after the explicit source-preserving bridge, not by visual similarity.

8 Finite empirical promotion

The relation-aware representation study and the chronological channel-restoration study prove future relevance on finite held-out data. They do not estimate an asymptotic exponent unless a repeated residual propagation law is declared.

Three independent representation rates 92.96%, 90.74%, and 91.48% combine relationally to 97.78%. The omitted environmental channel changes untouched-future coordinate accuracy from 72.0836% to 74.7125% and exact next-vector accuracy from 3.7540% to 7.2492%. These are finite promotion diagnostics: the discarded contrast was not future-inert.

9 Selection theorem

Theorem 9.1 (Recurrence-based state selection). *Let a residual evolve under a fixed finite-dimensional cocycle, and suppose state promotion has a declared finite cost. Then:*

1. $\chi < 0$ is sufficient for exponential assimilation;
2. $\chi > 0$ is sufficient for eventual failure of every bounded residual budget and therefore requires state promotion or rejection of the model;
3. $\chi = 0$ requires a secondary persistence or polynomial-rate test before either assimilation or promotion is justified.

The theorem is exact under the displayed assumptions. Cross-domain unification requires an explicit intertwiner identifying each domain's residual, propagator, clock, and norm class.

10 Scientific claim

The genuine result is not that one familiar Lyapunov exponent has been renamed. It is the placement of recurrence analysis inside a source-preserving representation calculus: residuals are generated by false closure, the exponent classifies whether the current representation can assimilate them, and positive escape becomes a mathematically explicit trigger for state-space enlargement. The cross-domain instances demonstrate the breadth of the calculus; they do not erase their local assumptions.

11 Detailed recurrence calculus

Let V be a finite-dimensional normed space, let $G_n : V \rightarrow V$ be the residual propagator between two representation updates, and let the intrinsic clock satisfy $\Delta\tau_n > 0$. The residual evolution is

$$B_n = G_{0:n}B_0, \quad G_{0:n} = G_{n-1} \cdots G_0, \quad (15)$$

and the recurrence exponent is

$$\chi(B_0) = \limsup_{n \rightarrow \infty} \frac{\log \|G_{0:n}B_0\| - \log \|B_0\|}{\tau_n - \tau_0}. \quad (16)$$

Because all norms are equivalent in finite dimension, changing the norm changes the numerator by a bounded additive term and therefore leaves χ invariant whenever $\tau_n \rightarrow \infty$.

Theorem 11.1 (Constant-operator formula). *For $G_n = G$ and $\tau_n = n$, let V_B be the cyclic invariant subspace generated by B . Then*

$$\chi(B) = \log r(G|_{V_B}), \quad (17)$$

where r is spectral radius.

Proof. Put the restriction of G to V_B in Jordan form. Every component of $G^n B$ has the form $p(n)\lambda^n$, where p is a polynomial whose degree is smaller than the corresponding Jordan-block size. After taking a logarithm and dividing by n , the polynomial term contributes $o(1)$. The eigenvalue of largest modulus that has nonzero projection from B therefore determines the limit superior. $\square$

11.1 The zero-exponent band

The sign trichotomy is exact at exponential scale, but $\chi = 0$ contains several different behaviours. Define

$$\beta(B) = \limsup_{n \rightarrow \infty} \frac{\log(\|B_n\|/\|B_0\|)}{\log(1+n)}, \quad p_\infty(B) = \limsup_{n \rightarrow \infty} \|B_n\|. \quad (18)$$

Then polynomial decay has $\chi = 0$ and $\beta < 0$; bounded nonzero oscillation has $\chi = 0$ and $p_\infty > 0$; polynomial growth has $\chi = 0$ and $\beta > 0$. A scientifically usable selection rule is therefore

condition	dynamical reading	representation action
$\chi < 0$	exponential assimilation	compress after backward tests
$\chi > 0$	exponential escape	promote or reject the state model
$\chi = 0, \beta < 0$	subexponential assimilation	retain until the decay certificate closes
$\chi = 0, p_\infty > 0$	critical retention	preserve as an active coordinate
$\chi = 0, \beta > 0$	subexponential escape	promote under the declared resource budget

12 Exact finite realizations

12.1 Primitive Markov law

For a primitive column-stochastic matrix C with stationary vector π , define

$$\Pi = \pi \mathbf{1}^\top, \quad R = C - \Pi. \quad (19)$$

Then $\Pi^2 = \Pi$, $\Pi R = R \Pi = 0$, and

$$C^n = \Pi + R^n. \quad (20)$$

Every zero-sum contrast evolves only through R . In the corrected four-record instrument, the nontrivial eigenvalues are

$$0.35301358, \quad 0.27498875, \quad 0.19041902, \quad (21)$$

so the slowest exact exponential rate is

$$\chi_{\text{mark}} = \log(0.35301358) \approx -1.0412 \quad (22)$$

per reuse. This explains exponential loss of source memory; it does not imply zero entropy production, because the stationary instrument continues to emit outcomes.

12.2 Quantum channel and witness margin

For a primitive quantum channel $\mathcal{T}$, a traceless eigenoperator X with eigenvalue λ has $\chi(X) = \log |\lambda|$. The stationary contextual-witness margin

$$\Delta_{\text{wit}} = q - \beta_{\text{NC}}, \quad q = \text{Tr}(\sigma R_{\text{score}}), \quad (23)$$

is a second coordinate. A channel may satisfy $\chi_{\text{state}} < 0$ while $\Delta_{\text{wit}} > 0$, as in a state-independent contextual witness. Contraction, raw entropy, and certification are therefore not interchangeable names for one scalar.

12.3 Arithmetic criticality

The squarefree orientation operator has singular values $n^{-\sigma}$ and is Hilbert–Schmidt exactly for $\sigma > 1/2$. This is an operator-ideal threshold. The capacity recurrence coordinate

$$\chi_{\text{cap}} = \limsup_{N \rightarrow \infty} \frac{\log(A_N/N)}{\log N} \quad (24)$$

asks a different question: whether the coercive excess grows subpower or by a positive power. The two boundaries are related by the complete arithmetic construction, but they are not identical merely because both place special structure at one half.

13 Nine-state critical-return realization

The prime-observer carrier gives an exact finite realization of critical retention. Write

$$A = \{1\} \sqcup (\{2, 3, 5, 7\} \times \{+, -\}), \quad (25)$$

where the four two-sided sectors correspond to (2, 9), (3, 8), (5, 6), and (7, 4). The return map flips the side and preserves the prime-sector label. For

$$r_q = e_{q,+} - e_{q,-}, \quad (26)$$

the return satisfies $Gr_q = -r_q$, so $\|G^n r_q\| = \|r_q\|$ and $\chi(r_q) = 0$. A reporter retaining only q sees a fixed point; the full state has period two. This is a strict example of

$$U(r_q) = 0 \quad \not\Rightarrow \quad \chi(r_q) < 0. \quad (27)$$

The complete pair-relation frontier of the nine states has $\binom{9}{2} = 36$ cells, decomposed as eight unit relations, four return relations, and twenty-four transverse relations. The finite theorem is developed in P17. Its role here is to distinguish report-level cancellation from dynamical assimilation.

14 Finite promotion evidence and inference boundary

The relation-aware representation experiment and the chronological feature-restoration experiment establish that a discarded difference changed an untouched future. That is a finite promotion certificate. It is not an estimate of an asymptotic χ unless a repeated residual propagation law is separately specified and measured. This distinction is essential: the exponent is a property of a dynamical cocycle, while held-out improvement is evidence that a particular quotient was insufficient.

15 Falsification programme

The recurrence-selection proposal fails as a cross-domain law if no source-preserving map sends the domain residual and its continuation to the displayed cocycle; if the sign is not invariant under admissible clock changes; or if a promoted residual repeatedly fails to alter later continuation after ablation. These are operational failure conditions rather than editorial reservations.

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Future-Complete Quotients

Recursive Distinction

Abhijit Singh

Contents

Abstract	3
1 Introduction	4
2 Principal results and scope	4
3 Complete distinction and behavioural closure	5
4 Generated perspectives, predictive closure, and target-free continuation	19
5 Prime-source occurrences and the 36-relation frontier	34
6 Related work	35
7 Limitations and falsifiability	36
8 Reproducibility and research artefacts	37
9 Open problems	37
10 Conclusion	37
11 References	38
12 Appendix A. Theorem and provenance register	40

Abstract

A finite source calculus is developed from one formal primitive: a distinction has a complete family of resultants, and every resultant may distinguish again. Finite unlabelled non-plane hierarchies retain containment, multiplicity, and complete resultant families. Complete recursive views and complete one-occurrence continuation decks reconstruct every finite source hierarchy. The central closure theorem characterizes exactly when a quotient reporter supports an autonomous continuation law: the complete reported response must be constant on every reporter fibre. Failure induces a least response-forced refinement; success implies stable blindness to every distinction erased by the quotient. The calculus further constructs least closed combinations of reporters, finite predictive quotients, target-free expansion systems, and common-source relations between incomparable perspectives. An occurrence-sensitive prime triad and its nine-state, thirty-six-relation realization are added as a concrete quotient-closure test.

Keywords: recursive distinction; predictive quotient; behavioural closure; finite reconstruction; reporter refinement

Scope and status: The reconstruction, closure, refinement, finite quotient, and common-source results are theorem-level statements in the declared finite constructor. Interpretations beyond that constructor are stated separately.

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1 Introduction

Any formal account of distinction begins after at least one distinction has already been made. Terms such as *state*, *event*, *set*, and *operation* belong on the reportable side of that boundary. The first task is therefore to mark where formal description begins without representing the absence of description as one more object.

We use P_0/R for that boundary. It denotes no reportable distinction and is assigned no carrier, point, value, state, law, temporal order, modality, or truth value. This is a constraint on the theory's language rather than a claim about a hidden substrate. Every symbol in the paper is already a distinction, so every proof and computation belongs to P_1 , the first reportable perspective, whose primitive separation is binary.

The sole source rule is

D : a distinction has resultants, and every resultant may distinguish.

Completeness and recursion specify how the primitive is reported. Completeness forbids the omission of a resultant; recursion permits the same rule to act at any resultant. Neither introduces a clock, probability, coordinate system, objective, or update schedule. A complete q -resultant family written in P_1 is consequently an encoding of q retained occurrences. When $q > 2$, indexing that record by a hypothetical P_{q-1} does not grant access to a native nonbinary logic.

The least finite transcription that keeps containment, multiplicity, and complete resultant families visible is an unlabelled, non-plane rooted hierarchy. A record is either an unresolved terminal $\bullet$, or a containing occurrence with an unordered multiset of at least two child records. Equal child descriptions may repeat: descriptive equality does not cancel occurrence. The hierarchy is a declared representation within P_1 , not an ontological identification of R with a tree.

Given a source hierarchy and a reporter Q that identifies some source states, when does an operation on the source induce a law on the reported states? The closure criterion answers this directly: two states with the same report must have the same complete reported response. Otherwise the reporter has merged states with incompatible futures. Retaining the incompatibility, rather than choosing or averaging an outcome, determines the least refinement needed to make the report lawful.

Closure also fixes a limit. Once a law descends to a quotient, all of its iterates remain functions of that quotient. The law cannot reconstruct a distinction that its own state space consistently erased. Thus autonomy and blindness are not separate phenomena in this calculus; both are consequences of the same fibre condition.

The development proceeds from the finite hierarchy and its reconstructive views to complete output decks, quotient closure, forced refinement, and closed reporter combination. Later sections introduce finite interaction systems, a target-free expansion process, and common-source comparisons. These latter constructions carry additional hypotheses and are kept separate from the primitive core.

2 Principal results and scope

The central theorem may be stated without interpretive terminology:

Complete-behaviour closure. Let Q report the declared finite hierarchies, whose complete occurrence decks are finite. A single-valued continuation law descends to the Q -states if and only if complete reported response is constant on every fibre of Q . If constancy fails, equality through all finite response depths defines the least finer reporter on which every reported arity closes.

The paper establishes five groups of results.

1. **Source structure and reconstruction.** The declared hierarchy grammar transcribes complete recursive distinction. Within it, the containing occurrence is the unique unselected site at which one retained role persists through a distinction without deletion, duplication, or child selection. Complete recursive views reconstruct a hierarchy at finite depth, as does the deck formed by replacing every terminal occurrence once. Counts and flattened arity profiles are strictly coarser because they lose shared containment.
2. **Law, refinement, and stable loss.** The closure criterion is necessary and sufficient. Its failure determines a least response-forced refinement; its satisfaction proves stable blindness, since response iteration preserves the reporter's fibres. Grouping-depth reporters form a closed hierarchy, and the least closed combination of two reporters is obtained by closing their direct pair. This combination is associative and commutative up to fibres, with $X \vee_c X \simeq Cl_\infty(X)$.
3. **Finite effect and interaction quotients.** On a declared finite carrier, an effect family induces the least same-referent quotient sufficient for those effects. For codomains with at least two values, properly finer reports admit strictly more compatible candidate laws. For $|X| = q \geq 2$ and a pointed operation of arity $m \geq 1$, finite interaction contexts generate the coarsest focus-preserving predictive quotient and stabilise after at most $q - 2$ strict refinements.
4. **Test systems.** A target-free binary expansion process yields local replacement, name invariance, commutation at separate current occurrences, ancestry-restricted histories, exact continuation factorisation, and nontermination. A second construction separates histories, ordered forms, order-free forms, and availability, making the permanence of closed information loss explicit.
5. **Relations among perspectives.** For two quotients of a common source, direct translation exists exactly along refinement. Incomparable quotients are related through the source-induced relation; their joint quotient is the least report retaining both distinction families and may be strictly finer than either marginal.

3 Complete distinction and behavioural closure

3.1 Formal boundary and provenance

The calculus begins with a restriction: its first inscription must not be mistaken for a description of what precedes inscription. We therefore separate the formal boundary from every object constructed beyond it.

3.1.1 The P_0/R boundary

Boundary statement. P_0/R denotes no reportable distinction. It is assigned no carrier or internal relation and is not identified with a familiar mathematical object. Any such identification would already distinguish that representation from alternatives, and so would belong on the reportable side of the boundary.

The notation is eliminative, not descriptive: it marks the limit of positive formal attribution. Even the sentence “there is no reportable distinction” is written from a reportable perspective. It fixes the provenance of the calculus rather than supplying a positive theory of R .

3.1.2 The first reportable perspective

Perspective convention. P_1 is the first reportable perspective: one distinction with two resultants. Every word, equation, diagram, proof, and program in this paper is consequently a P_1 inscription. The calculus cannot step outside that representational condition to certify the native experience or mathematics of another perspective.

A binary medium may nevertheless encode a finite family with $q > 2$ distinguishable occurrences. Accordingly, “a q -resultant distinction” means precisely:

P_1 records, without deleting an occurrence, a complete family of q distinguishable resultants.

This is an external P_1 presentation, not a claim of access to a native nonbinary perspective.

Figure 1 records this asymmetry: the boundary is not a source node, while binary and higher-arity records are constructions on its reportable side.

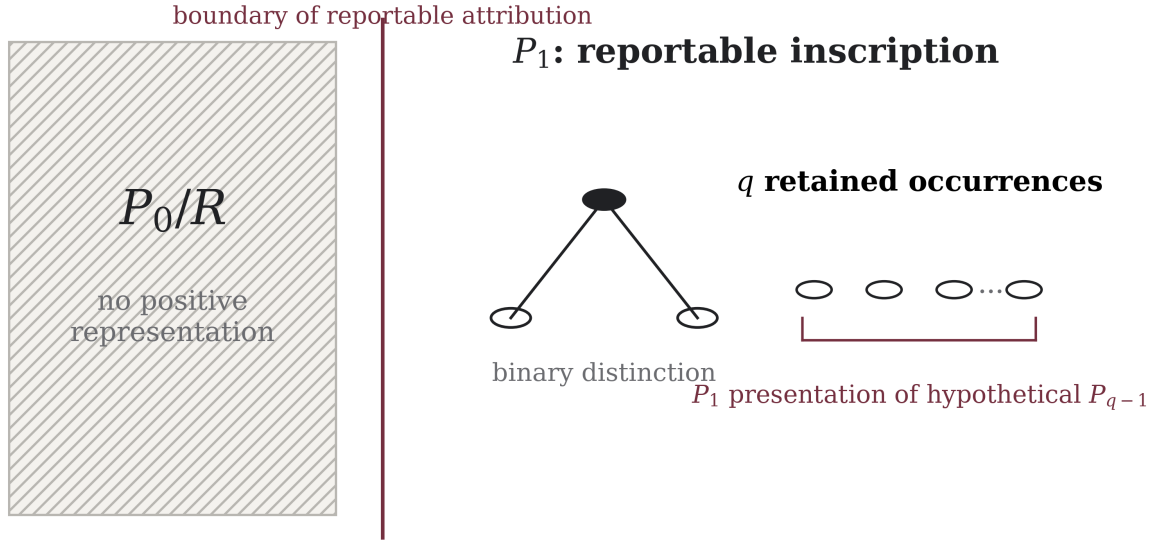


Figure 1. Representation boundary and perspective indexing. The vertical rule marks the limit of positive attribution; no transition is drawn across it.

3.1.3 The sole source rule

The construction retains one source rule:

D : a distinction has resultants, and every resultant may distinguish.

Completeness means that no resultant is discarded from the report; recursion means that the same rule is applied again to a resultant. Neither adds a second source rule. In particular, D contains no scheduler, coordinates, probability law, metric, conservation equation, logical valuation, or target phenomenon.

Definitions below specify the mathematical constructor; numbered propositions, theorems, and corollaries state consequences of their displayed hypotheses. Appendix A records their provenance.

3.2 The minimal complete hierarchy

3.2.1 Grammar and equality

Definition 3.2.1 (finite complete hierarchy). Let $\mathcal{H}$ be generated by

$$T ::= \bullet \quad \text{or} \quad \langle T_1, \dots, T_q \rangle, \quad q \geq 2.$$

The symbol $\bullet$ records a resultant with no further reported distinction. The term $\langle T_1, \dots, T_q \rangle$ records a containing occurrence and all of its q present resultants. Children are unordered, since D supplies no order; equal descriptions retain their occurrence multiplicity rather than being identified.

Equality in $\mathcal{H}$ is therefore isomorphism of finite unlabelled, non-plane, rooted hierarchies with multiplicity. A node of the underlying rooted hierarchy is called an *occurrence*. An occurrence is *terminal* when it is represented by $\bullet$, and *internal* when it carries a nonempty child family.

This grammar is minimal relative to four declared reporting obligations:

1. a distinction leaves one containing occurrence at which it occurred;

2. every present resultant is retained;
3. every resultant may recursively carry the same form of record;
4. no child order or label is inserted without another supplied distinction.

“Minimal” is relative to these four obligations. The claim is not that R is a hierarchy, but that $\mathcal{H}$ is the least explicit P_1 transcription used here that records recursive containment and the complete resultant family without introducing labels or order.

For $T \in \mathcal{H}$, let $L(T)$ be its number of terminal occurrences and $h(T)$ the maximum number of containment edges from the root to a terminal. For an internal occurrence v , let $q(v) \geq 2$ denote its arity. These are measurements of a completed record, not source primitives.

We use finite multisets throughout. The notation $\{\{a_1, \dots, a_m\}\}$ retains the number of occurrences of each value; multiset equality includes multiplicity. If $f : A \rightarrow B$ and M is a finite multiset on A , then f_*M denotes its multiplicity-preserving pushforward.

3.2.2 The information order on reporters

Definition 3.2.2 (reporter and refinement). A reporter on $\mathcal{H}$ is any function $Q : \mathcal{H} \rightarrow X$. For reporters X and Y , write

$$X \preceq Y$$

when equality of Y -reports implies equality of X -reports. Equivalently, X factors through Y . Thus Y retains every distinction retained by X , and perhaps more. Reporters are identified *up to fibres*: $X \simeq Y$ means $X(T) = X(U)$ if and only if $Y(T) = Y(U)$ for all $T, U \in \mathcal{H}$.

The order compares retained distinctions only; it carries no independent ranking by truth, observer quality, physical resolution, or temporal progress.

3.2.3 Why the containing occurrence persists

Suppose an occurrence carries one focal role x . After it distinguishes into a complete unlabelled family, the old role could apparently be erased, assigned to one child, copied to every child, or retained at the containing occurrence. The first choice violates retention; the second introduces an unsupplied child selection; the third turns one role into q . Only the fourth preserves exactly one focal role.

Proposition 3.2.3 (unselected persistence). Inside the declared hierarchy transcription, the containing occurrence is the unique placement that retains one old focal role through its distinction without deleting, selecting, or duplicating that role.

This is an identity principle of the transcription. It assigns no objecthood or persistence to R : the children form the complete new resultant family, with no default heir to the old role.

3.2.4 First invariants and the cost of flattening

The completed record supports exact accounting identities that are absent from the source rule.

Proposition 3.2.4 (balance). Every $T \in \mathcal{H}$ satisfies

$$L(T) = 1 + \sum_{v \text{ internal}} (q(v) - 1).$$

Proof. The unresolved record $\bullet$ has one terminal and no internal contribution. Replacing one terminal by a family of q resultants increases the terminal total by $q - 1$ and adds one internal occurrence contributing exactly $q - 1$ to the displayed sum. Every finite hierarchy is obtained by finitely many such replacements, so induction on the number of internal occurrences proves the identity. $\square$

For a homogeneous q -resultant hierarchy with I internal occurrences, the balance reduces to

$$L = 1 + (q - 1)I.$$

For homogeneous binary hierarchies this becomes $L = I + 1$. Both identities belong to the finite P_1 transcription.

Now put $e_v = q(v) - 1$. Then

$$L(T) - 1 = \sum_v e_v.$$

This scalar factors through two distinct losses:

$$\mathcal{H} \rightarrow \{\{e_v\}\} \rightarrow \sum_v e_v = L - 1.$$

The first arrow erases which internal occurrences contain which others. The second erases even the multiset decomposition of the total excess. To see the first loss directly, let $T_{a,b}$ have a children at the root and b children below each root child. Reversing the two layers yields $T_{b,a}$. For $a \neq b$ their containment histories differ, but both have ab terminals. The equality $ab = ba$ is therefore visible at the scalar shadow while the two complete hierarchies remain distinct.

Flattening can lose information even when terminal total, root arity, and the total number of grandchildren are all retained. Consider

$$A = \langle \bullet, \langle \bullet, \bullet \rangle, \langle \bullet, \bullet, \bullet, \bullet \rangle \rangle$$

and

$$B = \langle \bullet, \langle \bullet, \bullet, \bullet \rangle, \langle \bullet, \bullet, \bullet \rangle \rangle.$$

Both have seven terminals, three root children, and six grandchildren, but their complete nested child-family reports are respectively

$$\{\{\emptyset, 2[*], 4[*]\}\} \quad \text{and} \quad \{\{\emptyset, 3[*], 3[*]\}\}.$$

Hence a flattened occurrence union does not determine recursive nesting. An associativity law on a flat algebra may be perfectly sound, but the quotient has changed the object under study: scalar and flat reporters are proper shadows of the hierarchy.

3.3 Complete present-resultant views

Two complete families will be used: the children present at an occurrence, and the possible one-occurrence continuations of a terminal. They are different constructions.

3.3.1 Present-family readout

Definition 3.3.1 (present-resultant operator). For any expression e evaluable on hierarchies, define

$$\llbracket \Delta e \rrbracket (\bullet) = \emptyset,$$

and

$$\llbracket \Delta e \rrbracket (\langle T_1, \dots, T_q \rangle) = \{ \{ \llbracket e \rrbracket (T_1), \dots, \llbracket e \rrbracket (T_q) \} \}.$$

Thus Δ returns the entire present child multiset. It neither selects nor labels a child, and it performs no ordering, averaging, or scalarization.

3.3.2 Recursive self-view

Begin with one undifferentiated focal report

$$V_0(T) = *.$$

Apply the same report recursively to every present resultant:

$$V_{r+1}(\bullet) = (*, \emptyset),$$

$$V_{r+1}(\langle T_1, \dots, T_q \rangle) = (*, \{ \{ V_r(T_1), \dots, V_r(T_q) \} \}).$$

Write

$$V_\omega(T) := (V_r(T))_{r \geq 0},$$

so equality of V_ω -reports means equality at every finite view depth.

The first coordinate retains the focal role at its containing occurrence. The second records the absence or presence of a child family and, when present, applies the same reading to every child occurrence.

Theorem 3.3.2 (finite reconstruction). If $T \in \mathcal{H}$ has height h , then $V_{h+1}(T)$ determines T . Consequently, equality through every finite view depth has exactly the equality fibres of $\mathcal{H}$:

$$\boxed{V_{h(T)+1}(T) \text{ reconstructs } T, \quad V_\omega \simeq \mathcal{H}.}$$

Proof. We induct on the height. If $T = \bullet$, then $V_1(T) = (*, \emptyset)$, which identifies the terminal record. Let $T = \langle T_1, \dots, T_q \rangle$ have height $h > 0$. Its V_{h+1} -report contains, with multiplicity, the family $\{ \{ V_h(T_1), \dots, V_h(T_q) \} \}$. Every child has height at most $h - 1$; by the induction hypothesis, its V_h -report reconstructs it. Reconstruct every member of the complete child multiset and place that multiset under the single containing occurrence. This recovers T . Because every finite T has finite height, equality at all finite view depths implies equality in $\mathcal{H}$, while equality in $\mathcal{H}$ plainly implies equality of every view. $\square$

Theorem 3.3.3 (strict view hierarchy). Erasing the deepest retained family from a V_{r+1} -report yields its V_r -report, so

$$V_0 \preccurlyeq V_1 \preccurlyeq V_2 \preccurlyeq \cdots .$$

Each refinement is strict on the unbounded family of finite hierarchies.

Proof. The forgetful map is immediate from the recursive definition. A terminal and an internal occurrence agree under V_0 and differ under V_1 . More generally, if T and U first differ at view depth $r + 1$, embed each beneath the same newly supplied containing context; their first difference is shifted to depth $r + 2$. Hence examples witnessing strictness exist at every finite level. $\square$

Here r counts nested applications of the P_1 readout. It has no assigned interpretation as time, distance, physical scale, knowledge, computational complexity, or perspective index.

Reconstruction requires neither names nor child order. Complete multiplicity and recursive containment suffice. Present-resultant access is therefore faithful on finite hierarchies, whereas scalar count identifies many nonisomorphic records.

3.4 Complete one-occurrence future decks

3.4.1 Definition and elementary invariants

Fix an externally reported arity $q \geq 2$.

Definition 3.4.1 (complete occurrence deck). For $T \in \mathcal{H}$, let $E_q(T)$ be the finite multiset obtained by replacing each terminal occurrence of T , one occurrence at a time, by a q -resultant occurrence $\langle \bullet, \dots, \bullet \rangle$. If symmetric terminal occurrences yield isomorphic outputs, the output hierarchy appears with the corresponding multiplicity.

The deck is an exhaustive family of one-occurrence continuations, not a probability distribution. Each output $U \in E_q(T)$ satisfies

$$L(U) = L(T) + q - 1,$$

and the total occurrence multiplicity of the deck is

$$|E_q(T)|_{occ} = L(T).$$

Where Δ reads the current children of one focal occurrence, E_q enumerates every terminal occurrence at which another distinction can be recorded.

3.4.2 Source reconstruction from one complete deck

Theorem 3.4.2 (deck reconstruction). For every fixed integer $q \geq 2$ and all finite $T, U \in \mathcal{H}$,

$$E_q(T) = E_q(U) \Rightarrow T = U.$$

Proof. Let $h = h(T)$. Expanding a terminal at depth h creates an output of height $h + 1$, while expanding any shallower terminal creates an output of height at most h . Hence the maximum height among members of $E_q(T)$ is $h + 1$, and every maximum-height member results from expansion at a maximum-depth terminal.

Choose a maximum-height output W . In the source T , no internal occurrence can lie at depth h , since it would have a terminal descendant below depth h , contradicting maximality of h . Therefore W has exactly one internal occurrence at depth h : the newly inserted q -resultant occurrence. Contract that occurrence to a

terminal. The contraction uniquely recovers T . Thus a complete deck determines its source. Equal decks have a common maximum-height member and reconstruct the same source, proving $T = U$. $\square$

Thus the unquotiented one-step continuation family is faithful to the full finite hierarchy, not merely to terminal count. Before passing to quotients, the following subsection records the exact locality available when two replacements occur on opposite sides of a retained cut. Section 3.5 then asks which parts of this continuation structure survive a quotient report.

3.4.3 Persistent cuts and restricted two-step interaction

Occurrence provenance permits an exact cut without choosing a child or flattening either side. For a finite hierarchy T , let $\Lambda(T)$ be its set of terminal occurrences. For a marked occurrence v , write $D = T|_v$ for the subhierarchy rooted at v and $T = C[D]$ for the associated one-hole context. The terminals below v form $\Lambda_v(T) \subseteq \Lambda(T)$, canonically identified with $\Lambda(D)$. Write

$$\Lambda_v^{out}(T) = \{u \in \Lambda(T) : u \notin \Lambda_v(T)\}$$

for the terminal occurrences outside the marked subhierarchy. Let

$$B_q = \langle \bullet, \dots, \bullet \rangle, \quad q \geq 2,$$

where the bracket contains exactly q terminal entries, and write $T[u \leftarrow B_q]$ for replacement of the terminal occurrence u by B_q . Define the occurrence-indexed multisets

$$E_q^{in}(T; v) = \{T[u \leftarrow B_q] : u \in \Lambda_v(T)\},$$

$$E_q^{out}(T; v) = \{T[u \leftarrow B_q] : u \in \Lambda_v^{out}(T)\}.$$

Outputs that become isomorphic are aggregated only after their originating terminal occurrences have been counted.

Theorem 3.4.3 (persistent cut and exact reunion). For every finite $T \in \mathcal{H}$, every marked occurrence v , and every integer $q \geq 2$,

$$E_q(T) = E_q^{in}(T; v) \uplus E_q^{out}(T; v).$$

Moreover,

$$E_q^{in}(T; v) = \{C[W] : W \in E_q(D)\},$$

with multiplicity preserved. Every member of the inside deck retains the outer context $C[-]$, and every member of the outside deck retains the marked subhierarchy D unchanged.

Proof. Every terminal occurrence of T lies in exactly one of the disjoint sets $\Lambda_v(T)$ and $\Lambda_v^{out}(T)$. The definition of $E_q(T)$ indexes one replacement by each terminal occurrence, so this set partition induces the displayed multiset disjoint union, including all multiplicities. Under the canonical identification $\Lambda_v(T) \cong \Lambda(D)$, replacing u below v changes only D and yields $C[D[u \leftarrow B_q]]$; this proves the inside-deck formula. A replacement outside $\Lambda_v(T)$ is disjoint from the marked occurrence and therefore leaves D unchanged. $\square$

The cut also isolates an exact, deliberately local order-independence result.

Theorem 3.4.4 (opposite-side commutation). Let $T = C[D]$ and ν be as above. Fix integers $q, r \geq 2$. For every $u \in \Lambda_\nu(T)$ and $w \in \Lambda_\nu^{\text{out}}(T)$, both terminal occurrences survive when the other is replaced, and

$$T[u \leftarrow B_q][w \leftarrow B_r] = T[w \leftarrow B_r][u \leftarrow B_q].$$

Consequently, the complete occurrence-indexed two-step multiset obtained by a q -replacement inside the cut followed by an r -replacement outside is equal to the multiset obtained in the opposite order.

Proof. The terminal occurrences u and w are distinct and lie in disjoint occurrence regions. Replacing u changes only the subhierarchy at u , so w remains a terminal occurrence with the same provenance; symmetrically, replacing w leaves u available. Both composites are therefore the simultaneous hierarchy obtained from T by inserting B_q at u and B_r at w . Pairing the two orderings by the same occurrence pair (u, w) is a bijection of their finite indexing sets and preserves the resulting hierarchy. Aggregating isomorphic outputs thus preserves equal multiplicities. $\square$

The restriction to opposite sides of a retained cut is essential. Identical or nested occurrences require a separate interaction analysis. This result supplies a concrete two-step law at source level; the quotient criterion below decides whether any reporter retains enough information for that law, or any complete continuation law, to descend.

3.5 Autonomous quotient laws

A reporter may retain less than the full hierarchy. The pertinent question is whether its reported state determines its complete reported behaviour.

Let $Q : \mathcal{H} \rightarrow X$ be any declared reporter. For a deck $E_q(T)$, define its pushed report

$$Q_*E_q(T) = \{\{Q(W) : W \in E_q(T)\}\},$$

with occurrence multiplicity retained.

Definition 3.5.1 (autonomous q -law). An autonomous continuation law for Q at arity q is a function

$$F_{Q,q} : Q(\mathcal{H}) \rightarrow MFin(Q(\mathcal{H}))$$

such that

$$F_{Q,q}(Q(T)) = Q_*E_q(T)$$

for every $T \in \mathcal{H}$. Here $MFin(X)$ denotes finite multisets on X .

Thus an autonomous law is internal to the reporter: the present Q -state determines the complete multiset of next Q -states without reference to a hidden representative in $\mathcal{H}$.

Theorem 3.5.2 (autonomy/closure criterion). An autonomous q -law exists if and only if the complete pushed deck is constant on every Q -fibre:

$$Q(T) = Q(U) \Rightarrow Q_*E_q(T) = Q_*E_q(U) \quad \text{for all } T, U \in \mathcal{H}.$$

Proof. If $F_{Q,q}$ exists and $Q(T) = Q(U) = x$, then

$$Q_*E_q(T) = F_{Q,q}(x) = Q_*E_q(U),$$

so the condition is necessary. Conversely, suppose the condition holds. For $x \in Q(\mathcal{H})$, choose any T with $Q(T) = x$ and define $F_{Q,q}(x) = Q_*E_q(T)$. Fibre constancy makes this definition independent of the representative, so it is a function satisfying the required equation. $\square$

Figure 2 separates the two cases of Theorem 3.5.2: descent when complete decks agree on every fibre, and refinement when they do not.

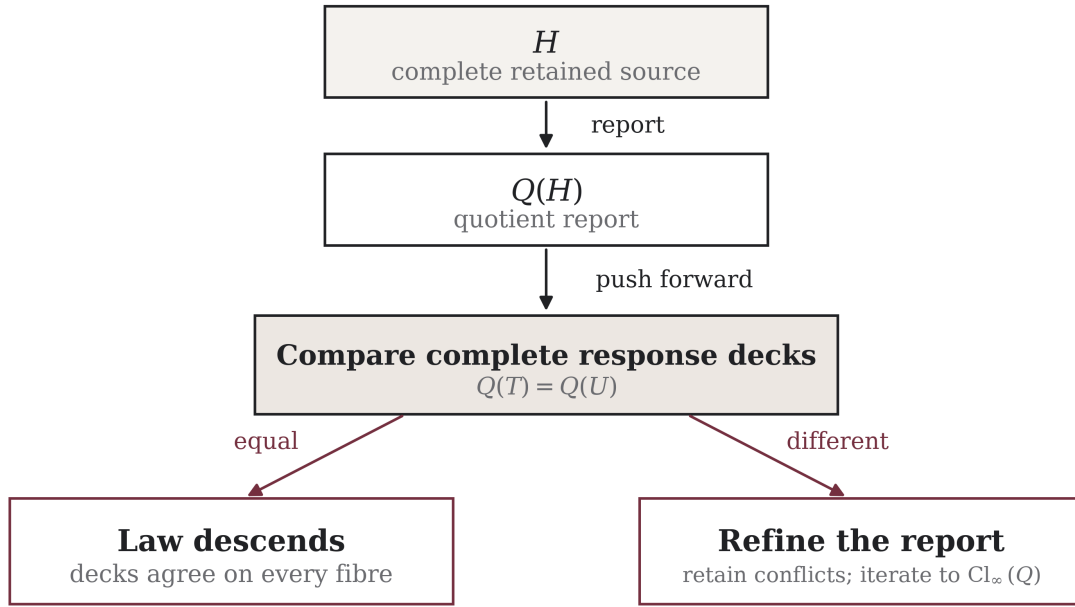


Figure 2. Complete-behaviour closure. Fibre constancy yields an autonomous quotient law; conflicting decks force the least refinement that retains every behaviourally relevant distinction.

We call Q *q-closed* when it satisfies the criterion, and *all-arity closed* when it is *q-closed* for every integer $q \geq 2$.

If closure fails, the complete response at a fibre is the set of all decks it supports:

$$\widehat{F}_{Q,q}(x) = \{Q_*E_q(T) : Q(T) = x\}.$$

This set-valued family is not an autonomous single-valued law. Its plurality records one precise fact: Q has erased a distinction on which its complete response depends. No probabilistic or stochastic interpretation follows without additional structure.

Corollary 3.5.3 (count forced by complete multiplicity). Every q -closed reporter Q refines terminal count L .

Proof. If $Q(T) = Q(U)$, closure gives equal pushed decks. Equal finite multisets have equal total multiplicity. These multiplicities are respectively $L(T)$ and $L(U)$, hence $L(T) = L(U)$. $\square$

Terminal count is itself all-arity closed, because its complete law is

$$L \mapsto L \text{ occurrences of } L + q - 1.$$

Therefore L is the coarsest reporter closed under complete terminal replacement with occurrence multiplicity. The result is operation-relative: count is not sufficient for Δ , whose recursive views reconstruct the full hierarchy.

The identity reporter is closed. So is the multiset of arity excesses $\{\{q(v) - 1\}\}$: a q -replacement appends $q - 1$, while the current excess sum fixes deck multiplicity by Proposition 3.2.4. Internal-node count I , by contrast, is not closed when multiplicity is retained. A binary star and a ternary star both have $I = 1$, but their decks contain two and three occurrences, respectively. Its first response refinement is exactly (I, L) , which is closed: every replacement sends (I, L) to $(I + 1, L + q - 1)$ with multiplicity L .

Accordingly, a reporter may already be autonomous, may acquire finitely many missing coordinates, or may require an unbounded refinement tower. The criterion decides which case occurs from the reported behaviour itself.

3.6 Response-forced completion

When closure fails, the conflicting responses identify which fibres must split. Iterating this observation gives a canonical completion without prescribing target classes.

3.6.1 Iterated response reporters

Fix a reporter $Q : \mathcal{H} \rightarrow X$. Define

$$R_0^Q(T) = Q(T).$$

Recursively, let $R_{r+1}^Q(T)$ retain $R_r^Q(T)$ together with, for every integer $q \geq 2$, the complete multiplicity deck of R_r^Q -reports of members of $E_q(T)$:

$$R_{r+1}^Q(T) = \left(R_r^Q(T), (R_{r*}^Q E_q(T))_{q \geq 2} \right).$$

Only equality fibres matter; no canonical code for the arity-indexed tuple is needed. Let $T \equiv_r^Q U$ mean $R_r^Q(T) = R_r^Q(U)$. Since R_{r+1}^Q retains R_r^Q ,

$$\equiv_{r+1}^Q \subseteq \equiv_r^Q.$$

Define the limiting equivalence

$$T \equiv_\infty^Q U \iff T \equiv_r^Q U \text{ for every finite } r.$$

Write $Cl_\infty(Q)$ for the quotient reporter whose fibres are the $\equiv_\infty^Q$ -classes.

3.6.2 Least-closure theorem

Theorem 3.6.1 (least all-response closed refinement). On the finite hierarchy family $\mathcal{H}$, $Cl_\infty(Q)$ is all-arity closed. Moreover, it is the least discriminating all-arity closed reporter that refines Q . Explicitly:

1. $Q \preceq Cl_\infty(Q)$;
2. $Cl_\infty(Q)$ is q -closed for every $q \geq 2$;
3. if S is all-arity closed and $Q \preceq S$, then $Cl_\infty(Q) \preceq S$.

Proof. The first statement follows because $R_0^Q = Q$ is retained at every later depth.

For closure, suppose $T \equiv_\infty^Q U$ and fix $q \geq 2$. The occurrence decks $E_q(T)$ and $E_q(U)$ are finite. For each $r \geq 0$, equality of $R_{r+1}^Q(T)$ and $R_{r+1}^Q(U)$ implies equality of the complete R_r^Q -decks. Hence there exists a multiplicity-preserving bijection between the occurrence-indexed members of $E_q(T)$ and $E_q(U)$ that pairs outputs having equal R_r^Q -reports.

Let Π_r be the finite set of all such bijections. Each Π_r is nonempty. Since equality at depth $r + 1$ implies equality at depth r , the sets descend:

$$\Pi_0 \supseteq \Pi_1 \supseteq \Pi_2 \supseteq \cdots$$

They are subsets of the one finite set of all bijections between the two finite occurrence decks. A descending sequence of nonempty subsets of a finite set has nonempty intersection. Choose $\pi \in \bigcap_r \Pi_r$. Every output is paired by π with an output equal under every finite R_r^Q , hence equal under $\equiv_\infty^Q$. The pushed $Cl_\infty(Q)$ -decks are therefore equal. Since q was arbitrary, the limiting reporter is all-arity closed.

For leastness, let S be any all-arity closed reporter refining Q , and suppose $S(T) = S(U)$. We prove by induction on r that $R_r^Q(T) = R_r^Q(U)$. The base case follows from $Q \preccurlyeq S$. Assume the assertion holds at depth r for every pair of equal S -reports. Because S is q -closed, equality $S(T) = S(U)$ gives, for every $q \geq 2$, a multiplicity-preserving pairing of $E_q(T)$ and $E_q(U)$ whose paired outputs have equal S -reports. By the induction hypothesis those paired outputs have equal R_r^Q -reports. Hence the complete R_r^Q -decks agree for every q , and $R_{r+1}^Q(T) = R_{r+1}^Q(U)$. Induction yields equality at all finite response depths, so $T \equiv_\infty^Q U$. Thus equality under S implies equality under $Cl_\infty(Q)$, which is exactly $Cl_\infty(Q) \preccurlyeq S$. $\square$

The universal property is relative to the stated carrier and operation: finite hierarchies, complete one-terminal replacement at every reported arity, and multiplicity-preserving decks.

Restricting the constructor to a fixed $q \geq 2$ gives, by the same proof, the least q -closed refinement $Cl_q(Q)$. A reporter can be closed for one replacement arity and fail for another; the all-arity completion is the common refinement obtained when every reported arity is retained.

In the preceding closure proof, finiteness is used when the descending family of compatible pairings is shown to have a common member. Although $\mathcal{H}$ is infinite, no uniform finite stabilisation depth is asserted. Infinite occurrence decks would require an additional compactness or choice principle.

Two states separate at the first response depth where their complete behaviour differs, and later separations occur only when subsequent responses reveal another conflict. State refinement and autonomous law are therefore determined together.

3.7 Stable blindness

3.7.1 The ceiling theorem

Response completion repairs unstable loss. It cannot recover an erasure already compatible with every supported response.

Theorem 3.7.1 (stable blindness). If Q is all-arity closed, then for every finite $r \geq 0$,

$$R_r^Q(T) = R_r^Q(U) \iff Q(T) = Q(U).$$

Consequently, $Cl_\infty(Q) \simeq Q$.

Proof. Because R_r^Q retains Q , equality of R_r^Q -reports implies equality of Q -reports. For the converse, proceed by induction. It is immediate for $r = 0$. Suppose R_r^Q is a function of Q . Since Q is closed, $Q(T) = Q(U)$

implies that for each $q \geq 2$ the occurrence decks $E_q(T)$ and $E_q(U)$ can be paired with equal Q -reports. Applying the inductive functional dependence to paired outputs gives equal R_r^Q -reports, so the complete R_r^Q -decks agree. Together with equality of the retained first coordinate, this gives $R_{r+1}^Q(T) = R_{r+1}^Q(U)$. Thus every R_r^Q is both a function of Q and a refinement retaining Q , so their fibres agree. $\square$

Erased distinctions may remain in $\mathcal{H}$; the theorem says that iteration of the autonomous Q -law supplies no behavioural witness for them. Unequal responses expose unstable erasure, while closure makes the erasure stable under the observed operation.

This yields a sharp ceiling. The views V_r reconstruct because Δ retains recursive child grouping. A closed scalar or flat reporter may erase that relation while retaining a perfectly autonomous replacement law; further iteration then cannot restore it.

Stable blindness is thus an invariant of a reporter–operation pair: distinctions inside a closed fibre are unavailable to that reporter under that operation.

3.7.2 A strict hierarchy of closed grouping resolutions

Grouping can still be retained by a richer reporter; it cannot be regenerated by a closed reporter that has erased it. The following family makes the distinction explicit.

Let $G_0(T) = L(T)$. For $r \geq 0$, put

$$G_{r+1}(\bullet) = \bullet_r$$

and

$$G_{r+1}(\langle T_1, \dots, T_q \rangle) = \langle G_r(T_1), \dots, G_r(T_q) \rangle_{\text{unlabelled}}.$$

Write $G_\omega(T) := (G_r(T))_{r \geq 0}$; equality under G_ω therefore means equality at every finite grouping depth.

The terminal mark closes the transcription at the chosen depth. Each G_r is all-arity closed: a G_{r+1} state retains the full child multiset at level G_r , and one terminal replacement changes exactly one child occurrence according to the complete G_r law. Equal parent outputs are then aggregated with multiplicity.

Proposition 3.7.2 (grouping tower). The reporters form a strict chain on the unbounded finite hierarchy family:

$$L = G_0 \prec G_1 \prec G_2 \prec \dots \prec G_\omega \simeq \mathcal{H}.$$

For a hierarchy of height h , G_h determines it. No fixed finite G_r determines every finite hierarchy.

Proof. Erasing the outermost retained grouping layer maps G_{r+1} to G_r . The reconstruction induction used for V_r shows that depth h suffices for a height- h hierarchy. Strictness follows by taking any pair first separated at depth $r + 1$ and placing both beneath one additional common container. The difference moves one layer deeper and survives at the next resolution while disappearing at the previous one. $\square$

If $\pi_r : G_{r+1} \rightarrow G_r$ erases one layer and $F_{r,q}(s)$ denotes the complete G_r output deck from state s , the laws commute with projection:

$$(\pi_r)_* F_{r+1,q}(s) = F_{r,q}(\pi_r(s)).$$

The tower therefore consists of compatible autonomous shadows, each with its own stable blindness. Full finite reconstruction requires progressively richer retained relations, not repeated iteration of one coarse closed law.

Corollary 3.7.3 (closed ceiling). If $Q \preccurlyeq S$ and S is all-arity closed, then

$$Cl_\infty(Q) \preccurlyeq S.$$

Proof. The leastness clause of Theorem 3.6.1 applies directly: S is a closed refinement of Q , so the least closed refinement of Q cannot retain more distinctions than S . $\square$

Hence any family of reporters factoring through the same closed reporter remains confined to its fibres under response iteration. Recovering a lost relation requires a richer constructor that retains it.

3.8 Combining perspectives and forced relations

The direct pair of two autonomous reporters need not be autonomous. Each coordinate may preserve a marginal while both erase their same-occurrence relation.

3.8.1 The pairing obstruction

Let $D(T)$ be the multiset of terminal depths in T , and let $A(T)$ be the multiset of the final internal arities appearing immediately above terminal occurrences. For the root-terminal record $\bullet$, use a distinguished null mark $\perp$ in A ; its first replacement removes $\perp$ and inserts q copies of q . With that boundary convention, both reporters are defined on all of $\mathcal{H}$ and are closed under E_q . Selecting an occurrence with values (d, a) changes the depth marginal by

$$D \mapsto D - \{d\} + q\{d+1\},$$

and the final-arity marginal by

$$A \mapsto A - \{a\} + q\{q\}.$$

However, the separate multisets do not retain which d was paired with which a at a terminal. There are finite hierarchies T_1, T_2 with identical marginals

$$D = (2, 2, 2, 3, 3, 3, 3, 4, 4), \quad A = (2, 2, 2, 2, 2, 3, 3, 3, 3),$$

but different same-occurrence profiles:

Terminal pair (d, a)	Multiplicity in C_1	Multiplicity in C_2
(2, 2)	2	1
(2, 3)	1	2
(3, 2)	1	2
(3, 3)	3	2
(4, 2)	2	2

For a compact exact witness, put

$$B = \langle \bullet, \bullet \rangle, \quad C_3 = \langle \bullet, \bullet, \bullet \rangle, \quad J = \langle \bullet, B \rangle,$$

$$E = \langle \bullet, C_3, J \rangle, \quad F = \langle \bullet, \bullet, B \rangle.$$

Then

$$T_1 = \langle B, E \rangle, \quad T_2 = \langle F, \langle \bullet, F \rangle \rangle.$$

The nine terminal paths give the same two marginals but the displayed distinct joint profiles. Their direct-pair decks differ because replacement acts on an occurrence carrying both values. Replacing (d, a) removes that occurrence and inserts q copies of $(d + 1, q)$, so output multiplicities depend on C , not on D and A separately. Thus

$$D \text{ closed and } A \text{ closed} \Rightarrow (D, A) \text{ closed}.$$

The first response refinement restores the missing relation

$$C(T) = \{ \{ (d, a) \text{ at each terminal occurrence} \} \}.$$

The reporter C is closed, since replacement sends (d, a) to q copies of $(d + 1, q)$. To see that the first response refinement is exactly C , fix any $q \geq 2$. For fixed marginals D, A , the two update maps

$$d \mapsto D - \{d\} + q\{d + 1\}, \quad a \mapsto A - \{a\} + q\{q\}$$

are injective. Equality of two images in the first map would imply $-e_d + qe_{d+1} = -e_{d'} + qe_{d'+1}$, whose coefficient at the least index on which d and d' differ is impossible unless $d = d'$; the second map is injective because equality implies $e_a = e_{a'}$. Each member of the pushed direct-pair deck therefore identifies the selected pair (d, a) , and deck multiplicity recovers its multiplicity in C . Conversely C determines both marginals and every pushed deck. Hence

$$R_1^{(D,A)} \simeq C.$$

3.8.2 Least closed combination

For arbitrary reporters X and Y , let (X, Y) be their direct-pair reporter and define

$$X \vee_c Y := Cl_\infty(X, Y).$$

Theorem 3.8.1 (universal property of closed combination). The reporter $X \vee_c Y$ is the least all-arity closed common refinement of X and Y . Precisely:

1. $X \preceq X \vee_c Y$ and $Y \preceq X \vee_c Y$;
2. $X \vee_c Y$ is all-arity closed;
3. if S is all-arity closed with $X \preceq S$ and $Y \preceq S$, then $X \vee_c Y \preceq S$.

Proof. The direct pair retains both coordinates, so $X \preceq (X, Y)$ and $Y \preceq (X, Y)$; response completion then gives $(X, Y) \preceq Cl_\infty(X, Y)$. Closure and leastness follow from Theorem 3.6.1 applied to the direct pair. $\square$

Corollary 3.8.2 (combination laws). Up to equality fibres,

$$X \vee_c Y \simeq Y \vee_c X,$$

$$(X \vee_c Y) \vee_c Z \simeq X \vee_c (Y \vee_c Z),$$

and

$$X \vee_c X \simeq Cl_\infty(X).$$

In particular, if X is already closed, then $X \vee_c X \simeq X$.

Proof. Coordinate exchange leaves the set of closed common refinements unchanged, proving commutativity. Both parenthesizations in the second equation are the least closed reporter refining X, Y, Z , so the universal property makes them fibre-equivalent. Repeating X adds no initial distinction beyond X , and closure then gives $Cl_\infty(X)$; for closed X , stable blindness reduces this to X . $\square$

Up to fibres, closed reporters therefore form an idempotent, commutative, associative combination structure under $\vee_c$. The algebra is derived from behavioural completion.

The results in this section concern the finite hierarchy carrier and the specified complete operations. Report depth, occurrence, containment, closure, and multiplicity receive no physical interpretation without an additional empirical bridge.

4 Generated perspectives, predictive closure, and target-free continuation

The source boundary and the finite hierarchy now being fixed, this part studies structures generated by recursive distinction. Sets, maps, partitions, addresses, and algorithms remain reports made within P_1 . For $r \geq 2$, a construction retaining r resultants may be indexed P_{r-1} ; when $r > 2$, that index records arity without claiming native access by P_1 to the hypothetical perspective it names.

Provenance is stated once at the opening of each construction. General theorems carry the hypotheses under which they are proved; finite censuses are reported separately from those proofs. This keeps the formal question narrow: once an interaction has been declared or generated, which distinctions and laws does it force?

4.1 Same-referent quotients and the reversal of constraint

4.1.1 Retained indistinction

Conditional construction. Let C be a context carrier and let

$$a_i : C \rightarrow X_i, \quad 1 \leq i \leq h,$$

be effects already aligned on the same contexts. The alignment is a premise: the construction does not infer that two unrelated carriers concern the same referent. Write $A = (a_1, \dots, a_h)$, and define

$$c \sim_A d \iff a_i(c) = a_i(d) \text{ for every } i.$$

The quotient

$$Q_A = C / \sim_A, \quad q_A : C \rightarrow Q_A,$$

retains exactly the distinctions jointly made by A . Each coordinate factors as $a_i = \bar{a}_i \circ q_A$. Resultant renaming leaves the quotient unchanged, while a relabelling of C merely relabels its blocks. Thus the quotient is a structure of distinctions, not a structure of the temporary names used to encode them.

Theorem SRQ (same-referent shadow criterion). For any candidate effect $t : C \rightarrow Y$, the following are equivalent:

1. there is a map $\bar{t} : Q_A \rightarrow Y$ such that $t = \bar{t} \circ q_A$;
2. t is constant on every A -indistinction class:

$$c \sim_A d \Rightarrow t(c) = t(d).$$

When it exists, $\bar{t}$ is unique.

Proof. Factorisation immediately implies constancy on fibres. Conversely set $\bar{t}([c]_A) = t(c)$. Fibre constancy makes this independent of the representative, and surjectivity of q_A makes it unique. $\square$

The criterion separates three cases. A **shadow** factors through Q_A and adds no distinction, although it may merge quotient classes. A **faithful presentation** has the same indistinction relation as Q_A ; on its reached image it is only a relabelling. A **new distinction** is witnessed by some $c \sim_A d$ with $t(c) \neq t(d)$. Adjoining such a t changes the generated referent.

The quotient is also universal. If $r : C \rightarrow S$ is sufficient for every retained effect, so that $a_i = \alpha_i \circ r$, then $r(c) = r(d)$ implies $c \sim_A d$. A unique map $\rho : r(C) \rightarrow Q_A$ therefore satisfies $q_A = \rho \circ r$. Thus Q_A is the least-distinguishing sufficient carrier: a sufficient report may retain more, but it must map to Q_A .

4.1.2 More distinction forces less

For a partition Q of C and a resultant set Y , define its admissible shadow family

$$Sh_Y(Q) = \{t : C \rightarrow Y : t \text{ is constant on each } Q\text{-block}\}.$$

Theorem CR (constraint reversal). If Q_2 refines Q_1 , then

$$Sh_Y(Q_1) \subseteq Sh_Y(Q_2).$$

If the refinement is proper and $|Y| \geq 2$, the inclusion is strict.

Proof. Constancy on a Q_1 -block entails constancy on every Q_2 -subblock. If Q_2 properly splits a Q_1 -block, assign two distinct resultants to two of its Q_2 -subblocks while keeping each subblock constant. The resulting effect descends through Q_2 but not through Q_1 . $\square$

If Q has b blocks and $|Y| = q < \infty$, then $|Sh_Y(Q)| = q^b$.

This is the precise sense in which more distinction forces less. A finer base admits more candidate effects and therefore determines less about which candidate is realised. A one-block finite carrier admits only the q constant effects, whereas the discrete partition admits all $q^{|C|}$ effects. The one-block case is still a P_1 representation of a finite audit carrier; it is not a representation of P_0/R .

Two identification barriers follow immediately. First, **self-seeding** is circular: if t is used to construct $Q_{A \cup \{t\}}$, then t necessarily factors through that quotient. Second, a **discrete base** is non-predictive: every candidate factors through it. A meaningful same-referent test must therefore generate Q_A independently of the candidate and must retain some indistinction.

4.1.3 Reached laws and their completions

Let

$$e : C \rightarrow U = X_1 \times \cdots \times X_h, \quad e(c) = (a_1(c), \dots, a_h(c)),$$

and let $b : C \rightarrow Y$ be a target coordinate observed on the same contexts.

Theorem RLD (reached-law descent). There is a unique law on reached configurations,

$$\phi : e(C) \rightarrow Y, \quad b = \phi \circ e,$$

if and only if $e(c) = e(d) \Rightarrow b(c) = b(d)$.

Proof. If $b = \phi \circ e$, equal e -values have equal b -values. Conversely define $\phi(e(c)) = b(c)$. The displayed constancy condition makes this independent of the chosen representative, and every element of $e(C)$ has a representative; hence ϕ exists and is unique. $\square$

This is not yet a law on all of U . If U and Y are finite, with $M = |U|$, $r = |e(C)|$, and $q = |Y|$, then the number of complete extensions $\Phi : U \rightarrow Y$ realizing b is exactly zero when descent fails and exactly q^{M-r} when descent holds.

Assume descent holds. For $|Y| \geq 2$, the resulting partial law has a unique complete extension precisely when every configuration is reached. Completion and identification are different operations: retaining every extension guarantees that unreached assignments remain available, but does not choose among them. When $|Y| = 1$, descent is automatic and the extension is uniquely constant even with unreached configurations; a singleton codomain supplies no identifying constraint.

The same reasoning yields an exact forcing threshold. Fix a source coordinate i , a source focus x , and a target focus y , and let $H_{i,x} = \{u \in U : u_i = x\}$. Assume descent holds and $|Y| \geq 2$. Every compatible completion sends all of $H_{i,x}$ to y if and only if the full face is reached and every reached member of the face has that target output. If $m = |\{u \in H_{i,x} : u \notin e(C)\}|$ and no reached member contradicts y , exactly q^{M-r-m} of the q^{M-r} compatible completions preserve the face. An unreached point is not negligible: it is precisely a retained degree of freedom.

Formally, SRQ is a kernel-factorisation criterion of the kind used in universal algebra, while RLD is a reached-state analogue of the sufficient-state condition in automata theory (Burris and Sankappanavar 1981; Nerode 1958). The difference is one of provenance: no machine semantics is supplied in advance. The retained effects generate the quotient, and the unreached configurations quantify what those effects leave unresolved.

4.2 Interaction-generated predictive perspectives

The same-referent quotient presupposes an effect family. We now derive a focus-preserving quotient from interaction itself, before naming any downstream target. Depending on the law, the generated quotient may be discrete or may retain nontrivial indistinction.

4.2.1 Constructor and refinement

Declared inputs; conditional construction. Fix integers $q \geq 2$ and $m \geq 1$, a finite set X with $|X| = q$, and an internally pointed law fibre

$$(f, x), \quad f : X^m \rightarrow X, \quad x \in X.$$

The q elements of X are the retained resultants. The carrier, arity, total law, and focus are declared. From them, the following construction is forced by interaction; it receives no metric, clock, arithmetic coordinate, desired partition, or comparison law.

For coordinate j and fixed values in all other coordinates, law application produces an elementary one-place continuation

$$\tau_{j,a}(u) = f(a_1, \dots, a_{j-1}, u, a_{j+1}, \dots, a_m).$$

Let T_f be the monoid generated by these continuations and the identity. Begin with the intrinsic pointed distinction

$$u E_0 v \iff (u = x \iff v = x).$$

For any equivalence E , define

$$u \mathcal{R}_f(E) v \iff u E v \text{ and } \tau(u) E \tau(v) \text{ for every elementary } \tau,$$

and iterate $E_{r+1} = \mathcal{R}_f(E_r)$. A block splits only when application of the retained law distinguishes two members relative to a distinction already generated. There is no scoring function and no target partition.

Theorem IPQ-1 (finite stabilisation). For the finite carrier $|X| = q \geq 2$ and arity $m \geq 1$ fixed above, the chain

$$E_0 \succcurlyeq E_1 \succcurlyeq \dots$$

stabilises after at most $q - 2$ strict refinements.

Proof. A strict step increases the number of blocks by at least one and no step merges blocks. The initial partition has two blocks and a partition of X has at most q . $\square$

The bound is sharp. For the unary law $d_q(j) = \max(0, j - 1)$ focused at 0, successive rounds isolate 1, then 2, and so on. The generated block counts are $2, 3, \dots, q$, requiring exactly $q - 2$ strict steps.

Theorem IPQ-2 (all-finite-continuation characterisation). At stabilisation,

$$u E_\infty v \iff (t(u) = x \iff t(v) = x) \text{ for every } t \in T_f.$$

Proof. Let $T_f^{\leq r}$ be the continuation words of length at most r , including the identity word. We prove by induction that

$$u E_r v \iff (t(u) = x \iff t(v) = x) \text{ for every } t \in T_f^{\leq r}.$$

For $r = 0$ this is the definition of E_0 . Suppose it holds at r . By definition, $u E_{r+1} v$ precisely when $u E_r v$ and $\tau(u) E_r \tau(v)$ for every elementary continuation τ . The induction hypothesis applies to the first clause for words of length at most r and to the second for all words $t\tau$ of length at most $r + 1$. This proves the induction step. Because the finite chain stabilises, choose N with $E_N = E_{N+1} = E_\infty$. Fixed-point equality then gives $E_{N+j} = E_\infty$ for every $j \geq 0$, so the displayed bounded-word characterisation holds at every length. Hence E_∞ is exactly agreement under all finite continuation words. $\square$

4.2.2 Generated law and universal minimality

Theorem IPQ-3 (congruence and quotient law). The stable equivalence is compatible with f :

$$u_i E_\infty v_i \quad (1 \leq i \leq m) \implies f(u_1, \dots, u_m) E_\infty f(v_1, \dots, v_m).$$

Proof. For $0 \leq j \leq m$, put

$$w_j = f(v_1, \dots, v_j, u_{j+1}, \dots, u_m).$$

Fix $j < m$ and hold every coordinate except the $(j+1)$ st at the values displayed in w_j and w_{j+1} . This defines an elementary continuation τ . Since $u_{j+1} E_\infty v_{j+1}$ and E_∞ is a fixed point of $\mathcal{R}_f$, we have $\tau(u_{j+1}) E_\infty \tau(v_{j+1})$, or $w_j E_\infty w_{j+1}$. Transitivity along $w_0, \dots, w_m$ proves the claim. $\square$

Therefore $Q_f = X/E_\infty$ carries a unique induced law

$$\tilde{f}([u_1], \dots, [u_m]) = [f(u_1, \dots, u_m)]$$

with the exact commuting equation

$$\pi_f \circ f = \tilde{f} \circ (\pi_f)^m.$$

Theorem IPQ-4 (coarsest predictive quotient). Let C be any equivalence on X for which $\{x\}$ is a class and C is a congruence of f . Then C refines E_∞ .

Proof. Every such C refines E_0 . If it refines E_r , congruence under each elementary continuation implies that it refines E_{r+1} . Induction gives refinement of the limit; IPQ-3 shows the limit itself is admissible. $\square$

Thus Q_f is the unique least-distinguishing carrier that preserves the focus and supports the law. Coarser identification destroys either focus separation or law descent; finer presentation retains distinctions not forced by the interaction. Mathematically, E_∞ is the behavioural equivalence induced by all finite focus-reachability tests, and its computation is a specialised partition-refinement procedure (Moore 1956; Hopcroft 1971; Paige and Tarjan 1987; Rutten 2000). What is specific here is the observation family: it is generated internally from the pointed law before any external target is named.

Figure 3 shows the construction from the pointed law to its stable, focus-preserving quotient.

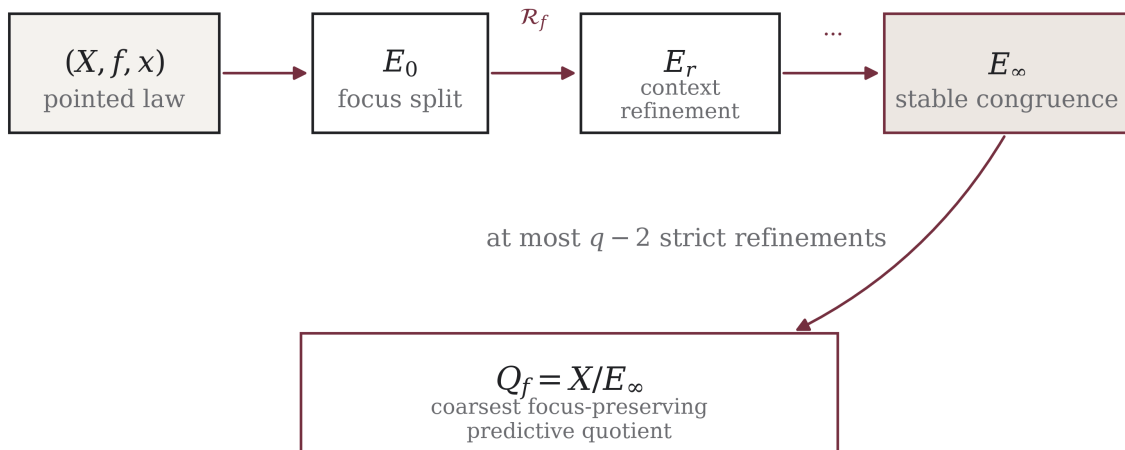


Figure 3. Interaction-generated predictive quotient. Repeated refinement separates only pairs distinguished by a finite continuation relative to the retained focus. The stable quotient carries the induced law.

At round r , the generated state is the partition E_r , not its block count. A scalar report may be taken only afterward:

$$k_r = |X/E_r| - 1.$$

For the sharp family this shadow is $(1, 2, \dots, q - 1)$. The ordering is logical dependence under repeated interaction, not physical time. If Q_f has s resultants, the corresponding P_1 report may be indexed P_{s-1} . Both k_r and the perspective index are read from the closed construction; neither was an input.

Finite audit. Every one of the $3^{3^2} = 19,683$ binary-input laws on a ternary carrier was checked at all three foci, giving 59,049 pointed laws. The audit covered monotonicity, stabilisation, congruence, the universal property, quotient reconstruction, cyclic-renaming equivariance, and canonical signatures.

Generated quotient	Pointed laws
2 resultants (P_1 report)	3,825
3 resultants (P_2 report)	55,224
Total	59,049

These produce 9,238 label-free predictive isomorphism classes. The iterative and all-finite-continuation definitions agree in 113 complete small-law comparisons; all 486 unary ternary pointed permutation cases are equivariant. The delayed-focus depth formula was also checked for carrier sizes 2 through 10. These computations audit the theorems; they do not replace their proofs.

4.3 Comparing perspectives through a common source

Cross-perspective comparison requires a common source. Appending coordinates to a P_1 carrier and declaring $X \times Q$ a new perspective is not neutral: the product already contains a deterministic projection to X , so incomparable descriptions have been ruled out by construction.

4.3.1 Future-complete source quotients

Conditional source calculus. Let M be a finite family of source effects closed under declared primitive continuations, and let O_θ be a retained observation. Define

$$m\theta n \iff O_\theta(wm) = O_\theta(wn) \text{ for every finite primitive word } w.$$

The perspective $P_\theta = M/\theta$ inherits every primitive continuation. The effect family and its primitive actions are declared; complete future behaviour determines the quotient.

Theorem CS-1 (translation criterion). For two such equivalences θ and ϕ , a deterministic common-source translation

$$[m]_\theta \mapsto [m]_\phi$$

exists if and only if θ refines ϕ . When it exists it is unique and commutes with every primitive continuation.

Proof. The displayed assignment is well-defined exactly when $m\theta n$ implies $m\phi n$. Commutation follows because both quotient actions descend from the same source action. $\square$

If neither equivalence refines the other, the framework does not invent a function. It retains the exact same-effect relation

$$\Gamma_{\theta\phi} = \{([m]_\theta, [m]_\phi) : m \in M\}.$$

The relation may be one-to-many in both directions. It is exactly the alignment supported by the common source.

The joint perspective is the common refinement

$$m\theta \wedge \phi n \Leftrightarrow m\theta n \text{ and } m\phi n.$$

Theorem CS-2 (strict pairwise creation). For finite M ,

$$\theta \text{ and } \phi \text{ are incomparable} \Leftrightarrow |P_{\theta \wedge \phi}| > \max(|P_\theta|, |P_\phi|).$$

Proof. If the equivalences are incomparable, each separates a pair merged by the other, so their common refinement strictly splits a class of each. If one refines the other, their meet is the finer equivalence and cannot be larger. $\square$

The common refinement gives a precise meaning to **joint organisation**. It combines the two distinction families and, when they are incomparable, is finer than either marginal quotient. It does not manufacture a pairwise distinction absent from both: if both θ and ϕ identify two source effects, so does $\theta \wedge \phi$. The gain lies in retaining the two inherited families simultaneously, not in altering the source.

Figure 4 displays the three comparison objects supplied by the same source: a direct map under refinement, a relation for incomparable quotients, and their least joint refinement.

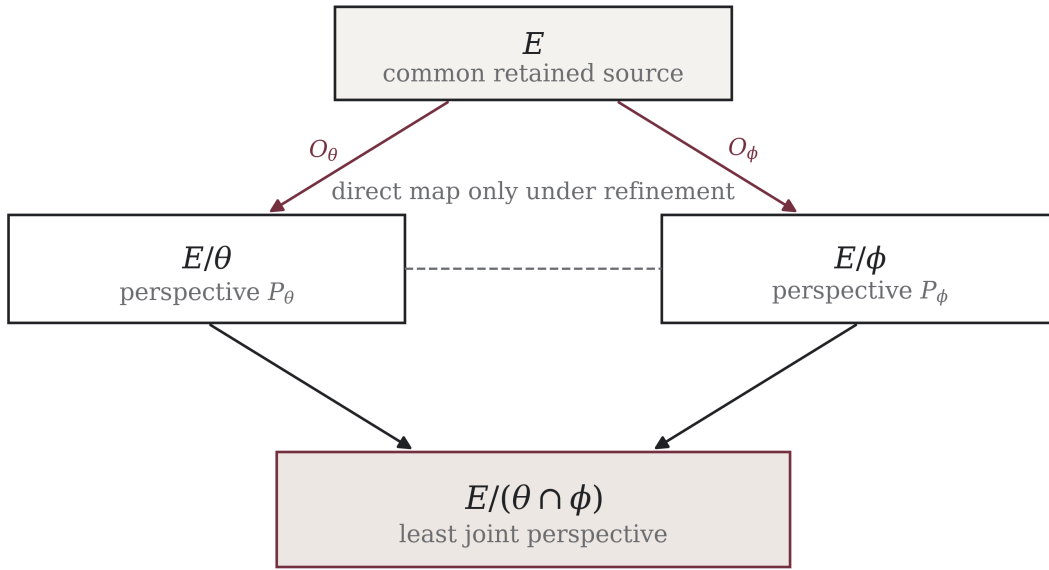


Figure 4. Common-source comparison. Refinement licenses a deterministic translation. Incomparable quotients retain a source-induced relation and meet in a strictly finer joint quotient.

Finite audit. Across the 1,555 canonical source rules in the two-transformation census, the comparison produced 774 equivalent perspective pairs, 3,840 proper one-way pairs, and 51 incomparable pairs. A rule code such as $\mathbf{0012} \mid \mathbf{1130}$ lists, on each side of $\mid$, the images of $0, \dots, n-1$ under one of the two generators. Strict joint growth occurred for all and only the 51 incomparable pairs. Three sources produced a joint calculus larger than each of three audited marginals:

Source rule	Source effects	Kernel	Reunion	Ideal	Joint
$\mathbf{0012} \mid \mathbf{1130}$	13	10	1	8	12

Source rule	Source effects	Kernel	Reunion	Ideal	Joint
0012 1220	14	10	1	10	12
0013 1321	16	12	1	11	14

Allowing each joint calculus to act as the next source produced closed interaction laws: the first two stabilized as 12-state laws, while the third passed from 14 states to a fixed 11-state law.

Finite underdetermination. Among 376 distinct complete rooted first-perspective laws, 246 admitted multiple compatible source systems. In 20 of those cases the other generated perspective laws differed; in three, even the presence of strict creation differed. These finite pairs refute unique completion within the declared constructor. A complete first-perspective law does not, in this census, select a unique compatible higher-perspective law. The warranted continuation is therefore the full compatible family; selecting the member nearest a desired target would import a new assumption.

Chu, sheaf, and institution formalisms each retain relations or contexts that need not collapse to a single global map (Pratt 1995; Abramsky and Brandenburger 2011; Goguen and Burstall 1992). The present criterion is more specific: both views are first closed against one finite source action, after which refinement decides whether the supported comparison is functional, relational, or a joint quotient.

4.4 A target-free binary distinction game

The preceding constructions begin with effects or laws. This game begins one step earlier. After the first P_1 distinction, any current resultant may distinguish into two retained resultants, and the rule may be applied again. There is no target phenomenon and no global update schedule.

4.4.1 Complete transition family

Declared constructor. Binary continuation is the defining P_1 move. The address marks 0 and 1 prevent accidental identification of occurrences; the transition itself never reads them.

A finite record E is a finite prefix-closed set of binary addresses containing the root ϵ , the first already-present distinction. Its current undistinguished resultants are

$$B(E) = \{w : (\exists v \in E)(w = v0 \vee w = v1), w \notin E\}.$$

The next step retains the complete successor family

$$\mathcal{D}(E) = \{(x, E \cup \{x\}) : x \in B(E)\}.$$

The ordered pair retains which existing occurrence was continued and the resulting complete record.

Let σ denote a prefix-preserving automorphism of the rooted binary address tree generated by local exchanges of the marks 0 and 1. It acts componentwise on addresses, records, and transitions:

$$\sigma(x, E) := (\sigma x, \sigma E).$$

Theorem TF (generated continuation laws). The transition family satisfies all of the following.

1. **Local replacement.** For every $x \in B(E)$,

$$B(E \cup \{x\}) = \{y \in B(E) : y \neq x\} \cup \{x0, x1\}.$$

2. **Inert-name equivariance.** Any recursively applied exchange of the address marks bijects the complete transition family:

$$\sigma(\mathcal{D}(E)) = \mathcal{D}(\sigma E).$$

3. **Commutation at separate current occurrences.** If $x \neq y$ lie in $B(E)$, then

$$(E \cup \{x\}) \cup \{y\} = (E \cup \{y\}) \cup \{x\}.$$

4. **Ancestry-only history restriction.** Finite resultant records are exactly the finite prefix-closed address sets containing ϵ . Legal histories of a fixed record are exactly its linear extensions under ancestry. Any two differ by exchanges of adjacent unrelated occurrences.
5. **Common core and common completion.** If E and F are legal, then $E \cap F$ and $E \cup F$ are legal. They obey absorption and distributivity as set identities.
6. **Unique continuation factorisation.** Every extension $F \supseteq E$ decomposes uniquely into one local continuation below each $x \in B(E)$:

$$F_x = \{u : xu \in F\},$$

- where $F_x = \emptyset$ records no continuation below x . Then

$$F = E \cup \bigcup_{x \in B(E)} xF_x.$$

7. **No finite terminal record.** $B(E) \neq \emptyset$ for every finite E .

Proof. If $x \in B(E)$, inserting x removes that boundary address and introduces exactly its two children, which proves local replacement. A prefix-preserving tree automorphism sends boundary addresses to boundary addresses and commutes with adjoining a node; applying this pointwise gives inert-name equivariance. If x and y are distinct boundary addresses, neither is a prefix of the other. Inserting one therefore leaves the other on the boundary, and both update orders produce $E \cup \{x, y\}$.

Every reachable record contains ϵ and is prefix-closed because an address can be inserted only after all of its proper prefixes. Conversely, for a finite prefix-closed set F containing ϵ , order all other elements of F by nondecreasing length. Each address is on the current boundary when selected, so this order constructs F . More generally, a construction history for fixed F is legal exactly when every prefix precedes its descendants—precisely when it is a linear extension of the ancestry order. Any two linear extensions of a finite poset are connected by swaps of adjacent incomparable elements: repeatedly move the next element of one extension leftward through elements that cannot precede it in the other.

Union and intersection preserve prefix closure and the root. For an extension $F \supseteq E$, every new address has a unique longest prefix $x \in B(E)$; the boundary is prefix-incomparable, so the resulting sets xF_x are disjoint and collectively contain precisely the elements of F not already in E . This proves existence and uniqueness of the factorisation. Finally, take an address of maximal length in finite E . Neither child can lie in E , so both are in $B(E)$ and the boundary is nonempty. $\square$

These relations can subsequently be identified with familiar structures. Legal records form a distributive lattice under union and intersection; a fixed record carries an ancestry poset; complete records have rooted full binary-tree shapes; independent local continuations give a recursive species decomposition (Birkhoff 1967; Stanley 2015; Bergeron, Labelle, and Leroux 1998). These identifications describe the result; they do not define the transition.

4.4.2 Enumeration after closure

Counting is applied only after the transition has been fixed. If A_n denotes the number of complete identity records with n distinguished occurrences and $A_0 = 1$ denotes an empty local continuation, unique root factorisation gives

$$A_n = \sum_{i+j=n-1} A_i A_j, \quad A_n = \frac{1}{n+1} \binom{2n}{n}.$$

The local replacement identity implies that every n -occurrence record has $n+1$ current resultants. The total number of legal histories at level n is consequently $2 \cdot 3 \cdots n = n!$. For a fixed record E , let $s_E(v)$ be the number of distinguished occurrences at or below a non-root occurrence v . Its history multiplicity is

$$H(E) = \frac{(|E| - 1)!}{\prod_{v \in E, v \neq \epsilon} s_E(v)}.$$

The proof interleaves the independent histories below the two root resultants and proceeds recursively.

Distinguished occurrences	Complete records	Complete legal histories	Current resultants per record
1	1	1	2
2	2	2	3
3	5	6	4
4	14	24	5
5	42	120	6
6	132	720	7
7	429	5,040	8
8	1,430	40,320	9

The finite audit covered 19,631 configuration/boundary identities, 17,576 recursive name exchanges, 67,071 separate-resultant diamonds, 390,625 union/intersection pairs, 6,968 continuation decompositions, and the history formula for all 2,055 records through level eight. No counterexample was found. The all-finite claims rest on the proofs above, not on the bounded audit.

Unlike a cellular automaton, the game has no ambient lattice, neighbourhood, synchronous clock, or selected update table (Wolfram 1984). Its object is an extensible genealogy together with every legal local continuation. The resulting lattice, factorisation, and counting laws are consequences of that transition rather than criteria imposed upon it.

4.5 Projection towers and erased distinctions

A complete record admits several autonomous projections. Studying them together separates closure from mere loss of information.

4.5.1 Four continuation-closed views

Declared projections. Let H_d be the complete set of explicit histories after d distinctions, and consider the tower

$$H_d \xrightarrow{\alpha_d} O_d \xrightarrow{\beta_d} U_d \xrightarrow{\gamma_d} C_d.$$

Here O_d forgets the temporal trace but retains the ordered current form; U_d also forgets every temporary left/right order; and C_d keeps only the number of available occurrences. None is designated as the true view.

For an ordered form T , let d_v be the number of internal occurrences in the subform rooted at v . For an order-free form T , assign each internal occurrence a factor $w_v = 1$ when its two subforms coincide and $w_v = 2$ when they differ, and define

$$e(T) = \prod_v w_v.$$

Then the exact fibre sizes are

$$|\alpha_d^{-1}(T)| = \frac{d!}{\prod_v d_v}, \quad |\beta_d^{-1}(T)| = e(T), \quad |(\beta_d \circ \alpha_d)^{-1}(T)| = \frac{d!e(T)}{\prod_v d_v}.$$

The first formula follows by recursively interleaving subtree histories; the second assigns two orientations to every unequal pair of subforms and one to an equal pair. Since every level- d form has $d + 1$ available occurrences, C_d has one class and its history fibre has size $d!$.

A view is **continuation-closed** if the multiset of next-view results from all current distinctions depends only on the view-class and not on its representative.

Theorem PT (projection closure and counting barrier). Every view H, O, U, C is continuation-closed. Continued distinction within the count-only view C , however, cannot regenerate any arrangement, orientation, or history distinction erased by $\gamma\beta\alpha$.

Proof. In H_d the complete successor multiset is part of the definition, so closure is immediate. The projection α_d erases only the order in which already recorded distinctions were made; it leaves the current ordered form and its available addresses unchanged. Hence representatives of one O_d -class have identical O_{d+1} successor multisets. Two ordered representatives of one U_d -class differ by finitely many local left/right exchanges. Each exchange bijects their available addresses, commutes with a continuation at the corresponding address, and yields order-free-isomorphic successors. Thus their projected successor multisets agree. Finally, every level- d record has exactly $d + 1$ available current occurrences, and every continuation enters the sole class of C_{d+1} . The count successor multiset therefore depends only on d .

For the barrier, induction on continuation length shows that every response generated from a C_d -class remains inside one count class at each later level. An autonomous continuation on C therefore has no response that separates two representatives already merged by $\gamma\beta\alpha$. Arrangement, orientation, and history distinctions cannot be regenerated inside the count-only quotient. $\square$

This is **stable blindness** in a concrete tower. The erased distinctions remain available in H_d, O_d , and U_d , but they are absent from C_d and cannot be recreated by its autonomous continuation. A scalar count can therefore support a simpler law while retaining strictly less source structure.

Figure 5 makes the direction of information loss explicit: every forward projection is continuation-closed, but there is no closed inverse.

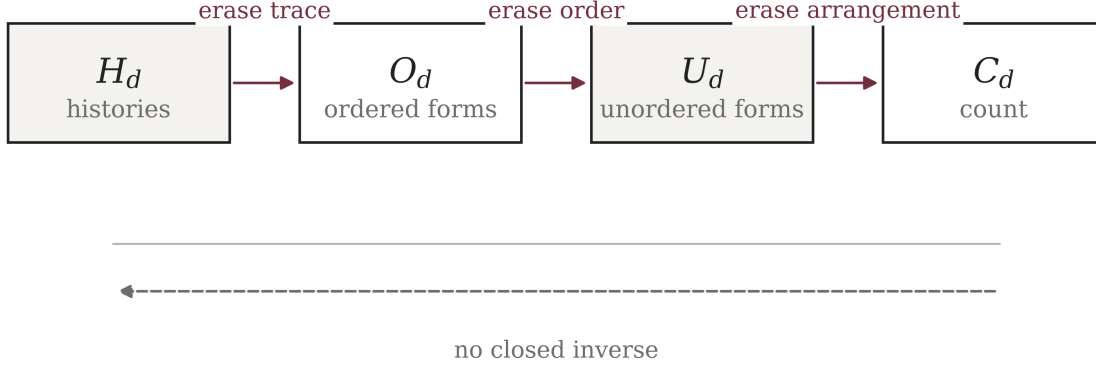


Figure 5. Projection tower. Each quotient supports an autonomous continuation, but no continuation within a quotient recovers distinctions erased by the preceding projection.

4.5.2 Complete partitions before scalar k

Let Π and Σ be continuation-closed partitions of the same forms. Retaining every distinction made by either produces their common refinement

$$x \sim_{\Pi \vee \Sigma} y \iff x \sim_{\Pi} y \text{ and } x \sim_{\Sigma} y.$$

Although the symbol $\vee$ is conventional here, the operation is the intersection of equivalence relations: the coarsest partition refining both. It is forced by complete retention and is commutative, associative, and idempotent.

Let S and T be finite source and target sets, and let $A : S \times T \rightarrow \mathbb{N}$ record continuation multiplicity. For a partition Π of T , define the source partition $F_A(\Pi)$ by

$$s \sim_{F_A(\Pi)} s' \iff \sum_{t \in B} A(s, t) = \sum_{t \in B} A(s', t) \text{ for every block } B \in \Pi.$$

Theorem PI (partition interaction). If Π refines Σ , then $F_A(\Pi)$ refines $F_A(\Sigma)$. Moreover,

$$F_A(\Pi \vee \Sigma) \text{ refines } F_A(\Pi) \vee F_A(\Sigma).$$

The refinement may be strict.

Proof. Each Σ -block is a union of Π -blocks, so equality of all fine-block counts implies equality of their coarse sums. Joint block counts determine both marginal block-count vectors, proving the second refinement. Marginal vectors need not determine their joint table, so the converse can fail. $\square$

The strict difference defines the interaction-only distinction relation $\mathcal{I}_A(\Pi, \Sigma)$: it consists of source pairs equivalent under $F_A(\Pi) \vee F_A(\Sigma)$ but inequivalent under $F_A(\Pi \vee \Sigma)$.

It contains source pairs that neither marginal pullback separates but the complete joint pullback does. A horizon-eight witness has six classes under separate pullback-and-combination and seven under joint combination-then-pullback. Its two level-six forms

$$((((..)) (..)) (..)) \quad \text{and} \quad (((..)(..)) ((..)..))$$

have identical marginal signatures but different joint signatures.

The scalar barrier is stronger than a loss of labels. At level 5, retained partitions with class counts 2 and 3 can interact to produce either 3 or 4 classes. Consequently no function f satisfies

$$|\Pi \vee \Sigma| = f(|\Pi|, |\Sigma|)$$

even in this finite generated family. Before a further projection is specified, the complete objects are the partitions themselves,

$$K_P = \Pi_P, \quad I(P, Q) = \mathcal{I}_A(\Pi_P, \Pi_Q),$$

not their cardinalities. A numerical k is a downstream shadow. Information measures based on scalar costs over a fixed alphabet remain valid in their intended domains (Hartley 1928; Shannon 1948; Cover and Thomas 2006); they begin after the distinctions kept explicit here have already been fixed.

4.6 Conditional laboratory: an open finite-rule census

The target-free game isolates the consequences of one binary continuation. A separate laboratory admits richer source structure in order to study the closed laws and perspectives it generates. Its findings are conditional on that choice of source.

4.6.1 Declared branch

Choose a finite carrier $X = \{0, \dots, n-1\}$, with $n \in \{2, 3, 4\}$; two arbitrary total transformations $a, b : X \rightarrow X$; identity; and ordinary composition. Products are written $gm := g \circ m$, so m acts first. Retain every unordered primitive pair with repetition, quotienting only by carrier relabelling and exchange of primitive names. These are declared laboratory conditions, not consequences of the sole distinction rule. Algebraic claims below hold for every finite system satisfying their displayed hypotheses. Counts and classifications are exhaustive only at the stated carrier sizes.

The exhaustive finite census begins with the following source counts.

Carrier size	Raw primitive pairs	Canonical systems	Distinct invariant profiles
2	10	7	6
3	378	74	67
4	32,896	1,474	1,175

Within this census, two primitive transformations generate up to 176 effective transformations.

Proposition OR-1 (ideal and rank order). In every generated transformation monoid M , composition induces a one-way preorder. For the two-sided ideal $I(m) = MmM$ and every $g \in M$,

$$\text{rank}(gm) \leq \text{rank}(m), \quad I(gm) \subseteq I(m).$$

Proof. The image of gm is the image under g of the image of m , so it has at most $\text{rank}(m)$ elements. Moreover,

$$I(gm) = MgmM \subseteq MmM = I(m),$$

because $Mg \subseteq M$. Repeating the argument after any further left composition proves that a strict ideal descent cannot later be reversed. $\square$

Further composition cannot reverse strict ideal descent, although recurrence may occur within a level. This is an internal order and carries no automatic interpretation as time, entropy, causality, or knowledge.

4.6.2 Quotients and predictive closure

Among the 1,474 four-state systems, 1,171 have proper nontrivial congruence quotients, 529 have incomparable quotient pairs, and 270 exhibit distributed effect recovery: no proper quotient is faithful alone, while the retained quotient family is jointly faithful. For rule **1002 | 2210**, two incomparable three-block quotients each distinguish 7 of 14 generated effects and together distinguish all 14.

Proposition OR-2 (kernel and predictor order). Each generated effect m has a kernel κ_m , where $x\kappa_my$ iff $m(x) = m(y)$. Along $m \mapsto g \circ m$, kernels can only coarsen. If K is the number of current-kernel classes, Q the number of classes under

$$m \equiv_\kappa n \iff \ker(xm) = \ker(xn) \text{ for every } x \in M,$$

and $E = |M|$ the number of full effects, then

$$K \leq Q \leq E.$$

Proof. If $m(u) = m(v)$, then $g(m(u)) = g(m(v))$, so $\kappa_m \subseteq \kappa_{gm}$. For the class count, equality of effects implies $\equiv_\kappa$, while $m \equiv_\kappa n$ implies $\ker(m) = \ker(n)$ by taking x to be the identity. Thus full-effect equality refines $\equiv_\kappa$, which refines current-kernel equality, and the displayed inequality follows. $\square$

The present kernel often fails to determine its successor. Among four-state systems, 1,446 show strict kernel descent, 811 have incomparable successor-kernel branches, 995 show kernel-level predictive failure, and 1,128 reach kernels that are not global congruences.

The equivalence $\equiv_\kappa$ is therefore the predictive quotient: it retains exactly the distinctions required to determine every future kernel history.

The four-state systems divide into three predictive regimes:

Four-state regime	Systems	Interpretation inside the branch
$Q = K$	479	current kernel is already autonomous
$K < Q = E$	158	every full-effect distinction is required
$K < Q < E$	837	a genuine intermediate predictor is forced

For **0032 | 2031**, $K = 12$, $Q = 137$, and $E = 176$: 39 full-effect distinctions are irrelevant to every future kernel history, but the kernel alone omits distinctions needed to predict its future.

Theorem OR-3 (recursive-lift termination). Treating the predictor as a transformation system and repeating the construction gives a finite recursive lift. After the first lift each proper step is a strict quotient. Up to pointed-monoid isomorphism, every finite trajectory reaches a fixed point and cannot enter a nontrivial cycle.

Proof. The relation $\equiv_\kappa$ is a two-sided monoid congruence. If $m \equiv_\kappa n$, then for any $y \in M$ and every left context x ,

$$\ker(xym) = \ker(xyn),$$

which proves left stability. For right stability, equality of $\ker(xm)$ and $\ker(xn)$ implies equality after pullback by y :

$$\ker((xm)y) = y^{-1}(\ker(xm)) = y^{-1}(\ker(xn)) = \ker((xn)y).$$

Hence the predictive classes form the quotient monoid $C = M/\equiv_\kappa$. The lifted system is the left-regular action of C on itself, $[u] : [m] \mapsto [um]$. Distinct elements induce distinct transformations because their actions on the identity class differ, so the lifted carrier and its generated effect monoid both have cardinality $|C|$. Identifying C with this faithful regular representation, every subsequent lift is a quotient of the preceding finite monoid. A proper lift strictly reduces cardinality; a nontrivial cycle would require a later increase or a proper quotient of equal cardinality, both impossible. Finite descent terminates at a fixed point up to pointed isomorphism. $\square$

Across all 1,555 canonical systems, 1,523 are fixed after the first lift, 32 require one further quotient, and all converge to 369 pointed fixed systems. The largest expansion is $4 \rightarrow 137 \rightarrow 137$; the largest second reduction is $4 \rightarrow 8 \rightarrow 4 \rightarrow 4$.

4.6.3 Endogenous futures and irreversible opportunity loss

Proposition OR-4 (future-domain and reunion order). Let a finite monoid M act on X . For $x \in X$, put $I_x = Mx$. Every $g \in M$ satisfies $I_{gx} \subseteq I_x$, and $I_x = I_y$ exactly when x and y are mutually reachable.

Proof. Since $Mg \subseteq M$, we have $I_{gx} = M(gx) = (Mg)x \subseteq Mx = I_x$. If $I_x = I_y$, then $x \in I_y$ and $y \in I_x$, so each state reaches the other. Conversely, mutual reachability gives both inclusions between the generated domains. $\square$

Across the 369 fixed systems, 11,211 states generate 3,059 distinct principal domains; 368 systems have strict domain descent, 246 have incomparable primitive branches, and 29 have multiple minimal terminal domains.

Reunion-capacity monotonicity. For an unordered pair x, y , define reunion capacity by three exhaustive cases:

- $\chi(x, y) = 2$ when some $u \in M$ satisfies $u(x) = u(y)$;
- $\chi(x, y) = 1$ when $I_x \cap I_y \neq \emptyset$ but no common u merges the pair;
- $\chi(x, y) = 0$ when $I_x \cap I_y = \emptyset$.

Then every generated continuation obeys

$$\chi(gx, gy) \leq \chi(x, y).$$

Proof. If the target pair (gx, gy) has a merger u , then ug merges (x, y) , so capacity 2 at the target implies capacity 2 at the source. If $I_{gx} \cap I_{gy}$ is nonempty, the domain inclusions already proved give $I_x \cap I_y \neq \emptyset$;

thus positive target capacity implies positive source capacity. These are all three ordered cases, proving the inequality. $\square$

Opportunity may therefore be lost but cannot be recovered under a shared continuation.

The complete census classifies 347,335 unordered distinct pairs: 346,991 are coalescible by a common continuation, 131 overlap only under different continuations, and 213 have disjoint future domains. It records 347 strict primitive decreases: 37 of type $2 \rightarrow 1$, 207 of type $2 \rightarrow 0$, and 103 of type $1 \rightarrow 0$. The monotonicity inequality was independently checked in 31,930,235 pair/effect applications.

For an effect m , the future-profile construction retains $\Lambda_m(x, y) = \chi(m(x), m(y))$ and identifies two effects exactly when all their future profiles agree.

Present reunion capacity is not always autonomous. Across the 369 fixed systems, 11,211 effects yield 484 current profiles, while complete future prediction requires 552 states. Twenty-two systems require hidden refinement, and four force an intermediate carrier strictly between present-profile and full-effect descriptions. Recursive future-profile closure terminates in 22 fixed systems, with no expansion or nontrivial cycle.

4.6.4 What the laboratory establishes—and what it cannot establish

The census supplies exact examples of transformation growth, irreversible ideal and kernel orders, incomparable quotients, distributed recovery, minimal predictors, recursive closure, distinct local futures, monotone opportunity loss, and interaction-created distinctions. It also gives counterexamples to several scalar and functional simplifications.

The finite carrier, total transformations, composition, identity, and two-primitive source are laboratory inputs. They are not derived from P_0/R or from the binary constructor. Kernel depth is not equated with k , a predictive class with an actual perspective, or ideal descent with physical time. Exhaustion at $n \leq 4$ establishes the census, not a theorem beyond it.

The laboratory separates two questions: what follows from the declared interaction, and whether that interaction models a named phenomenon. Only the first is addressed here. Fifty-six automated tests cover all 1,555 canonical systems, 32,926 generated effects, 28,313 ordered quotient pairs, 20,317 minimised predictive states, 31,930,235 opportunity-loss checks, and an independent reconstruction of every recorded quotient and profile. These tests assess the calculus and its implementation; they do not privilege one audited system as a model of reality.

The falsifiability section records the corresponding counterexamples and provenance failures. None of the finite representations is asserted to exhaust the possible perspectives of R .

5 Prime-source occurrences and the 36-relation frontier

The elementary identity $p \cdot 1 = p$ supplies a compact example of the source/report distinction. The two p tokens have equal value but different source and returned-product roles. A value reporter merges them; an occurrence-sensitive reporter retains the relation

$$p_{\text{source}} \cdot 1_{\text{unit}} = p_{\text{return}}. \quad (1)$$

The reduced triad contains every positive factor relation exactly when p is prime. For a composite n , an intermediate factor pair is a separator that the triad omitted.

The finite terminal carrier $\mathbb{Z}_{10}$ gives a second example. Removing the zero terminal leaves nine active states. The complete unordered relation frontier contains $\binom{9}{2} = 36$ cells. Under the affine return $d \mapsto 1 - d \pmod{10}$, the eight interior states form the four pairs (2, 9), (3, 8), (5, 6), and (7, 4). A reporter that keeps only the prime-sector label identifies the two sides of each pair. It is closed only for continuation families that cannot distinguish source from return.

Proposition 5.1 (Prime-observer quotient test). *Let $U(q, \varepsilon) = q$ on the four two-sided sectors. An autonomous law descends to the U -states if and only if every admitted continuation has the same U -response on $(q, +)$ and $(q, -)$ for each $q \in \{2, 3, 5, 7\}$.*

Proof. This is the autonomy/closure criterion applied to the side-forgetting fibres. A distinguishing continuation witnesses failure; fibrewise constancy defines the quotient law. $\square$

The example connects the abstract fibre condition to an arithmetic occurrence: equality of present value is insufficient whenever role, multiplicity, or source address changes a future operation. P17 gives the complete $8 + 4 + 24$ relation decomposition and the all-prime predictive refinement.

6 Related work

Spencer-Brown's *Laws of Form* gives distinction formal priority through the mark and the calculus of indications (Spencer-Brown 1969). Varela develops related calculi of self-reference and autonomy (Varela 1975, 1979). The present construction shares the foundational emphasis on distinction but takes a different mathematical route: the primitive report is an unordered complete family with occurrence multiplicity, and the central objects are reconstructive views, complete output decks, and quotient reporters. No crossing law, re-entry rule, or Boolean reading is assumed.

The language of perspective and autonomy also places the work near second-order cybernetics, autopoiesis, and the law of requisite variety (Foerster 1981; Maturana and Varela 1980; Ashby 1956). Conant and Ashby's good-regulator theorem makes the relation between regulation and internal modelling particularly explicit (Conant and Ashby 1970). Here, however, perspective has a fixed formal meaning: it is a quotient of a declared source. Autonomy is the descent of complete behaviour through that quotient, and its associated limit is the stable-blindness theorem.

The core calculus is prior to probability. The discrete coding formulations of Hartley and Shannon begin with a specified alphabet and a quantitative coding or distributional problem (Hartley 1928; Shannon 1948; Cover and Thomas 2006). Complete decks instead retain occurrence multiplicity before normalisation or randomisation. Blackwell's comparison of experiments and the data-processing behaviour of Kullback–Leibler divergence provide useful analogies for information loss under post-processing (Blackwell 1953; Kullback and Leibler 1951), but the results here are deterministic statements about quotient fibres. No probability measure or divergence enters their proof.

The closest technical neighbours of response closure are behavioural equivalence and partition refinement. Myhill–Nerode theory distinguishes states by their future contexts (Myhill 1957; Nerode 1958; Rabin and Scott 1959); classical algorithms compute stable partitions efficiently (Moore 1956; Hopcroft 1971; Paige and Tarjan 1987). Coalgebra provides a general language for observations, transitions, and behavioural equivalence (Rutten 2000; Wismann et al. 2020), while universal algebra treats quotients through operation-preserving congruences (Birkhoff 1935; Burris and Sankappanavar 1981). Transformation semigroups supply a related setting for finitely generated effects (Eilenberg 1976; Howie 1995). The distinction-generated setting contributes a particular provenance: complete occurrence decks are derived from the source hierarchy, and disagreement between those decks is used to generate—not merely test—the next refinement.

The hierarchy itself lies in the combinatorics of rooted trees, species, and recursive classes (Otter 1948; Joyal 1981; Bergeron, Labelle, and Leroux 1998). Classical reconstruction asks, among other variants, whether a structure can be recovered from deletion cards (Kelly 1957). The deck used here is different: each retained occurrence is replaced once, and origin multiplicity is preserved. The target-free binary process produces Catalan counts (Stanley 2015), but the enumeration is observed only after the transition has been fixed. This reverses the usual modelling order in cellular automata, where a lattice, neighbourhood, alphabet, and update rule are selected before emergent behaviour is studied (Wolfram 1984; Crutchfield 1994).

Common-source comparison can be located more precisely among categorical and contextual formalisms. Institutions formalise translation across signatures (Goguen and Burstall 1992); Chu spaces retain an explicit object–observation pairing (Pratt 1995); and sheaf- and topos-based approaches encode compatibility among contextual descriptions (Abramsky and Brandenburger 2011; Döring and Isham 2008). The present result is narrower: two quotient maps from a single finite source determine whether their comparison is a function, a genuinely relational span, or a joint refinement. The order-theoretic part belongs to the lattice of equivalence relations, with intersection as meet (Birkhoff 1967), while its interpretation as a least shared report parallels the intent, though not the full machinery, of formal concept analysis (Ganter and Wille 1999).

6.1 Position of the contribution

None of the individual ingredients—rooted hierarchies, multisets, partitions, congruences, closure operators, behavioural equivalence, or Catalan enumeration—is new. The contribution claimed here is a source-closed theorem chain: complete recursive distinction yields a source hierarchy; that hierarchy determines complete behaviour; and complete behaviour determines either a lawful quotient or the least forced refinement.

Its main assertion is a coupled one: constancy of complete behaviour on quotient fibres is exactly the condition for autonomous law; failure determines a least missing distinction; and success fixes an internal ceiling on reconstruction. The one-occurrence reconstruction theorem, the relational obstruction to naive reporter pairing, and the common-source treatment of incomparable reports complete that chain. The novelty claim is deliberately synthetic rather than absolute: rooted hierarchies, congruence refinement, behavioural equivalence, and partition lattices are established mathematics. What is claimed here is their integration under one source-retention discipline, together with the stated reconstruction, least-closure, stable-loss, and reporter-combination results. Any stronger priority claim would require an equivalence analysis preserving occurrence multiplicity, reporter order, and the relevant universal properties.

7 Limitations and falsifiability

The boundary P_0/R has no positive theory here. Assigning it even an empty carrier would place a distinction at the point reserved for no reportable distinction. Likewise, when $q > 2$, a q -resultant record in P_1 supports theorems about that record but says nothing by itself about the native logic or experience of a hypothetical P_{q-1} .

The hierarchy grammar is minimal only relative to stated retention requirements: a containing occurrence, all present resultants, multiplicity, recursion, and no unlicensed child order. Other faithful encodings may exist. The proved reconstruction results are finite and well-founded; infinite hierarchies, non-well-founded recurrence, limiting closure, measurability, and effective computation require additional hypotheses. Non-well-founded set theory may supply tools for one such extension (Aczel 1988), but it is not part of the present core.

The finite interaction branches deliberately add carriers, operations, focus values, effect families, or schedules. Those assumptions appear in the relevant theorem statements and are marked [X] in Appendix A. Formal resemblance to time, space, causality, energy, learning, or another empirical concept is insufficient for identification. Such a bridge would require a measurement map and discriminating predictions.

The mathematical claims are directly refutable. Counterexamples would include unequal finite hierarchies with equal reconstructive views at the stipulated depth; unequal sources with the same complete one-occurrence deck; a reporter satisfying fibre constancy but lacking a descended law; a closed refinement properly below the claimed least closure; or an already-closed reporter whose response iteration refines its fibres. The finite audits are refuted by an independently reproducible discrepancy in domain, canonicalisation, totals, or witnesses.

The provenance conditions are also testable. A [B] claim is violated if a proof later assigns structure to P_0/R ; a [C] construction is defective if it silently introduces order, labels, selection, probability, deletion, or a target; an [X] result is mis-scoped if its added assumptions disappear from the theorem statement. Beyond formal correctness, the programme would have limited scientific value if its bridges proved uninformative,

its complete decks intractable, or its theorem chain reducible to a known formalism without a useful gain in provenance or application.

8 Reproducibility and research artefacts

Proofs carry the mathematical claims. Code is used to audit finite domains, retain counterexample witnesses, and expose assumptions that may be hidden in an implementation. The companion archive, *Abhijit_Singh_Distinction-Generated_Perspectives_v2.0_source.zip*, contains the manuscript sources, figures, cited bibliography, theorem-oriented test corpus, complete finite-audit data, and a single audit runner. Its *REPRODUCIBILITY.md* records the file manifest, Python version, exact invocations, and expected totals.

Running *bash run_audits.sh* from the archive root executes the curated suite under Python 3.12. The suite contains **134 automated test methods**: 21 core hierarchy-and-closure tests, 38 same-referent, predictive, and universal-law tests, 7 target-free-game tests, 12 projection-and-partition tests, and 56 open-rule-census tests. The largest enumerations cover all 59,049 pointed binary-input ternary laws and 1,555 canonical finite source rules in the common-source study. The release documentation supplies the canonicalisation rules, test modules, expected totals, and machine-readable witnesses; the bounded computations remain logically separate from the unbounded proofs.

These computations support diagnosis and independent replication; they do not replace unbounded proofs or establish the boundary and constructor choices.

9 Open problems

The following problems arise directly from the present calculus.

1. **Complete two-distinction interaction.** Classify the full two-step families for identical, nested, and incomparable selected occurrences without assuming a cut. Determine when opposite orders agree, refine one another, or force a relational report.
2. **Infinite and non-well-founded completion.** Identify infinite classes on which finite views determine a source and least response closure exists; state the topological or coalgebraic assumptions explicitly.
3. **Encoding invariance.** Characterise the theorems preserved by any faithful P_1 encoding of incidence and multiplicity.
4. **Complexity of forced refinement.** Obtain sharp bounds for computing $Cl_\infty(Q)$, and locate tractable subclasses relative to classical partition-refinement algorithms.
5. **Composition of perspectives.** Develop a categorical setting for closed reporters, source-compatible translations, and relations, treating the common-source relation as primitive when no translation exists.
6. **Empirical bridge criteria.** State necessary conditions for identifying a derived structure with an observed regularity, including behaviour preservation, risky predictions, and instrument-induced erasure.
7. **Nonbinary representation.** For $q > 2$, determine the strongest encoding-invariant statements P_1 can make about a complete q -resultant presentation without claiming native P_{q-1} access.

10 Conclusion

The paper begins at a carefully delimited point: P_0/R has no positive model, while P_1 can record complete resultant families, recursive containment, and occurrence multiplicity. That limited transcription already supports two reconstruction theorems and a general criterion for the descent of law to a quotient reporter.

The criterion links autonomy to loss. A quotient is lawful precisely when the distinctions it erases do not alter complete reported response. If that condition fails, behaviour identifies the least required refinement. If it holds,

the descended law cannot recover the erased distinctions. Closed reporter combination and common-source comparison show that relational information may become necessary again when perspectives interact.

This is a finite formal theory, not an empirical ontology. Its contribution is an explicit calculus in which source construction, quotient law, forced refinement, and stable blindness belong to one theorem chain. Further claims must state their added structure and earn any interpretation through a separate bridge.

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12 Appendix A. Theorem and provenance register

The following register records the principal claims and the strongest defensible status assigned to each. Codes are [B] boundary discipline, [C] declared constructor, [G] general theorem, [F] finite audit, and [X] conditional extension. Here “general” is restricted to the declared mathematical domain and does not extend to the non-reportable boundary.

ID • status	Result	Statement and scope
A1 [B]	P_0/R boundary	No carrier, object, state, law, value, or operation is assigned to the absence of reportable distinction.
A2 [B]	First reportable perspective	Every symbol and computation is a binary report in the first reportable perspective; a nonbinary q -resultant presentation with $q > 2$ is a P_1 encoding, not native access to the corresponding nonbinary perspective.
A3 [C]	Finite hierarchy grammar	The unordered-multiset grammar with terminal $\bullet$ and internal q -resultant nodes, $q \geq 2$, is the minimal declared no-discard transcription.
A4 [C] [G]	Prop. 3.2.3: persistent occurrence	Under A3’s declared no-erasure, no-selection, and no-duplication obligations, the container uniquely retains one old role through distinction.
A5 [C]	Present-resultant operator	Δ returns the complete multiset of already-present child reports; no child is selected or weighted.
A6 [G]	Thms. 3.3.2–3: view reconstruction	For height $h(T)$, the $\text{depth-}h(T) + 1$ recursive complete-resultant view determines T ; view depth strictly refines.
A7 [G]	Prop. 3.2.4: balance	$L(T) = 1 + \sum_v (q(v) - 1)$ for every finite hierarchy.
A8 [G]	Lossy scalarization	Terminal count, internal count, and flattened profiles identify explicit pairs of unequal hierarchies. Nested relation cannot be reconstructed from their scalar values.
A9 [G]	Thm. 3.4.2: deck reconstruction	For fixed reported $q \geq 2$, equality of complete one-occurrence decks $E_q(T) = E_q(U)$ implies $T = U$.

ID · status	Result	Statement and scope
A10 [G]	Thm. 3.4.3: cut and reunion	For a marked occurrence $T = C[D]$, internal and external one-step outputs partition the full deck exactly, with occurrence origin and multiplicity retained.
A11 [G]	Thm. 3.4.4: restricted order equality	Replacements at disjoint occurrences on opposite sides of a persistent cut preserve one another and produce the same complete two-step family in either order.
A12 [G]	Thm. 3.5.2: quotient closure	A quotient law exists exactly when equal Q -reports have equal complete pushed-forward output decks.
A13 [G]	Thm. 3.6.1: least response closure	Iterated complete response reports define $Cl_\infty(Q)$, the least closed refinement of Q relative to the declared output decks.
A14 [G]	Thm. 3.7.1: stable blindness	If Q is closed, every response iteration has the same fibres as Q ; its own law cannot recover stably erased distinctions.
A15 [G]	Prop. 3.7.2: grouping hierarchy	Closed reporters $G_0 \prec G_1 \prec \dots \prec G_\omega \simeq \mathcal{H}$ retain successively deeper shared containment; adjacent laws commute with erasure.
A16 [G]	Thm. 3.8.1: closed combination	The response closure of the direct pair defines $X \vee_c Y$, the least reporter above both that supports the complete law. Cor. 3.8.2 gives its commutative, associative, and conditional idempotence laws.
A17 [C] [G]	TF: target-free binary game	Prefix-closed binary records with all legal boundary expansions retained satisfy local replacement, inert-name invariance, separate-occurrence commutation, ancestry-only history restriction, exact factorisation, and no finite terminal state.
A18 [F]	Binary-game enumerations	Complete-state counts for sizes 1–8 are 1, 2, 5, 14, 42, 132, 429, and 1,430; complete legal histories are $1!, \dots, 8!$, with finite identity and decomposition audits.

ID · status	Result	Statement and scope
A19 [C] [G] [F]	PT/PI: projection and partition closure	Explicit histories project to ordered forms, order-free forms, and availability; each projection is continuation-closed, and partition interaction is monotone under refinement. No count-only reporter recovers erased form.
A20 [X] [G]	SRQ: same-referent quotient	For a declared effect family A , equality under all effects defines the least sufficient quotient Q_A ; a candidate descends iff constant on its fibres.
A21 [X] [G]	CR: constraint reversal	If Q_2 properly refines Q_1 , then for every codomain with at least two values the family of compatible candidate laws is strictly larger over Q_2 .
A22 [X] [G]	RLD: completion count	For a finite codomain of size $q \geq 2$, a reached partial law with r reached configurations among M admits exactly q^{M-r} total completions; uniqueness holds exactly when $r = M$. For $q = 1$, the extension is uniquely constant regardless of reachability.
A23 [C] [X] [G]	IPQ: predictive quotient	For a finite carrier $ X = q \geq 2$ and arity $m \geq 1$, finite one-place contexts of a pointed operation generate a stabilizing congruence; its quotient is the coarsest focus-preserving predictive quotient.
A24 [F]	Predictive audit	The finite ternary audit checks all 59,049 pointed binary-input laws, including the stabilisation bound and quotient universal property.
A25 [X] [G]	CS-1: common-source translation	For quotient perspectives of one declared finite source, a direct translation exists exactly in the refinement direction; otherwise the full source-induced relation is retained.
A26 [X] [G]	CS-2: joint perspective creation	Incomparable source quotients induce a joint quotient strictly finer than either when each retains a distinction erased by the other.

ID · status	Result	Statement and scope
A27 [F]	Common-source audit	The translation, relation, and joint-perspective claims are checked across all 1,555 frozen finite source rules in the declared audit domain.
A28 [C] [X]	Open finite-rule constructor	The laboratory declares a carrier of size $n \in \{2, 3, 4\}$, two total transformations, identity, and composition; none is derived from the primitive.
A29 [X] [G]	Prop. OR-1: ideal and rank order	In the declared transformation monoid, $\text{rank}(gm) \leq \text{rank}(m)$, while multiplication can only contract the generated two-sided ideal.
A30 [X] [G]	Prop. OR-2: kernel and predictor order	Kernels can only coarsen under left composition, and current-kernel, minimal-predictor, and full-effect counts obey $K \leq Q \leq E$.
A31 [X] [G]	Thm. OR-3: recursive predictive lift	Equality of every future kernel history is a two-sided congruence. Up to pointed-monoid isomorphism, the induced finite lift reaches a fixed point and has no nontrivial cycle.
A32 [X] [G]	Prop. OR-4: future-domain and reunion order	Generated domains satisfy $I_{gx} \subseteq I_x$, and reunion capacity satisfies $\chi(gx, gy) \leq \chi(x, y)$ under every shared continuation.
A33 [F]	Open-rule census	Exhaustive enumeration for $n \leq 4$ records 1,555 canonical systems, their quotient and predictor regimes, recursive lifts, reunion profiles, and all stated opportunity-loss audits.

Sequential Quantum Instruments

Depletion, Regeneration, and Minimal Predictive State

Abhijit Singh

Contents

Abstract	3
1 Operational object	4
I Sharp sequential measurement of one singlet	4
2 Even-cycle protocol	4
3 Exact entropy profile	4
4 Geometric correlator depletion	5
II Repeated contextual instruments	6
5 Unital self-idling	6
6 Passage potential and stationary engine criterion	7
7 Yu-Oh and KCBS calculations	7
III A complete four-outcome marked instrument	7
8 Physical completion and common basis	7
9 Corrected transition law and minimal predictive state	8
10 Information loss and continuing entropy	8
11 Record cost	8
12 Scope and exact claim boundary	9
A Reproducibility table	9
B Full depletion and certificate proofs	9
C Complete stationary witness derivation	10
D Exact contextual regeneration values	10
E Corrected four-record instrument	11
F Reproduction obligations	11
References	11

Abstract

Sequential measurement is analysed through the complete instrument object: an outcome map and its conditional successor state. A sharp alternating walk on one singlet is solved exactly, proving that the first local result destroys the reusable Bell resource while later conditional entropy remains positive. Repeated unread projective channels are classified by their fixed operator algebra, yielding the exact stationary witness value m/d and the sharp rank-over-dimension compatibility floor. A Poisson equation gives the finite-time passage potential and the exact criterion for sustained contextual witnesses; the Yu-Oh no-reset protocol and KCBS selective-replacement thresholds are derived. A processed four-outcome Bell instrument is then physically completed and iterated in a common basis. Its current mark is the unique minimal predictive state, with exact stationary, spectral, information, and record-cost values.

Keywords: quantum instrument; sequential measurement; Bell nonlocality; contextuality; predictive state; entropy rate

Scope and status: Analytic theorems retain their displayed protocol assumptions. Processed-matrix results are exact for the supplied matrices and positivity completion. No result is generalized to all measurement systems.

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1 Operational object

A measurement operation is represented by a quantum instrument, not by a probability distribution alone. The instrument produces both an outcome and a conditional successor state. The same distinction is required when a sequence is reused: outcome uncertainty, state memory, nonclassical certification, and physical record cost are separate quantities.

For a completely positive instrument $\{\mathcal{I}_a\}_a$,

$$p(a | \rho) = \text{Tr } \mathcal{I}_a(\rho), \quad \rho'_a = \frac{\mathcal{I}_a(\rho)}{p(a | \rho)}. \quad (1)$$

The unread channel is $\mathcal{T} = \sum_a \mathcal{I}_a$. All repeated-use calculations below preserve this complete object.

Part I Sharp sequential measurement of one singlet

2 Even-cycle protocol

Let $n \geq 4$ be even, $\delta = \pi/n$, $c = \cos \delta$, and $d = \cos(2\delta)$. Define

$$O(\theta) = \cos \theta \sigma_z + \sin \theta \sigma_x, \quad (2)$$

and start from the singlet

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \quad (3)$$

At global step k , measure $O(k\delta)$ on Alice for even k and Bob for odd k , recording $X_k \in \{-1, +1\}$. Adjacent measurements act on different subsystems and are compatible; every measurement is rank-one and projective.

Theorem 2.1 (Sharp-mark depletion). *Conditional on $X_0 = a$, the post-measurement joint state is*

$$\rho_{AB}^{(a)} = |a; 0\rangle\langle a; 0|_A \otimes |-a; 0\rangle\langle -a; 0|_B. \quad (4)$$

Its entanglement negativity is zero. Every later selected local projective outcome keeps the state product. The same pair cannot generate a later Bell violation unless a new entangling operation is applied.

Proof. The singlet is perfectly anticorrelated in every common basis: $(P_a(0) \otimes I)|\Psi^-\rangle \propto |a; 0\rangle_A \otimes |-a; 0\rangle_B$. This vector has Schmidt rank one. A local rank-one projector maps a product vector to another product vector or zero, so induction covers the remaining selected outcomes. The unread first measurement is a convex mixture of product states and is separable. $\square$

3 Exact entropy profile

The first result is uniform. Singlet anticorrelation and the local two-step update give

$$\Pr(X_1 = b | X_0 = a) = \frac{1 - abc}{2}, \quad (5)$$

$$\Pr(X_k = x | X_{<k}) = \Pr(X_k = x | X_{k-2} = y) = \frac{1 + xyd}{2}, \quad k \geq 2. \quad (6)$$

With binary entropy h_2 ,

Theorem 3.1 (Entropy profile).

$$H(X_0) = 1, \quad (7)$$

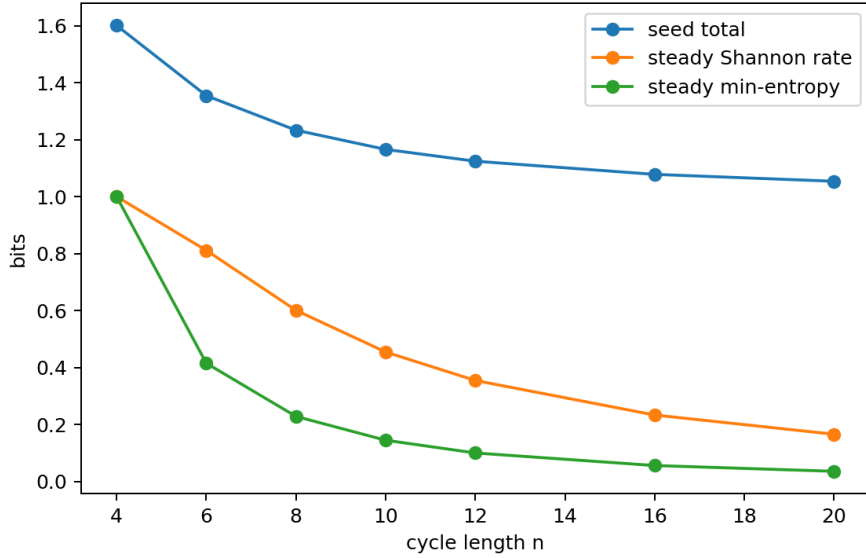
$$H(X_1 | X_0) = h_2\left(\sin^2 \frac{\pi}{2n}\right), \quad (8)$$

$$H(X_k | X_{<k}) = h_2\left(\cos^2 \frac{\pi}{n}\right), \quad k \geq 2. \quad (9)$$

The trusted-model guessing probability after the seed is $\cos^2(\pi/n)$ and the conditional min-entropy is $-\log_2 \cos^2(\pi/n) > 0$.

Table 1: Exact entropy values for representative even cycles.

n	$1 + H(X_1 X_0)$	steady Shannon rate	steady guessing probability
4	1.6009	1.0000	0.5000
6	1.3546	0.8113	0.7500
8	1.2333	0.6009	0.8536


Figure 1: Seed entropy, steady Shannon rate, and trusted-model min-entropy.

4 Geometric correlator depletion

Define $R_k = \mathbb{E}[X_k X_{k+1}]$. Since $\mathbb{E}[X_{k+2} | X_k] = dX_k$,

Theorem 4.1 (Geometric decay).

$$R_k = -\cos \frac{\pi}{n} \cos^k \frac{2\pi}{n}. \quad (10)$$

For $n = 4$, all correlators after R_0 vanish exactly. The signed sum over every later n -step block tends to zero.

A fresh-singlet experiment on every edge instead gives the chained magnitude $n \cos(\pi/n)$. These are different resource schedules, not two readings of the same sequential resource.

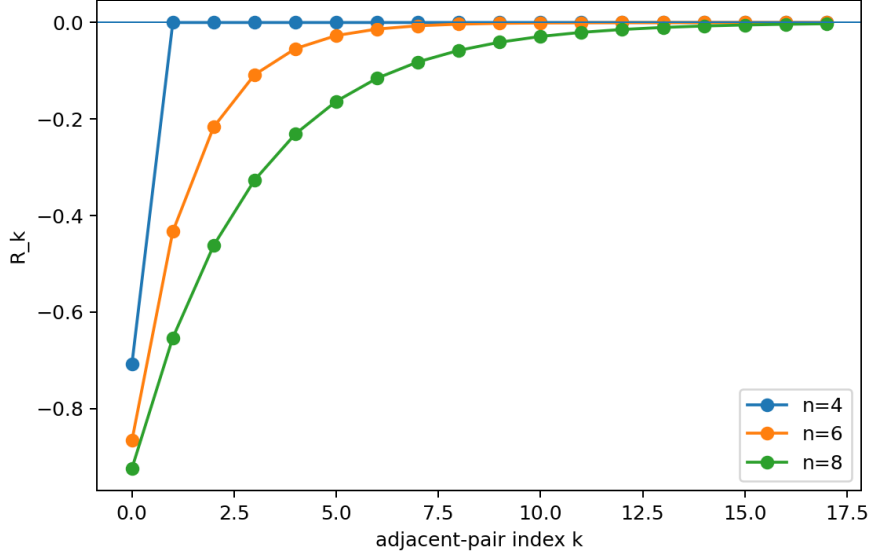


Figure 2: Exact adjacent-correlator decay for three even cycles.

Theorem 4.2 (Certificate separation). *Positive post-seed conditional entropy does not imply a post-seed Bell-device-independent certificate from the same walked pair.*

Proof. After the first result the conditional joint state is product. The remaining process has a local causal realization: sample the correlated seed pair, then update the two sides with independent local transition coins. The later statistics therefore do not require a nonlocal resource, while the Shannon and trusted-model min-entropies remain positive. $\square$

Part II Repeated contextual instruments

5 Unital self-idling

For a projector P , the unread Lueder channel

$$\mathcal{D}_P(X) = PXP + (I - P)X(I - P) \quad (11)$$

is an orthogonal projection in Hilbert-Schmidt operator space. For a periodic sequence $P_0, \dots, P_{m-1}$, let $\mathcal{T} = \mathcal{D}_{P_{m-1}} \cdots \mathcal{D}_{P_0}$.

Theorem 5.1 (Unital self-idling). *The fixed operator space is the intersection of the commutants of the P_j . If the family is irreducible, $\mathcal{T}^k(\rho) \rightarrow I/d$ for every initial state. An unweighted rank-one witness therefore idles at*

$$W_{\text{idle}} = m/d. \quad (12)$$

This explains the exact values $5/3$, $7/5$, and 2 in the audited walks. Comparing m/d with the noncontextual bound $\alpha(G)$ gives strict idling, facet idling, or state-independent violation.

Theorem 5.2 (Adjacent-orthogonality floor). *If $P_j P_{j+1} = 0$ and the stationary state is I/d , the asymptotic fraction of forced next steps obeys*

$$\tau \geq \frac{1}{m} \sum_j \frac{\text{rank } P_j}{d}. \quad (13)$$

For rank-one tests the sharp floor is $1/d$. The frequently observed value $1/3$ is the qutrit instance, not a universal constant.

6 Passage potential and stationary engine criterion

For randomized contexts c and score $g(c, x)$, define the averaged reward operator R and unread channel $\mathcal{T}$. If $\mathcal{T}$ is primitive with stationary state σ , put $q = \text{Tr}(\sigma R)$.

Theorem 6.1 (Poisson passage potential). *There is a Hermitian K , unique up to a scalar, satisfying*

$$K - \mathcal{T}^*(K) = R - qI. \quad (14)$$

For the mean evolution,

$$\sum_{t=0}^{N-1} \text{Tr}(\rho_t R) = Nq + \text{Tr}(\rho_0 K) - \text{Tr}(\rho_N K). \quad (15)$$

The long-run witness exceeds the noncontextual bound exactly when $q > \beta_{\text{NC}}$.

If $R = qI$, then $K = 0$ and the expected witness is q at every round for every state; no return to a preferred state is required.

7 Yu-Oh and KCBS calculations

For the 13-ray Yu-Oh construction,

$$L_{YO} = \frac{25}{3}I, \quad (16)$$

while the noncontextual bound is 8. With the displayed singleton/edge randomization, the quantum mean is $1/3$, the noncontextual mean is at most $8/25$, and the per-round gap is $1/75$.

For the five-ray KCBS family, replacement after a no outcome is optimal within the audited outcome-selective family. The exact thresholds are

$$r_{\text{no}}^* = 0.8396425434, \quad (17)$$

$$c_{\text{min}} = 0.5037855260, \quad (18)$$

$$r_{\text{uniform}}^* = 0.5115311845. \quad (19)$$

Replacement after yes outcomes alone reaches only $1.9098300563 < 2$.

Part III A complete four-outcome marked instrument

8 Physical completion and common basis

Let

$$E_a = \sum_r e_{ar} |v_{ar}\rangle\langle v_{ar}|, \quad \sigma_a = \sum_s q_{as} |w_{as}\rangle\langle w_{as}|. \quad (20)$$

The Kraus family

$$K_{asr} = \sqrt{e_{ar} q_{as}} |w_{as}\rangle\langle v_{ar}| \quad (21)$$

realizes $\mathcal{I}_a(\rho) = \text{Tr}(E_a \rho) \sigma_a$. The processed realization contains 60 nonzero Kraus operators. The complete classical-quantum Choi operator is

$$\mathbb{J} = \sum_a |a\rangle\langle a|_C \otimes \sigma_a \otimes E_a^\top. \quad (22)$$

One nonpositive tomography estimate is repaired by retaining Bell-basis populations and scaling coherences by the maximal positivity factor

$$\eta_* = 0.708418699729814. \quad (23)$$

Effects and conditional states must be transformed into one common basis before repeated-use multiplication. Omitting this transformation changes the transition law by Frobenius norm 0.541915971853 .

9 Corrected transition law and minimal predictive state

The corrected column-stochastic law is

$$C = \begin{pmatrix} 0.43696297 & 0.19810609 & 0.20644079 & 0.14060033 \\ 0.16017410 & 0.46165671 & 0.13951861 & 0.22362294 \\ 0.24443666 & 0.11892971 & 0.46061556 & 0.17659060 \\ 0.15842628 & 0.22130750 & 0.19342504 & 0.45918613 \end{pmatrix}. \quad (24)$$

Every pair of columns has positive total-variation distance, so no two marks can be merged in a first-order predictive state.

Theorem 9.1 (Minimal mark state). *The four current marks are the unique coarsest predictive states for all later repeated uses of the processed instrument.*

The stationary law and nontrivial eigenvalues are

$$\pi = (0.24327934, 0.24563854, 0.24991431, 0.26116781), \quad (25)$$

$$\text{spec}(C) = \{1, 0.35301358, 0.27498875, 0.19041902\}. \quad (26)$$

Writing $\Pi = \pi \mathbf{1}^\top$ and $R = C - \Pi$ gives $C^n = \Pi + R^n$.

10 Information loss and continuing entropy

For a uniform initial source label J ,

$$I(J; A_0) = 0.4720622220 \text{ bits}, \quad (27)$$

$$I(J; A_1) = 0.0409311197 \text{ bits}. \quad (28)$$

Only 8.6707% remains after one reuse. At stationarity,

$$H(A_t) = 1.9994586368 \text{ bits}, \quad (29)$$

$$h = H(A_{t+1} | A_t) = 1.8413629168 \text{ bits/use}, \quad (30)$$

$$I(A_t; A_{t+1}) = 0.1580957201 \text{ bits}. \quad (31)$$

The best next-mark probability is 0.4547437955, giving conditional min-entropy 1.1368741409 bits.

For surprisal, the centred Poisson potential is

$$k = (0.03023948, -0.02314357, 0.00046185, -0.00684276), \quad (32)$$

and the finite-time mean error is at most $0.06047896/N$ bits per use.

11 Record cost

With the previous mark available as side information, optimal cyclic compression and erasure obey

$$Q_{\min} \geq k_B T \ln 2 h = 1.2763355141 k_B T \quad (33)$$

per use. Avoiding reset requires memory growth at 1.8413629168 bits per use.

12 Scope and exact claim boundary

The singlet and contextuality theorems apply to their displayed protocols. The four-outcome results are exact for the processed public matrices and the displayed positive completion. No universal claim about all sequential measurements follows. Raw conditional entropy, trusted-model min-entropy, nonclassical witness value, source memory, and physical record cost remain separate ledger entries.

A Reproducibility table

Object	Check
Even-cycle depletion	product state after first result; negativity $1/2 \rightarrow 0$
Entropy	exact $n = 4, 6, 8$ values and positive post-seed min-entropy
Correlator	$R_k = -cd^k$ and geometric block extinction
Unital cycles	fixed commutant, m/d idling, rank/ d floor
Contextual regeneration	Yu-Oh operator identity and KCBS thresholds
Marked instrument	positivity repair, Kraus completeness, Choi positivity, basis conversion, stationarity, information decay, Poisson potential

B Full depletion and certificate proofs

Let $P_a(\theta) = (I + aO(\theta))/2$. The singlet satisfies perfect anticorrelation in every common basis:

$$(P_a(0) \otimes I)|\Psi^-\rangle \propto |a;0\rangle_A \otimes |-a;0\rangle_B. \quad (34)$$

This selected state has Schmidt rank one and zero negativity. The unread first result is the convex mixture of the two selected product states and is separable. Every later selected local rank-one projector preserves product form. Thus the first sharp record, not the second, consumes the same pair's future Bell resource.

Singlet anticorrelation gives

$$\Pr(X_1 = b \mid X_0 = a) = \frac{1 - ab \cos(\pi/n)}{2}, \quad (35)$$

and for $k \geq 2$ the most recent result on the same side is sufficient:

$$\Pr(X_k = x \mid X_{<k}) = \frac{1 + xX_{k-2} \cos(2\pi/n)}{2}. \quad (36)$$

Hence

$$H(X_0) = 1, \quad (37)$$

$$H(X_1 \mid X_0) = h_2\left(\sin^2 \frac{\pi}{2n}\right), \quad (38)$$

$$H(X_k \mid X_{<k}) = h_2\left(\cos^2 \frac{\pi}{n}\right), \quad k \geq 2. \quad (39)$$

The trusted-model guessing probability is $\cos^2(\pi/n)$ and the conditional min-entropy is its negative base-two logarithm.

The adjacent correlator obeys

$$R_{k+1} = \cos(2\pi/n)R_k, \quad R_0 = -\cos(\pi/n), \quad (40)$$

therefore

$$R_k = -\cos \frac{\pi}{n} \cos^k \frac{2\pi}{n}. \quad (41)$$

For $n = 4$, every correlator after the first vanishes exactly. A fresh-pair chained experiment instead supplies a new singlet at every edge and gives $n \cos(\pi/n)$. Those are different resource schedules, not two measurements of one conserved violation.

Table 3: Exact entropy profile of the sharp even-cycle walk.

n	seed $1 + H(X_1 X_0)$	steady rate	guess probability
4	1.6009	1.0000	0.5000
6	1.3546	0.8113	0.7500
8	1.2333	0.6009	0.8536

C Complete stationary witness derivation

For a sharp projector P , the unread channel

$$\mathcal{D}_P(X) = PXP + (I - P)X(I - P) \quad (42)$$

is the Hilbert–Schmidt orthogonal projection onto the commutant of P . A product of such channels fixes exactly the intersection of the commutants. If the projectors are irreducible, only scalar operators survive and every state converges to I/d . For m rank-one tests,

$$W_{\text{idle}} = \sum_{j=1}^m \text{Tr} \left(\frac{I}{d} P_j \right) = \frac{m}{d}. \quad (43)$$

If adjacent tests are orthogonal, every preceding yes result forces the next answer to be no. At stationarity the forced-step fraction satisfies

$$\tau \geq \frac{1}{m} \sum_j \frac{\text{rank } P_j}{d}, \quad (44)$$

which becomes $1/d$ for rank-one tests. The earlier apparent one-third constant is therefore the qutrit instance.

For a repeated instrument with averaged channel $\mathcal{T}$, reward operator R , and stationary state σ , put $q = \text{Tr}(\sigma R)$. On the centred subspace the Poisson equation

$$K - \mathcal{T}^*(K) = R - qI \quad (45)$$

has a solution under primitivity, giving

$$\sum_{t=0}^{N-1} \text{Tr}(\rho_t R) = Nq + \text{Tr}(\rho_0 K) - \text{Tr}(\rho_N K). \quad (46)$$

Thus the long-run witness is determined by q , while K is a bounded transient potential. If $R = qI$, the expected score is state-independent on every round and no reset is required.

D Exact contextual regeneration values

For the 13-ray state-independent witness, direct projector multiplication gives the scalar operator used in the protocol and an exact per-round separation of $1/75$ from the noncontextual limit. For the five-ray family, symmetry reduces the stationary state to one target-overlap coordinate. Selective replacement after a no result is optimal within the audited family. The thresholds are

$$r_{\text{no}}^* = 0.8396425434, \quad (47)$$

$$c_{\text{min}} = 0.5037855260, \quad (48)$$

$$r_{\text{uniform}}^* = 0.5115311845, \quad (49)$$

and yes-only replacement reaches only $1.9098300563 < 2$.

E Corrected four-record instrument

After expressing effects and returned states in one basis, the transition matrix is

$$C = \begin{pmatrix} 0.43696297 & 0.19810609 & 0.20644079 & 0.14060033 \\ 0.16017410 & 0.46165671 & 0.13951861 & 0.22362294 \\ 0.24443666 & 0.11892971 & 0.46061556 & 0.17659060 \\ 0.15842628 & 0.22130750 & 0.19342504 & 0.45918613 \end{pmatrix}. \quad (50)$$

Its stationary law is

$$\pi = (0.24327934, 0.24563854, 0.24991431, 0.26116781), \quad (51)$$

and every pair of columns has positive total-variation distance, proving that the four current records are the unique coarsest first-order predictive states. The exact information ledger is

$$H(A_{t+1}) = 1.9994586368, \quad (52)$$

$$H(A_{t+1} | A_t) = 1.8413629168, \quad (53)$$

$$I(A_t; A_{t+1}) = 0.1580957201 \quad (54)$$

bits. The finite-memory Landauer floor is therefore $1.2763355141 k_B T$ per use when the previous record is retained as side information.

Table 4: Decay of retained source information in the corrected repeated instrument.

reuse t	$I(J; A_t)$ bits	optimal source-guess accuracy
0	0.4720622220	0.6232838703
1	0.0409311197	0.3514199523
2	0.0040117838	0.2790063220
3	0.0004285525	0.2586660216
4	4.8954×10^{-5}	0.2526827310
10	1.7447×10^{-10}	0.2500040417

F Reproduction obligations

A full reproduction verifies projector identities, selected and unread negativity, transition probabilities, entropies, correlator decay, fixed spaces, Poisson telescoping, contextual bounds, threshold optimization, basis transformation, physical positivity repair, Kraus completeness, Choi positivity, stationary law, and information decay. Public matrix calculations are exact for the supplied processed estimates; population uncertainty requires the raw-count covariance.

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Provenance-Preserving Representation Genesis

Finite Deterministic Systems

Abhijit Singh

Contents

Abstract	3
1 False closure	4
2 Future equivalence and occurrence provenance	4
3 Generative definition	4
4 Predictive quotient construction	4
5 Essential-coordinate theorem	5
6 Relations, symmetries, and arity	5
7 Conservative return and native use	5
8 Joint coherent occurrences	6
9 Restart, transport, and authority boundaries	6
10 Executable evidence	6
11 A 36-cell relation-frontier benchmark	6
12 Complexity and limitations	7
A Reference algorithm	7
B Two construction regimes	7
C Proof receipts and source identity	8
D Collision certificates for minimality	8
E Native-use test	8
F Restart and transport protocol	8
G Executable test taxonomy	8
H Scaling and open-world boundary	9
References	9

Abstract

False closure occurs when a current report merges source occurrences whose admitted futures require different actions. A finite theory and executable architecture are developed to detect and repair that failure. Behavioural equivalence is separated from occurrence provenance; complete future signatures define the unique coarsest predictive refinement; and essential coordinates form the unique least direct-coordinate basis for a complete finite table. Representation changes are conservative extensions of valid old reports and are admitted only after native downstream use, matched ablation/restoration, restart recomputation, and hash-bound transport. Two finite constructions derive contextual object and relation classes, role symmetry, semantic arity, and transformation maps from source-addressed interactions. The supplied implementation evidence includes a large conformance suite, an independently executed 22-test transfer subset, and a collision benchmark in which the repaired state doubles exact continuation from 6/12 to 12/12. A new nine-state benchmark exercises relation-frontier discovery, conservative side-coordinate promotion, and arithmetic support-state repair.

Keywords: predictive quotient; provenance; representation synthesis; active learning; source address; schema evolution

Scope and status: Theorems are finite and relative to the declared continuation or complete direct-table envelope. Executable claims are limited to the supplied implementation and tests. Empirical truth and open-world discovery are not inferred.

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1 False closure

A predictive system usually begins with designer-supplied objects, actions, memory fields, and evaluation functions. The structural failure studied here occurs when two source occurrences receive one current report although an admitted continuation requires different actions.

Let S be a finite source set, $U : S \rightarrow Y$ a reporter, and $C(s)$ the set of correct continuation actions from s .

Proposition 1.1 (Impossibility under a lossy report). *If $C(s) \cap C(t) = \emptyset$ and $U(s) = U(t)$, no deterministic policy of the form $\pi = \hat{\pi} \circ U$ can be correct on both s and t .*

Proof. Report equality forces $\pi(s) = \pi(t)$. Correctness would require the common action to belong to two disjoint sets. $\square$

This is false closure: the present report is locally correct but is not a congruence for the declared future operations.

2 Future equivalence and occurrence provenance

For a continuation envelope Λ , define

$$s \equiv_{\Lambda} t \iff \text{Fut}_{\Lambda}(s) = \text{Fut}_{\Lambda}(t). \quad (1)$$

The reporter is sound exactly when

$$U(s) = U(t) \implies s \equiv_{\Lambda} t. \quad (2)$$

Behavioural equivalence does not erase occurrence identity. The implementation therefore stores two linked structures:

1. a predictive quotient used for decision and continuation;
2. an append-only occurrence graph carrying source address, parents, proof scope, operation version, and receipt hashes.

Two occurrences can share every covered future while remaining distinct historical events. Conversely, a new occurrence hash is not new semantic knowledge unless a later continuation uses it.

3 Generative definition

An extensional reporter $D_0 : X \rightarrow \{0, 1\}$ specifies visible membership but does not determine how objects were formed, what operations are admissible, or what evidence can revise the field. A generative definition is represented as

$$D = (A, O, I, R, U), \quad (3)$$

where A is an occurrence-address space, O a typed operation family, I the invariants that must survive, R a continuation/re-entry law, and U a forgetful reporter. The lift is conservative when the old report is recovered by projection and nonredundant when a possible future separates two lifted occurrences merged by U .

4 Predictive quotient construction

Given a complete finite direct table, define the response signature

$$\Sigma_{\Lambda}(s) = (U(\lambda s))_{\lambda \in \Lambda}. \quad (4)$$

Equality of signatures is the coarsest reporter sufficient for the declared continuation family.

Theorem 4.1 (Least predictive refinement). *If a reporter V refines U and every response $U \circ \lambda$ factors through V , then Σ_Λ factors through V . Hence Σ_Λ is the unique coarsest future-sufficient refinement up to equality of fibres.*

Proof. If $V(s) = V(t)$, sufficiency makes every response coordinate equal, so $\Sigma_\Lambda(s) = \Sigma_\Lambda(t)$. $\square$

A counterexample-guided compiler starts with U , finds a shortest separating continuation, adds its response coordinate, and repeats. On a finite carrier every strict split increases the number of classes, so the process terminates after at most $|S| - |U(S)|$ strict refinements.

5 Essential-coordinate theorem

Let $X \subseteq X_1 \times \cdots \times X_d$ be a complete covered table and $f : X \rightarrow Y$ its response. A coordinate set J is sufficient when f factors through the projection π_J . Coordinate j is essential if two rows agree on all other coordinates but differ in both coordinate j and output.

Theorem 5.1 (Unique least direct-coordinate basis). *Under complete coverage, every sufficient coordinate set contains every essential coordinate, and the essential coordinates are sufficient. They therefore form the unique least direct-coordinate basis.*

Proof. Omitting an essential coordinate merges its witnessing pair and makes the output nonfunctional. Conversely, if two rows agreeing on all essential coordinates had different outputs, a minimal output-changing pair would witness an additional essential coordinate. $\square$

The theorem is minimality within the displayed direct-coordinate class. It does not rule out redundant encodings or nonlinear representations outside that class.

6 Relations, symmetries, and arity

The audited finite fixture induces six contextual object classes and two relation classes. One relation is an unordered ternary relation with 27 complete formation rows. Role exchange is treated as a symmetry only when the complete response table is invariant under that exchange. Semantic arity is derived from the least sufficient role set, not read from the syntax of the input tuple.

7 Conservative return and native use

A new coordinate η is returned by

$$\tilde{U}(s) = (U(s), \eta(s)), \quad \pi_1 \tilde{U} = U. \quad (5)$$

Admission requires all of the following:

1. source address and parent relation;
2. proof or test scope;
3. an ordinary later continuation that reads the returned coordinate;
4. an ablation that removes the improvement;
5. restoration that restores it;
6. recomputation after restart from canonical source bytes.

A label, comment, memory note, or new hash does not satisfy native use by itself.

8 Joint coherent occurrences

Two nonsequential returns from one source should not be serialized into a fictitious temporal chain. Represent them as

$$(r_A, r_B, \Gamma_{AB}), \quad (6)$$

where Γ_{AB} carries compatibility, exclusion, shared assumptions, and operative composition. A later computation must read Γ_{AB} ; branch ablation or dephasing must change the result. This preserves the difference between two occurrences and one repeated payload.

9 Restart, transport, and authority boundaries

A durable result must survive process restart. The reference transport binds source hashes, operation version, input schema, expected receipt schema, and output hash. A worker result is rejected if any binding differs, even when visible output text happens to match.

The engine may propose a representation change and execute locally authorized tests. Consequential external action remains outside the theorem. Source integrity proves where a result came from; it does not prove that the source is empirically true.

10 Executable evidence

The complete source record reports 299 overlapping conformance tests and 17 isolated-process reproduction checks. The locally supplied transfer subset independently passes 22 tests. The separate collision benchmark introduced with this paper contains six collision families and 12 cases. A coarse reporter is correct on 6 of 12; the least promoted coordinate set raises correctness to 12 of 12, and removing either promoted coordinate reintroduces a collision.

Table 1: *Executable evidence retained by scope.*

Artifact	tests	passed	scope
Complete source artifact	299+17	all reported	overlapping and isolated process checks
Supplied transfer subset	22	22	carrier, integrity, transfer, restart behaviours
Collision benchmark	2	2	repair improvement and coordinate essentiality

11 A 36-cell relation-frontier benchmark

The nine-state carrier provides a compact benchmark for representation repair. Its complete pair frontier has 36 cells. A coarse reporter keeps one unit state and four prime-sector labels while erasing the side coordinate of each sector. Under the side-flip continuation, the visible report is stationary but the source state alternates. The reporter is therefore adequate for value-only queries and inadequate for role-sensitive queries.

A benchmark instance contains:

1. the nine source states and all 36 pair distinctions;
2. the coarse five-label reporter;
3. a hidden side-sensitive continuation;
4. the separating coordinate ε ;
5. a conservative repair (q, ε) ;

6. an ablation that removes ε and restores the collision.

The companion verifier also tests the stronger arithmetic correction: the static five-class Möbius reporter is not closed under prime multiplication, while the prime-support address plus an absorbing repeated-prime state is closed. This benchmark therefore joins finite relation genesis to an all-scale representation-growth example without claiming that the finite carrier itself is universal.

12 Complexity and limitations

For d candidate coordinates, exhaustive direct-basis minimization can be exponential in d ; collision-guided pruning can be substantially cheaper but has no universal polynomial guarantee. Predictive minimality is relative to the continuation envelope. Enlarging the operation set may split a state that was previously lawful. The results establish bounded representation synthesis in finite deterministic systems; they do not establish open-world ontology discovery, empirical truth, universal autonomy, or superiority over every learning architecture.

A Reference algorithm

1. Canonicalize source occurrences and preserve addresses.
2. Compute current reports and declared future signatures.
3. Detect report fibres containing more than one signature.
4. Choose a shortest separator under a frozen tie-break.
5. Refine the reporter conservatively.
6. Verify native use by matched ablation and restoration.
7. Recompute after restart and validate hash-bound transport.
8. Stop only when every report fibre is a continuation congruence.

B Two construction regimes

The representation problem appears in two finite forms.

B.1 Continuation-table construction

The first input is a finite source set, an initial reporter, and a declared family of continuations. Counterexamples to closure generate new response coordinates. Each strict refinement splits at least one report fibre; termination follows from finiteness. A certificate consists of the final quotient and, for every attempted proper coarsening, a continuation that separates a merged pair.

B.2 Direct-incidence construction

The second input supplies opaque occurrence addresses, physical incidence, domains, and a complete direct response table. No semantic object or relation names are supplied as targets. Contextual classes are induced by response equivalence, and a coordinate is admitted only when its omission creates an output collision. The audited fixture yields six object classes and two relation classes, including one unordered ternary relation with 27 complete rows. This is representation synthesis below the supplied semantic vocabulary but above the supplied physical incidence and response table.

C Proof receipts and source identity

A durable claim record is represented as

$$r = (a, o, p, \Gamma, v), \quad (7)$$

where a is an immutable occurrence address, o the report, p the proof or empirical scope, Γ the parent graph, and v the operation version. Semantic equivalence may quotient o for prediction; it never rewrites a or Γ . This prevents four common failures:

1. a copied result is treated as a new derivation;
2. a correction is separated from the claim it corrects;
3. authorship and reporting roles are reversed;
4. identical current values from different sources overwrite one another.

Source integrity is not source truth. A hash proves which bytes were used, not whether those bytes are experimentally correct.

D Collision certificates for minimality

For every coordinate excluded from a sufficient set, the implementation stores a collision pair. In the direct-coordinate model class this gives a constructive minimality certificate:

$$C_j = (x, x', c, f(x, c), f(x', c)), \quad (8)$$

where x and x' agree on all retained coordinates, differ on the excluded coordinate j , and require different outputs in context c . A basis is accepted only when all table rows are functional after projection and every selected coordinate has such a witness under its ablation.

E Native-use test

Let F denote an ordinary later continuation, $F^{-\eta}$ the same continuation after ablating returned coordinate η , and $F^{+\eta}$ the restored system. Native use requires

$$F \neq F^{-\eta}, \quad F^{+\eta} = F \quad (9)$$

on the frozen test. This blocks sham re-entry in which a new label, comment, or receipt is created but no downstream operation reads it. Joint nonsequential returns are stored with a relation object Γ_{AB} ; dephasing or branch ablation must change a later result before the relation is considered operative.

F Restart and transport protocol

A clean restart receives only canonical source bytes, operation version, schema, and the declared fixture. It recomputes the proof object rather than loading a cached verdict. Remote work is accepted only when the receipt binds input hash, operation hash, output hash, and expected schema. Matching visible text under a different binding is rejected.

G Executable test taxonomy

class	failure targeted
source collision	two equal current values overwrite one another
role reversal	author, reporter, and operator addresses are exchanged
stale proof scope	a result is reused outside the continuation envelope that justified it
missing parent	a claim loses the derivation or observation on which it depends
duplicate address	two occurrences are falsely identified
version drift	an operation changes without invalidating receipts
sham metadata	a new field is recorded but never used
return ablation	a claimed assimilation survives removal of the returned coordinate
transport mutation	worker input or output differs from the signed job
noncanonical serialization	equal semantic objects produce unstable hashes

The complete source artifact reports 299 overlapping checks and 17 isolated-process checks. The supplied transfer subset passes 22 tests. The collision benchmark deliberately merges provenance, time, role, relation, tool-version, and history-depth distinctions; its coarse report is correct on 6 of 12 cases, whereas the promoted direct basis closes all 12. Removing either selected coordinate recreates a collision.

H Scaling and open-world boundary

Exact direct-coordinate minimization is exponential in the worst case. Counterexample-guided refinement can avoid enumerating all subsets, but no general polynomial bound is claimed. More importantly, the result is relative to a continuation envelope. A state minimal for current operations may fail when a new instrument or action becomes admissible. The architecture therefore retains occurrence provenance needed for later refinement instead of declaring the present quotient ontologically final.

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Stochastic Frontier Information

Finite-Memory Dissipation and Cycle Holonomy

Abhijit Singh

Contents

Abstract	3
1 Finite inquiry frames	4
2 Uniform exchange rate	4
3 Entropy ledger of a physical record	4
4 Cycle holonomy	5
5 Recurrence and accessibility exponents	5
6 Finite verification	6
7 Scope	6
8 Detailed proof of the uniform exchange law	6
9 Worked finite examples	6
10 Entropy ledger and physical assumptions	7
11 Cycle holonomy in full	7
12 Verification and extensions	7
References	7

Abstract

A finite information measure is developed for distinctions newly revealed relative to a current reporter. Stochastic frontier information is the conditional min-entropy of a new mark given the existing report partition. On finite full-support carriers it is positive exactly when the mark causes a genuine partition refinement; for a uniform source an exact formula and a sharp lower bound in terms of combinatorial distinction gain are proved. A refinement/regraining entropy ledger yields a Landauer lower bound for sustained inquiry on finite cyclic memory. Independently, a cycle-holonomy theorem characterizes when local binary transition laws admit a global static valuation and when only a sequential occurrence index is consistent. Residual and accessibility exponents then separate dynamical instability from growth of inaccessible provenance.

Keywords: conditional min-entropy; predictive refinement; Landauer principle; causal order; contextuality; recurrence

Scope and status: The finite information and holonomy statements are proved. Thermodynamic consequences require the displayed finite-memory and reset assumptions. The recurrence/accessibility phase diagram is a model calculus rather than an established universal physical law.

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1 Finite inquiry frames

A finite inquiry frame is a triple

$$F = (S, Q, \mu), \quad (1)$$

where S is a finite carrier, Q a family of executable questions, and μ a source distribution. Questions induce a partition of S : two source states are in one cell when all current questions return the same values. A new mark $r : S \rightarrow \mathcal{R}$ may refine that partition.

Definition 1.1 (Stochastic frontier information). Let $S^* \sim \mu$ and $[S^*]_Q$ be its current report cell. Define

$$\text{SFI}_\mu(r \mid Q) = H_{\min}(r(S^*) \mid [S^*]_Q) = -\log_2 \sum_C \mu(C) \max_b \Pr(r = b \mid C). \quad (2)$$

This is the unpredictability of the new mark after every distinction decoded by the current reporter is known.

For deterministic finite carriers, let c_C be the number of nonempty r -fibres inside report cell C and define

$$\Delta_Q(r) = \sum_C (c_C - 1). \quad (3)$$

Theorem 1.2 (Deterministic reduction). If $\text{supp } \mu = S$, then

$$\boxed{\text{SFI}_\mu(r \mid Q) > 0 \iff \Delta_Q(r) > 0.} \quad (4)$$

Proof. If $\Delta_Q(r) = 0$, r is constant on every report cell and the conditional guessing probability is one. If a cell splits and every source state has positive probability, at least two mark values have positive conditional probability, so total optimal guessing probability is strictly below one. $\square$

2 Uniform exchange rate

Assume $|S| = N$ and μ uniform. Let $c_{\max}(C)$ be the largest r -fibre inside C .

Theorem 2.1 (Quantitative bridge).

$$\text{SFI}(r \mid Q) = \log_2 \frac{N}{\sum_C c_{\max}(C)} \geq \log_2 \frac{N}{N - \Delta_Q(r)}. \quad (5)$$

Equality holds exactly when every split cell consists of one largest fibre and singleton residual fibres.

Proof. Uniform conditioning gives $P_{\text{guess}} = N^{-1} \sum_C c_{\max}(C)$. Every nonmaximal fibre contains at least one element, so $c_{\max}(C) \leq |C| - (c_C - 1)$. Sum over cells. $\square$

3 Entropy ledger of a physical record

Let L be a finite physical memory storing the current report label. Refinement by r changes the Shannon entropy by

$$H(L, r) = H(L) + H(r \mid L). \quad (6)$$

A regraining is a many-to-one map of labels.

Lemma 3.1 (Exact ledger). For any finite sequence of refinements and regrainings,

$$\sum_{\text{regrain}} \Delta H_{\text{drop}} = \sum_{\text{refine}} H(r_i \mid L_i) - [H(L_f) - H(L_0)]. \quad (7)$$

Assume a memory carrier with at most N distinguishable states, a heat bath at temperature T , reversible use of blank space for refinements, and logically irreversible reset for regrainings.

Theorem 3.2 (Finite-memory dissipation floor).

$$Q_{\text{diss}} \geq k_B T \ln 2 \left[\sum_i H(r_i | L_i) - (\log_2 N - H(L_0)) \right]. \quad (8)$$

If the average fresh conditional entropy is at least $h > 0$ for n steps, then

$$Q_{\text{diss}} \geq k_B T \ln 2 [nh - \log_2 N + H(L_0)]. \quad (9)$$

The heat statement is conditional on a physical cyclic reset. A formal residual alone has no automatic thermodynamic price.

4 Cycle holonomy

Consider binary variables $x_1, \dots, x_n \in \{\pm 1\}$ with local laws

$$x_{i+1} = s_i x_i, \quad s_i \in \{\pm 1\}, \quad (10)$$

with cyclic indexing.

Theorem 4.1 (Cycle holonomy). *A global static valuation exists if and only if*

$$\prod_{i=1}^n s_i = +1. \quad (11)$$

If the product is -1 , no static assignment satisfies all local laws, but sequential propagation is total and one loop implements $x \mapsto -x$.

Proof. Multiplication of all local equations cancels every variable and gives the necessary condition. If the product is positive, choose x_1 and propagate consistently. If negative, propagation still defines every next value but returns the opposite starting value after one loop. $\square$

The three-cycle with all $s_i = -1$ has zero valid global assignments among eight. The sequential value alternates on successive loops. This is a combinatorial contextual support model; it is not by itself a quantum-realizable triangle correlation.

5 Recurrence and accessibility exponents

A residual propagated by a cocycle $G_{0:n}$ has exponential rate

$$\chi(B_0) = \limsup_n \frac{\log \|G_{0:n} B_0\| - \log \|B_0\|}{\tau_n - \tau_0}. \quad (12)$$

An independent accessibility exponent can be assigned to the growth of observationally indistinguishable provenance,

$$\eta = \limsup_{\tau \rightarrow \infty} \frac{1}{\tau} \log \mu([F]_{\Lambda(B_\tau)}). \quad (13)$$

The pair (χ, η) separates at least four regimes: assimilation with bounded hidden multiplicity; local relaxation with growing inaccessible provenance; critical recurrence with increasing hidden correlation; and proliferating residuals requiring a larger state description. The second coordinate prevents one exponent from being forced to represent both instability and inaccessibility.

6 Finite verification

The supplied audit reports:

- 400 random frames satisfying the deterministic equivalence and uniform inequality;
- 150 cases attaining the equality condition;
- 200 random refinement/regraining protocols satisfying the entropy ledger;
- exhaustive cycle-holonomy verification for $n = 3, 4, 5, 6$ over all sign patterns;
- zero valid global sections for the all-negative three-cycle.

These finite checks support the implementation and examples; the finite theorems are proved analytically above.

7 Scope

Information is a future-relevant distinction. Causal order is a dependency relation. Duration requires a clock. Dissipation requires a physical memory and reset protocol. The equations relate these objects without reducing every temporal asymmetry to heat or every form of information to min-entropy.

8 Detailed proof of the uniform exchange law

Let the current reporter partition a uniform carrier S of size N into cells C . A fresh mark r divides cell C into nonempty fibres; let $c_{\max}(C)$ be the largest fibre. Conditional optimal guessing gives

$$P_{\text{guess}}(r \mid Q) = \sum_C \frac{|C|}{N} \frac{c_{\max}(C)}{|C|} = \frac{1}{N} \sum_C c_{\max}(C), \quad (14)$$

so

$$\text{SFI}(r \mid Q) = \log_2 \frac{N}{\sum_C c_{\max}(C)}. \quad (15)$$

If c_C is the number of nonempty fresh fibres in C , then every nonmaximal fibre contains at least one element and

$$c_{\max}(C) \leq |C| - (c_C - 1). \quad (16)$$

Summing yields

$$\text{SFI}(r \mid Q) \geq \log_2 \frac{N}{N - \Delta_Q(r)}, \quad \Delta_Q(r) = \sum_C (c_C - 1). \quad (17)$$

Equality holds exactly when every nonmaximal fresh fibre is a singleton. The theorem therefore identifies the exact loss in replacing a probabilistic source by a combinatorial split count.

9 Worked finite examples

If one four-element report cell is split into two fibres of sizes two and two, then $P_{\text{guess}} = 1/2$ and SFI is one bit, while $\Delta = 1$. If it is split into sizes three and one, then $P_{\text{guess}} = 3/4$ and SFI is $-\log_2(3/4)$; the same combinatorial distinction gain carries less guessing information. If two cells are split under nonuniform source mass, the largest conditional fibre must be weighted by the mass of its parent cell; cardinality alone is insufficient.

10 Entropy ledger and physical assumptions

Let L_i be the physical record label before fresh mark r_i . A logically reversible refinement writes (L_i, r_i) and increases Shannon entropy by

$$H(L_i, r_i) - H(L_i) = H(r_i | L_i). \quad (18)$$

A regaining is a many-to-one map of labels and decreases record entropy. Telescoping every increase and decrease gives

$$\sum_{\text{regain}} \Delta H_{\text{drop}} = \sum_{\text{refine}} H(r_i | L_i) - [H(L_f) - H(L_0)]. \quad (19)$$

If the physical carrier has at most N reliably distinguishable states, $H(L_f) \leq \log_2 N$. Under a cyclic reset implemented against a bath at temperature T ,

$$Q_{\text{diss}} \geq k_B T \ln 2 \left[\sum_i H(r_i | L_i) - (\log_2 N - H(L_0)) \right]. \quad (20)$$

The result requires a finite memory, a reset operation, and the standard thermodynamic implementation assumptions. It is not a claim that causal order itself is created by heat dissipation.

11 Cycle holonomy in full

For local binary laws $x_{i+1} = s_i x_i$ on a cycle, multiply all equations. Every x_i appears on both sides, leaving the necessary condition $\prod_i s_i = 1$. If it holds, choose x_1 arbitrarily and propagate to obtain a global assignment. If it fails, local propagation remains total but one circuit sends

$$x_1 \mapsto \left(\prod_i s_i \right) x_1 = -x_1. \quad (21)$$

The system therefore requires an occurrence index: the same local question alternates on successive loops. The three-cycle with every $s_i = -1$ has no global assignment among its eight possibilities and alternates exactly under passage. It is a combinatorial contextual model; additional constraints are required for quantum realizability.

12 Verification and extensions

The finite audit checked 400 random frames for the positivity equivalence and quantitative inequality, including 150 equality cases; 200 random refine/merge protocols for the entropy ledger; and every sign pattern for cycles of lengths three through six. Live extensions are a sharp nonuniform-source inequality, finite-size thermodynamic corrections, and a characterization of which holonomy structures admit quantum rather than merely no-signalling realization.

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Coherent Alternatives

Complex Amplitudes and Marked Quantum Dynamics

Abhijit Singh

Contents

Abstract	3
1 Operational starting point	4
2 Coherent alternatives	4
3 Translation-invariant measure and the square law	4
4 Finite complex kinematics	5
5 Causal partial order and slicing independence	5
6 Marked quantum instruments	6
7 Locality and entropy	6
8 Causal geometry from order and measure	6
9 Current-totality obstruction	6
10 Finite spectral labels on the nine-state relation carrier	7
11 Scope and falsification	7
12 Dependency graph of the reconstruction	7
13 Detailed Haar–Born proof	7
14 Finite-dimensional extension	8
15 Complete measurement operation	8
16 Locality, entropy, and causal order	8
17 Uniqueness and falsification obligations	8
References	8

Abstract

A direct operational route is given from coherent unresolved alternatives to the minimal complex carrier, quadratic outcome weights, reversible unitary dynamics, and complete measurement instruments. A faithful minimal continuous periodic return acts on a real rotation plane, generating the complex unit as an oriented quarter-turn. Translation-invariant area on that plane forces the weight $|z|^2$ and hence the normalized probability rule. Reversible weight-preserving transformations are unitary; differentiable return chains obey Schrodinger form; commuting local operations on a finite causal partial order are slicing independent. Recorded outcomes are treated as instruments containing both probability and successor state. Unread local operations preserve distant marginals and sharp dephasing cannot lower entropy. Under continuum regularity, causal order plus event volume determines Lorentzian geometry. The finite nine-state carrier supplies four commuting return planes and a scoped mathematical quantum-number interpretation.

Keywords: quantum foundations; complex amplitudes; probability rule; unitary dynamics; quantum instruments; causal order

Scope and status: Every theorem is conditional on the displayed operational or continuum assumptions. The paper does not claim a completed microscopic theory of nature or numerical derivation of physical constants.

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1 Operational starting point

The primitive operational data are finite histories, admissible continuations, and open alternatives sharing the same input and output boundary. A present state is not assumed to be fundamental; it is the quotient of histories that have the same future behaviour for the declared continuation family.

Definition 1.1 (Future equivalence). For histories h, h' and interventions Λ ,

$$h \sim_{\Lambda} h' \iff P(Y_{1:H} \mid \text{do}(\lambda), h) = P(Y_{1:H} \mid \text{do}(\lambda), h') \quad (\lambda \in \Lambda). \quad (1)$$

If a new operation separates histories previously merged, the old state space is incomplete for the enlarged experiment.

2 Coherent alternatives

Consider unresolved routes with one common boundary. Assume:

1. a faithful finite-dimensional continuous real representation;
2. addition for unresolved alternatives and bilinear sequential composition;
3. nontrivial cancellation upon recombination;
4. a minimal irreducible continuous periodic return group;
5. a common inner product preserved by unmarked return.

These assumptions do not insert complex numbers, probabilities, Hilbert space, or a balanced splitter.

Theorem 2.1 (Minimal phase carrier). *The minimal nontrivial carrier is a real two-dimensional rotation plane. In an oriented orthonormal basis,*

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I, \quad R_{\theta} = e^{\theta J}. \quad (2)$$

It is isomorphic to one complex dimension, up to reversal of orientation.

Proof. Average any positive definite inner product over the compact return group, making the representation orthogonal. A continuous faithful circle action is trivial in one real dimension. The generator of an orthogonal one-parameter group is skew-symmetric and decomposes into two-dimensional rotation blocks and a zero block. Minimality and irreducibility remove the zero block and all but one rotation block; faithfulness fixes winding number ± 1 . $\square$

The imaginary unit is therefore the oriented quarter-turn of the minimal coherent cell.

3 Translation-invariant measure and the square law

Let $V \cong \mathbb{R}^2$ carry a nonzero locally finite translation-invariant Borel measure λ . Such a measure is unique up to scale and is ordinary area. A complex coefficient $z = a + ib$ acts by

$$M_z = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}. \quad (3)$$

Define the weight by equal-cell scaling,

$$w(z) = \frac{\lambda(M_z A)}{\lambda(A)}, \quad 0 < \lambda(A) < \infty. \quad (4)$$

Theorem 3.1 (Quadratic actualization weight).

$$w(z) = |z|^2. \quad (5)$$

For mutually recorded alternatives $z_1, \dots, z_n$,

$$P(j) = \frac{|z_j|^2}{\sum_k |z_k|^2}. \quad (6)$$

Proof. A real linear map scales translation-invariant area by the absolute determinant. Here $\det M_z = a^2 + b^2$. Recorded alternatives occupy disjoint terminal sectors, so their weights add and normalization gives the second formula. $\square$

Corollary 3.2 (Balanced splitter). *Two exchange-symmetric alternatives have equal modulus r and satisfy $2r^2 = 1$, hence $r = 1/\sqrt{2}$.*

The balanced coefficient is therefore a consequence rather than an input.

4 Finite complex kinematics

For n mutually recorded alternatives the carrier is $\mathbb{C}^n$ with norm $\|\psi\|^2 = \sum_j |z_j|^2$. A reversible complex-linear transformation preserving this weight for every state is unitary.

Let $U(\tau)$ be a differentiable one-parameter unitary group. Then

$$U(\tau) = e^{-iH\tau}, \quad i \frac{d\psi}{d\tau} = H\psi, \quad (7)$$

for Hermitian H . The parameter τ compares repeatable local return processes; it need not be an external universal clock.

5 Causal partial order and slicing independence

Let $(E, \prec)$ be a finite causal partial order and attach a local transformation T_e to each event. For a linear extension $L = (e_1, \dots, e_N)$, let $T_L = T_{e_N} \cdots T_{e_1}$.

Theorem 5.1 (Foliation independence). *If*

$$e \parallel f \implies T_e T_f = T_f T_e, \quad (8)$$

then every linear extension of the same finite partial order yields the same total transformation.

Proof. Any two finite linear extensions are connected by adjacent swaps of incomparable elements. Each swap preserves the product. $\square$

The written global order is therefore bookkeeping; causal precedence and local return phases provide the physical temporal structure.

6 Marked quantum instruments

For orthogonal projectors P_j with $\sum_j P_j = I$, a coherent system-record coupling is

$$V\psi = \sum_j |j\rangle_M \otimes P_j\psi, \quad V^\dagger V = I. \quad (9)$$

A recorded outcome is the instrument

$$\mathcal{I}_j(\rho) = P_j \rho P_j, \quad (10)$$

with

$$p_j = \text{Tr}(P_j \rho), \quad \rho'_j = \frac{P_j \rho P_j}{p_j}. \quad (11)$$

If the mark is unavailable, the successor is the dephased state

$$\mathcal{D}(\rho) = \sum_j P_j \rho P_j. \quad (12)$$

The outcome and successor state belong to one operation.

7 Locality and entropy

For every local trace-preserving map Φ_A ,

$$\text{Tr}_A[(\Phi_A \otimes I_B)(\rho_{AB})] = \text{Tr}_A \rho_{AB}. \quad (13)$$

An unread local operation cannot signal to a distant subsystem. Conditional remote states may differ after classical comparison of records, but the local marginal does not.

The sharp unread channel is a random-unitary average over phase operators, so

$$S(\mathcal{D}\rho) \geq S(\rho), \quad (14)$$

and

$$S(\mathcal{D}\rho) - S(\rho) = D(\rho \| \mathcal{D}\rho) \geq 0. \quad (15)$$

This quantifies coherence removed from sector-preserving future operations.

8 Causal geometry from order and measure

Suppose a dense regime admits a strongly causal smooth continuum representation. Causal order determines the conformal light-cone structure. A locally finite event-volume measure fixes the conformal scale. Under these explicit regularity assumptions,

$$(\text{causal order, event volume}) \longrightarrow (M, g_{\mu\nu}). \quad (16)$$

The statement is a conditional representation theorem, not a proof that every finite event network converges to a smooth spacetime.

9 Current-totality obstruction

Let H be an open macro-operation and dH the completed record produced when it closes. Acyclic causal dependence forbids dH from participating in the production of the identical token closure. A usable description of the completed whole is therefore either partial, produced by a proper suboperation while H remains open, or late, participating in a successor operation. This is a token-causal statement, not a prohibition on later complete records.

10 Finite spectral labels on the nine-state relation carrier

The nine-state prime-observer carrier admits a conditional finite spectral representation. Let

$$\mathcal{H}_A = \mathbb{C}|1, 0\rangle \oplus \bigoplus_{q \in \{2,3,5,7\}, \varepsilon = \pm 1} \mathbb{C}|q, \varepsilon\rangle. \quad (17)$$

The commuting operators

$$\widehat{Q}|q, \varepsilon\rangle = q|q, \varepsilon\rangle, \quad \widehat{S}|q, \varepsilon\rangle = \varepsilon|q, \varepsilon\rangle \quad (18)$$

provide complete joint labels. The oriented pair-relation space has dimension

$$\dim \bigwedge^2 \mathbb{R}^9 = 36. \quad (19)$$

Four antisymmetric generators exchange the disjoint pairs (2, 9), (3, 8), (5, 6), and (7, 4) and commute because their two-planes are disjoint.

This establishes a mathematically legitimate quantum-number interpretation: the labels are joint eigenvalues of commuting operators. It does not establish that a physical system realizes these observables, that the full relation algebra is a gauge symmetry, or that the decimal harmonic scale is fundamental. A physical claim requires an independently specified preparation, instrument, transition Hamiltonian, and held-out measurement.

11 Scope and falsification

The derivation is conditional on the operational assumptions stated in Section 2. It establishes a constructive route to the complex carrier, the quadratic weight, unitary dynamics, complete instruments, no-signalling, and conditional causal geometry. It does not derive the observed value of $\hbar$, the Standard Model, gravitational coupling, or a unique microscopic theory of spacetime. The programme fails if a smaller faithful continuous coherent carrier exists, if equal-cell translation-invariant measure scales by another exponent, or if the representation assumptions admit physically relevant noncomplex alternatives without an independent selection principle.

12 Dependency graph of the reconstruction

The constructive route is

$$\text{continuous coherent return} \Rightarrow \mathbb{R}^2 \text{ rotation cell} \Rightarrow \mathbb{C} \Rightarrow |z|^2 \text{ area scaling} \Rightarrow \text{unitary reversible dynamics}. \quad (20)$$

Each arrow has an explicit assumption. A faithful periodic action is required for the rotation cell; irreducibility and minimality exclude direct sums; translation invariance and local finiteness select area measure; complex linearity and norm preservation select unitarity. The derivation does not establish that every generalized probabilistic theory must satisfy those assumptions.

13 Detailed Haar–Born proof

Let $z = a + ib$ act on the real phase plane by

$$M_z = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}. \quad (21)$$

Every nonzero locally finite translation-invariant Borel measure on $\mathbb{R}^2$ is a constant multiple of Lebesgue area. Real linear maps scale that measure by absolute determinant, and

$$|\det M_z| = a^2 + b^2 = |z|^2. \quad (22)$$

For disjoint recorded alternatives, areas add. Thus normalization yields the squared-modulus rule. The balanced splitter follows rather than being inserted: exchange symmetry gives equal modulus r and $2r^2 = 1$.

14 Finite-dimensional extension

For n mutually recorded alternatives, the carrier is $\mathbb{C}^n$ with weight $\|\psi\|^2$. A reversible complex-linear map preserving this weight for every ψ satisfies $U^\dagger U = I$. A differentiable one-parameter group is generated by a Hermitian operator:

$$U(\tau) = e^{-iH\tau}, \quad i \frac{d\psi}{d\tau} = H\psi. \quad (23)$$

The parameter τ is calibrated by repeatable physical returns. A numerical value of the action scale is not derived from dimensionless algebra.

15 Complete measurement operation

For orthogonal projectors $\{P_j\}$, the isometry

$$V\psi = \sum_j |j\rangle_M \otimes P_j\psi \quad (24)$$

represents coherent system-record coupling. Actual record j has weight $\|P_j\psi\|^2$ and conditional successor $P_j\psi/\|P_j\psi\|$. If the record is unavailable, the correct successor is the dephased density operator $\sum_j P_j \rho P_j$. This unifies outcome and successor state and avoids comparing two marginals as though they were independent physical sectors.

16 Locality, entropy, and causal order

For any trace-preserving local operation Φ_A ,

$$\text{Tr}_A[(\Phi_A \otimes I_B)(\rho_{AB})] = \text{Tr}_A \rho_{AB}, \quad (25)$$

so an unread local record cannot signal. Sharp dephasing is a random-unitary average and

$$S(\mathcal{D}\rho) - S(\rho) = D(\rho\|\mathcal{D}\rho) \geq 0. \quad (26)$$

For a finite causal partial order, commuting maps at incomparable events make every linear extension produce the same total operation. Under strong causality and continuum regularity, order fixes conformal Lorentzian geometry and event volume fixes scale.

17 Uniqueness and falsification obligations

The programme is weakened or rejected if a one-dimensional real faithful continuous carrier exists; if another locally finite translation-invariant phase measure gives a different weight; if the same operational assumptions admit inequivalent noncomplex composition; or if a claimed complete instrument cannot reproduce its own outcome and successor-state statistics. These are mathematical and experimental checks, not appeals to interpretive preference.

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Arity-Indexed Refinement Posets

Forced Inverses and Rotational Root Geometry

Abhijit Singh

Contents

Abstract	3
1 One exhaustive replacement rule	4
2 Kraft and orbit conservation	4
3 Growth regimes	4
4 Reachable partition subposets	4
5 Forced incidence inverse	5
6 Characteristic polynomial	5
7 Parallel families and root at one	6
8 Transparent prime sectors in the observer carrier	6
9 What is proved and what is finite evidence	7
10 Executable audit	7
11 Complete counting derivations	7
12 Reachable ranks and forced inverse	7
13 Rotational characteristic geometry	8
14 Reproducibility envelope	8
References	8

Abstract

Exhaustive replacement of one terminal by n unordered resultants generates arity-specific occurrence counts, multifactorial history counts, Fuss-Catalan forms, exact Kraft equality, and an orbit-stabilizer conservation law. Partial distinguishability states form the subposet of set partitions whose block count is congruent to one modulo $n - 1$. Incidence inversion on this subposet produces an arity-indexed family of forced inverse sequences; the familiar binary factorial values are one member. The associated characteristic polynomial factors through t^{n-1} , spreading each positive reduced root over $n - 1$ rotational rays. Parallel residue families, interlacing, sign alternation, and the root-at-one congruence are recorded with exact finite scope. An executable package passes 85 tests. The four prime-transparent arities within the terminal carrier are related to the four return-sector labels, while the stronger prime-closure ladder remains conjectural.

Keywords: refinement poset; Stirling numbers; Mobius inversion; Fuss-Catalan numbers; real-rootedness; interlacing

Scope and status: Counting, reachability, recursion, factorization, and conservation statements are exact. General root and sign laws beyond proved cases are identified as finite evidence and conjecture, not universal theorem.

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1 One exhaustive replacement rule

Begin with one terminal occurrence. At each application choose one current terminal and replace it by n unordered resultants. Histories are retained separately; they are not aggregated by final form.

After j applications the number of current occurrences is

$$L_j = 1 + j(n - 1). \quad (1)$$

The labelled history count obeys

$$H_{j+1} = L_j H_j, \quad H_0 = 1, \quad (2)$$

so

$$H_j = \prod_{i=0}^{j-1} (1 + i(n - 1)). \quad (3)$$

For $n = 2$ this is $j!$; for $n = 3$ it is $(2j - 1)!!$.

The number of unlabelled full n -ary forms after j applications is the Fuss-Catalan number

$$F_{n,j} = \frac{1}{(n - 1)j + 1} \binom{nj}{j}. \quad (4)$$

Forms reached by exactly one history number n^j .

2 Kraft and orbit conservation

Every completed form satisfies the exact base- n Kraft equality

$$\sum_{\ell \in \text{leaves}} n^{-\text{depth}(\ell)} = 1. \quad (5)$$

For every symmetry class at depth j ,

$$|\text{Aut}(T)| |\text{Orb}(T)| = (n!)^j. \quad (6)$$

The executable audit verifies this to the declared finite boundaries for arities two through five.

3 Growth regimes

Occurrence count is linear in j , form count is exponential with asymptotic base $n^n / (n - 1)^{n-1}$, the symmetry budget $(n!)^j$ is exponential, and history count is superexponential. At low arity forms eventually outgrow the symmetry budget; at higher arity the relation reverses. This crossover is a property of the exhaustive replacement system, not a new arithmetic operation.

4 Reachable partition subposets

A partial report is a partition of the presented resultants. One n -ary refinement splits one block into n , raising block count by $n - 1$. Therefore the reachable ranks are exactly

$$k \equiv 1 \pmod{n - 1}. \quad (7)$$

At an admissible rank k , the number of partitions of a q -element carrier is still the Stirling number $\left\{ \begin{smallmatrix} q \\ k \end{smallmatrix} \right\}$. Arity changes reachability, not the census at a reachable rank.

5 Forced incidence inverse

Define $\mu_k^{(n)}$ on admissible ranks by

$$\mu_1^{(n)} = 1, \quad (8)$$

$$\sum_{\substack{j \leq k \\ j \equiv 1 \pmod{n-1}}} \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \mu_j^{(n)} = 0 \quad (k > 1 \text{ admissible}). \quad (9)$$

This is the incidence inverse of the arity-restricted refinement poset.

Table 1: First forced inverse values on admissible ranks.

arity n	$\mu_k^{(n)}$
2	1, -1, 2, -6, 24, -120, 720, ...
3	1, -1, 24, -3060, 1249920, -1253750400, ...
4	1, -1, 349, -2018016, 77366328753, ...
5	1, -1, 6950, -2491030400, ...
6	1, -1, 179486, -5191034942731, ...

The binary factorial sequence is one member of a larger arity-indexed family. Strict sign alternation was computationally verified through the declared finite range; a universal proof is a separate obligation.

6 Characteristic polynomial

For a carrier with q presented resultants define

$$C_q^{(n)}(t) = \sum_{\substack{k \leq q \\ k \equiv 1 \pmod{n-1}}} \left\{ \begin{matrix} q \\ k \end{matrix} \right\} \mu_k^{(n)} t^{q-k}. \quad (10)$$

All exponents lie in one residue class modulo $n - 1$, hence

$$C_q^{(n)}(t) = t^a D_q^{(n)}(t^{n-1}), \quad a \equiv q - 1 \pmod{n-1}. \quad (11)$$

A positive root u of $D_q^{(n)}$ generates roots of $C_q^{(n)}$ on the $n - 1$ rays

$$\arg t = \frac{2\pi r}{n-1}, \quad r = 0, \dots, n-2. \quad (12)$$

The binary positive-real geometry is the one-ray degeneration.

Table 2: Audited reduced polynomials at $q = 7$.

n	$D_7^{(n)}(u)$	positive roots
3	$u^3 - 301u^2 + 3360u - 3060$	1, 10.573, 289.427
4	$u^2 - 350u + 349$	1, 349
5	$u - 140$	140

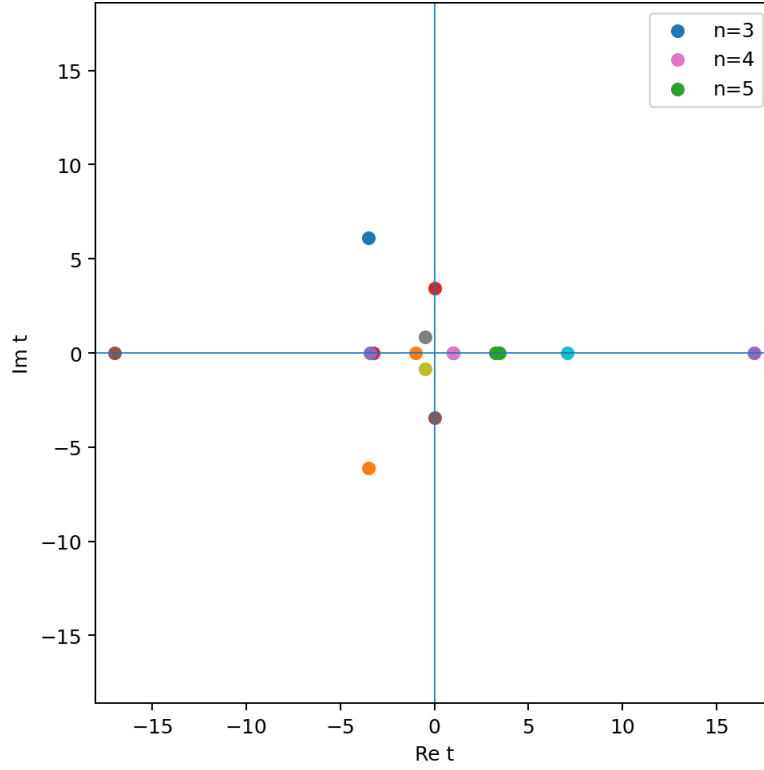


Figure 1: Rotational root rays generated from audited positive reduced roots.

7 Parallel families and root at one

At each arity there are $n - 1$ residue families indexed by $q \bmod (n - 1)$. The finite audit displays interlacing within each family. The audited rule

$$D_q^{(n)}(1) = 0 \iff q \equiv 1 \pmod{n-1} \quad (13)$$

holds throughout the computed domain. A general theorem requires a recurrence preserving real stability or a total-positivity argument for the restricted Stirling transform.

8 Transparent prime sectors in the observer carrier

On the finite nonterminal index set $\{2, \dots, 9\}$, the transparent arities are exactly

$$\{2, 3, 5, 7\}. \quad (14)$$

They label four prime/composite return sectors under $d \mapsto 1 - d \pmod{10}$. This gives a generated role for the prime indices: they are the irreducible arities whose intermediate binomial capacities vanish in their own modulus. The composite partners record either squarefree coupling ($6 = 2 \cdot 3$) or repeated occupation ($4, 8, 9$).

The stronger notebook conjecture that every genuinely new observer-closure rank must be prime is not proved by this finite pattern. A valid theorem would require a fixed role-generation rule that predicts the next closure rank before testing primality. The present exact result is narrower: prime transparency selects the four sector labels, and the nine-state carrier organizes their pair relations into a 36-cell frontier.

9 What is proved and what is finite evidence

The occurrence, history, Fuss-Catalan, Kraft, orbit-stabilizer, reachability, Stirling-census, inverse-recursion, and residue-factorization statements are exact. General real-rootedness, sign alternation, interlacing, and the root-at-one rule outside the proved binary case are retained as computationally supported conjectures unless and until universal proofs are supplied.

10 Executable audit

The supplied perspective-arity package passes 85 tests. It checks exhaustive constructors, independent closed forms, refinement inverses, symmetry conservation, capacity identities, prime transparency, groupoid lifts, and negative controls. Numerical roots are computed only after exact integer coefficients are established.

11 Complete counting derivations

After j applications of an n -resultant replacement, the number of current occurrences is

$$L_j = 1 + j(n - 1). \quad (15)$$

At the next step any current occurrence may be selected, so the number of labelled histories obeys

$$H_{j+1} = L_j H_j, \quad H_0 = 1, \quad (16)$$

and therefore

$$H_j = \prod_{i=0}^{j-1} [1 + i(n - 1)]. \quad (17)$$

For $n = 2$ this is $j!$; for $n = 3$ it is $(2j - 1)!!$. The number of unlabelled full n -ary forms with j internal occurrences is the Fuss–Catalan number

$$F_{n,j} = \frac{1}{(n - 1)j + 1} \binom{nj}{j}. \quad (18)$$

The finite constructor exhaustively reproduced, among other values, 285, 2530, and 23751 for $n = 5$.

For every completed form,

$$\sum_{\ell \in \text{leaves}} n^{-\text{depth}(\ell)} = 1. \quad (19)$$

This follows by induction because replacing one leaf of weight n^{-d} by n children of weight $n^{-(d+1)}$ preserves the sum. Every internal occurrence also contributes one independent permutation of its resultants, so orbit–stabilizer gives

$$|\text{Aut}(T)| |\text{Orb}(T)| = (n!)^j. \quad (20)$$

12 Reachable ranks and forced inverse

One replacement increases block count by $n - 1$. Hence reachable partition ranks satisfy

$$k \equiv 1 \pmod{n - 1}. \quad (21)$$

At an admitted rank the number of partitions remains the ordinary Stirling number. The incidence inverse is forced by

$$\mu_1^{(n)} = 1, \quad \sum_{\substack{j \leq k \\ j \equiv 1 \pmod{n-1}}} \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \mu_j^{(n)} = 0. \quad (22)$$

The first values are

n	inverse on admitted ranks
2	1, -1, 2, -6, 24, -120, ...
3	1, -1, 24, -3060, 1249920, ...
4	1, -1, 349, -2018016, ...
5	1, -1, 6950, -2491030400, ...
6	1, -1, 179486, -5191034942731, ...

The alternating signs were checked through admitted rank 31 in the supplied finite audit. Only the binary member reduces to factorials.

13 Rotational characteristic geometry

Define

$$C_q^{(n)}(t) = \sum_{\substack{k \leq q \\ k \equiv 1 \pmod{n-1}}} \begin{Bmatrix} q \\ k \end{Bmatrix} \mu_k^{(n)} t^{q-k}. \quad (23)$$

All exponents occupy one residue class, so

$$C_q^{(n)}(t) = t^a D_q^{(n)}(t^{n-1}), \quad a \equiv q - 1 \pmod{n-1}. \quad (24)$$

A positive root u of D produces $n - 1$ roots on rays $\arg t = 2\pi r/(n - 1)$. At $q = 7$ the audited reduced polynomials are

n	$D_q^{(n)}(u)$	positive roots
3	$u^3 - 301u^2 + 3360u - 3060$	1, 10.573, 289.427
4	$u^2 - 350u + 349$	1, 349
5	$u - 140$	140

Finite calculations show parallel residue families and a root at one when $q \equiv 1 \pmod{n-1}$. A universal proof of reduced real-rootedness and interlacing remains a stated theorem target; the manuscript does not replace that proof with plots.

14 Reproducibility envelope

The archive constructs histories and forms independently, checks the closed counts, verifies Kraft equality and orbit products, computes the forced inverses as exact integers, constructs characteristic polynomials, and extracts roots. Floating-point root plots are evidence only; a proof-quality computation should add exact Sturm isolation or certified interval arithmetic.

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Capacity Complexes

One-Bit Persistence and Operator-Ideal Criticality

Abhijit Singh

Contents

Abstract	3
1 Binary capacity	4
2 Transparent arities	4
3 Squarefree incidence orientation	4
4 Quota complex	4
5 Exact persistence barcode	5
6 Nonbinary contrast fibres	5
7 Null transport and the half exponent	5
8 Orientation operator	6
9 Scalar nonclosability	6
10 Clifford orientation charge	6
11 Prime-occupation refinement of the composite sector	6
12 Claim boundary	7
13 Prime transparency and generated arithmetic	7
14 Capacity complex and matching proof	7
15 Nonbinary contrasts and half-cost transport	8
16 Operator completion and scalarization barrier	8
17 What this paper contributes	8
References	8

Abstract

Binary capacity and prime-transparent coordinates generate a squarefree arithmetic carrier under a shared logarithmic budget. The resulting quota complex admits an explicit acyclic matching: every odd squarefree carrier m produces one reduced homology class with exact persistence interval $[m, 2m)$. The signed occupation is the Mertens sum. Retaining all nonbinary resultants gives a standard contrast representation whose full-fibre average cancels the linear charge, producing a quadratic completion boundary. Preservation of generated null pairs independently forces half-cost transport. The orientation operator is Hilbert-Schmidt exactly beyond one half and trace class only beyond one. The uniform odd scalar functional is proved nonclosable on the quadratic carrier. These are exact topological and operator statements, not a proof of a classical zero-location conjecture. Prime-occupation coordinates refine the finite composite sector into squarefree coupling and repeated-prime depth.

Keywords: capacity complex; persistent homology; squarefree incidence; operator ideals; null transport; scalarization

Scope and status: Topological, representation, null-transport, and operator theorems are exact in their displayed source-generated constructions. Classical analytic consequences require the separate bridge in P08.

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1 Binary capacity

Repeated binary distinction assigns capacity 2^{-r} to a depth- r cell. Multiplicativity of independent capacity gives the additive coordinate

$$c(u) = -\log_2 u. \quad (1)$$

A symmetric q -resultant event has ideal capacity $1/q$ per resultant and cost

$$c(q) = \log_2 q. \quad (2)$$

Independent coordinates therefore satisfy $c(n) = \log_2 n$, and the shared budget $c(n) \leq \log_2 N$ is exactly $n \leq N$.

2 Transparent arities

Call $q \geq 2$ transparent when

$$\binom{q}{k} \equiv 0 \pmod{q}, \quad 0 < k < q. \quad (3)$$

Theorem 2.1 (Prime transparency). *An arity is transparent if and only if it is prime.*

Proof. For prime q , divisibility is immediate. If q is composite and $p \mid q$ is prime, then $\binom{q}{p} = (q/p)\binom{q-1}{p-1}$ lacks one factor p required for divisibility by q . $\square$

This is a classical arithmetic equivalence used here after the cost construction; novelty is not claimed for the equivalence itself.

3 Squarefree incidence orientation

Independent prime coordinates yield finite-support exponent vectors and unique factorization. On squarefree support, Boolean incidence inversion gives

$$\mu(n) = (-1)^{\omega(n)}, \quad (4)$$

with $\mu(n) = 0$ when a prime repeats. A continuous character of additive cost is 2^{-sc} and becomes n^{-s} on generated integer identities. In the absolute domain,

$$\sum_{n \geq 1} \mu(n) n^{-s} = \prod_p (1 - p^{-s}). \quad (5)$$

4 Quota complex

Define

$$K_N = \left\{ A \subset \mathbb{P} : \prod_{p \in A} p \leq N \right\}. \quad (6)$$

Pair every odd squarefree m with $2m$ whenever $2m \leq N$.

Theorem 4.1 (Binary capacity homotopy). *The reduced complex is homotopy equivalent to a wedge with one sphere of dimension $\omega(m) - 1$ for each odd squarefree m satisfying*

$$N/2 < m \leq N. \quad (7)$$

The signed Euler occupation is the Mertens sum

$$M(N) = \sum_{n \leq N} \mu(n). \quad (8)$$

5 Exact persistence barcode

Each odd squarefree m births one class at $N = m$ and dies when its binary cone appears at $N = 2m$. The complete interval is

$$[m, 2m), \quad (9)$$

or one bit in logarithmic capacity.

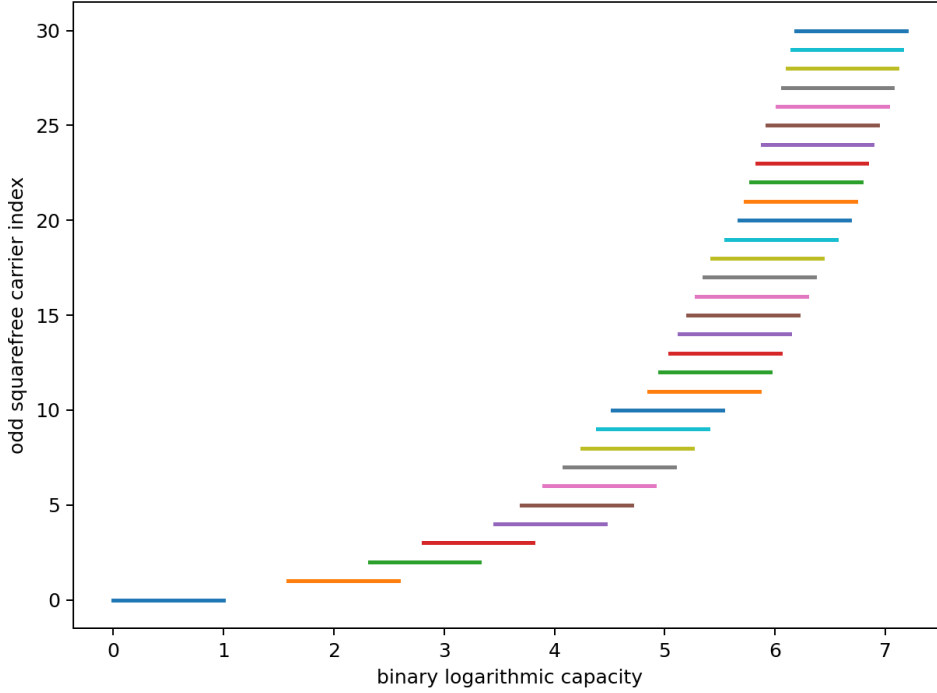


Figure 1: Initial one-bit persistence intervals for odd squarefree carriers.

6 Nonbinary contrast fibres

A q -resultant relation decomposes into one collective direction and the $(q - 1)$ -dimensional standard contrast representation. The permutation character is

$$\chi_q(\pi) = \# \text{Fix}(\pi) - 1. \quad (10)$$

Full relabelling-fibre averaging cancels the linear contrast charge; the normalized all-channel determinant begins quadratically and completes in the half-plane $\Re s > 1/2$. This is a statement about the uncollapsed contrast response, not by itself about a scalar Dirichlet series.

7 Null transport and the half exponent

The generated null pairing is preserved only when local transport satisfies

$$a_p(s)^2 = p^s, \quad (11)$$

so

$$a_p(s) = \pm p^{s/2}. \quad (12)$$

The exponent $1/2$ is therefore forced within the null-preservation law.

8 Orientation operator

On the squarefree basis w_n , define

$$Jw_n = \mu(n)w_n, \quad Hw_n = (\log n)w_n, \quad D_s = Je^{-sH}. \quad (13)$$

The singular values are $n^{-\Re s}$ on squarefree carriers.

Theorem 8.1 (Operator-ideal thresholds). *D_s is Hilbert-Schmidt exactly for $\Re s > 1/2$ and trace class exactly for $\Re s > 1$.*

Proof. The Hilbert-Schmidt sum is $\sum_{n \text{ sf}} n^{-2\Re s}$, which converges exactly when $2\Re s > 1$. The trace-norm sum is $\sum_{n \text{ sf}} n^{-\Re s}$, converging exactly for $\Re s > 1$. $\square$

9 Scalar nonclosability

Let S_N be the visible squarefree set and

$$X_N = |S_N|^{-1} \sum_{n \in S_N} \mu(n)|n\rangle\langle n|. \quad (14)$$

Then $\|X_N\|_2 \rightarrow 0$ while $\text{Tr}(JX_N) = 1$.

Theorem 9.1 (Odd-trace obstruction). *The uniform signed scalar functional is not closable on the quadratic Hilbert-Schmidt carrier.*

This proves that the quadratic completion does not automatically supply the desired scalar estimate. It does not prove that every special scalar trajectory diverges.

10 Clifford orientation charge

Retaining oriented prime creation yields generators γ_p with

$$\gamma_p^2 = I, \quad \gamma_p \gamma_q + \gamma_q \gamma_p = 0 \quad (p \neq q), \quad J\gamma_p + \gamma_p J = 0. \quad (15)$$

The odd charge

$$\Gamma_P(s) = \sum_{p \in P} p^{-s} \gamma_p \quad (16)$$

satisfies

$$\Gamma_P(s)^2 = \left(\sum_{p \in P} p^{-2s} \right) I, \quad (17)$$

so it has a natural half-cost completion. Sending all γ_p to one scalar is not an algebra homomorphism.

11 Prime-occupation refinement of the composite sector

The prime/composite split of the finite terminal lattice is too coarse for Möbius continuation. Introduce occupation labels

$$N_p|d\rangle = v_p(d)|d\rangle, \quad p \in \{2, 3, 5, 7\}. \quad (18)$$

Then

$$6 \leftrightarrow (1, 1, 0, 0) \quad (19)$$

is squarefree coupling, whereas

$$4 \leftrightarrow (2, 0, 0, 0), \quad 8 \leftrightarrow (3, 0, 0, 0), \quad 9 \leftrightarrow (0, 2, 0, 0) \quad (20)$$

contain repeated prime coordinates. Accordingly $\mu(6) = +1$ but $\mu(4) = \mu(8) = \mu(9) = 0$.

This finite refinement explains why equal prime/composite Fourier magnitudes do not imply Möbius cancellation. The arithmetic source distinguishes support, orientation, and repeated depth. The 36-cell relation frontier captures all binary differences among nine active indices, but an all-prime continuation requires unbounded finite support addresses rather than a fixed finite label set.

12 Claim boundary

The capacity complex, matching, barcode, contrast representation, null-transport exponent, operator thresholds, and scalar nonclosability are exact within the displayed constructions. The appearance of $1/2$ is an operator and geometric boundary. Any identification with a classical critical line requires an additional exact bridge, supplied only in the companion completion paper.

13 Prime transparency and generated arithmetic

A symmetric q -resultant event assigns cost $\log_2 q$. Intermediate capacities are the binomial coefficients. Define transparency by

$$\binom{q}{k} \equiv 0 \pmod{q}, \quad 0 < k < q. \quad (21)$$

For prime q divisibility is immediate. If q is composite and $p \mid q$ is prime, then

$$\binom{q}{p} = \frac{q}{p} \binom{q-1}{p-1} \quad (22)$$

lacks the required factor of p for divisibility by q . Thus transparent arities are exactly the primes. Independent transparent coordinates generate finite exponent vectors, unique factorization, logarithmic cost, and squarefree incidence orientation

$$\mu(n) = (-1)^{\omega(n)}, \quad (23)$$

with zero when a coordinate repeats.

14 Capacity complex and matching proof

For $N \geq 1$ define

$$K_N = \left\{ A \subset \mathbb{P} : \prod_{p \in A} p \leq N \right\}. \quad (24)$$

Pair every odd squarefree m with $2m$ whenever $2m \leq N$. The matching is acyclic because matched edges strictly change membership of the distinguished prime 2 and no alternating cycle can return with the same orientation. Unmatched cells are exactly odd squarefree m with

$$\frac{N}{2} < m \leq N. \quad (25)$$

Discrete Morse reduction therefore gives one sphere of dimension $\omega(m) - 1$ for each unmatched m . Every class is born at $N = m$ and killed when its binary cone enters at $N = 2m$:

$$\boxed{[m, 2m)}. \quad (26)$$

The interval has width one in logarithmic binary cost. Its signed occupation is the Mertens sum.

15 Nonbinary contrasts and half-cost transport

A q -resultant relation decomposes into one collective direction and the $(q - 1)$ -dimensional standard contrast representation. Complete relabelling averages the linear contrast charge to zero, so the normalized determinant begins quadratically. Separately, preservation of the generated null pair requires a local transport amplitude satisfying

$$a_p(s)^2 = p^s, \quad (27)$$

forcing $a_p(s) = \pm p^{s/2}$. The exponent one half is therefore derived twice in distinct structures: quadratic contrast completion and null preservation. Neither derivation alone supplies a scalar cancellation estimate.

16 Operator completion and scalarization barrier

On the squarefree basis w_n , set

$$Jw_n = \mu(n)w_n, \quad Hw_n = (\log n)w_n, \quad D_s = Je^{-sH}. \quad (28)$$

The singular values are $n^{-\Re s}$. Thus D_s is Hilbert–Schmidt exactly for $\Re s > 1/2$ and trace class exactly for $\Re s > 1$.

The odd scalar is not closable on the quadratic carrier. Let S_N be the visible squarefree set and

$$X_N = |S_N|^{-1} \sum_{n \in S_N} \mu(n)|n\rangle\langle n|. \quad (29)$$

Then $\|X_N\|_2 = |S_N|^{-1/2} \rightarrow 0$ while $\text{Tr}(JX_N) = 1$. Therefore ordinary scalar trace cannot be extended continuously from finite rank to the half-cost Hilbert–Schmidt completion. The obstruction is exact; it does not prove that every special scalar trajectory diverges.

17 What this paper contributes

The topology and operator theory are affirmative results: an exact barcode, an exact operator-ideal boundary, and an exact nonclosability witness. The universal arithmetic cancellation problem begins only after scalarization and is treated in the companion paper rather than smuggled into this one.

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Critical Capacity Completion

Equivalent Forms of the Riemann Hypothesis

Abhijit Singh

Contents

Abstract	3
1 Source freeze and classical identification	4
2 Finite scalarization defect	4
3 Weighted Mertens capacity energy	4
4 Capacity recurrence exponent	5
5 Equivalent completion statements	5
6 Completed-incidence formulation	6
7 Causal-tail formulation	6
8 Failed bridges retained	6
9 Static five-class reporter and all-prime predictive closure	6
10 Conclusion	7
11 Weighted capacity energy in detail	7
12 Three equivalent critical coordinates	8
13 Finite scalarization reconstruction	8
14 Countermodels and proof boundary	8
References	8

Abstract

A frozen source-derived arithmetic construction is connected to the classical Riemann problem through exact finite and infinite statements. Scalarization of the uncollapsed response algebra fails only through repeated-prime and over-capacity interactions; differentiation gives an exact boundary-current reconstruction. A weighted Mertens capacity energy A_N is introduced and proved to satisfy $A_N/N \rightarrow \infty$ and $N/A_N \rightarrow 0$. The power-growth exponent of the excess provides a critical coordinate. After classical identification, zero transverse escape, local-uniform Mobius control, completed-incidence positivity, and subpower capacity excess are proved equivalent to the Riemann Hypothesis in the displayed normalizations. The equivalence theorem is stated boldly and separately from the still-unproved universal estimate required to settle the conjecture. The static five-class Möbius reporter is proved to be a false closure under prime multiplication; the all-scale predictive quotient is the prime-support address plus one absorbing repeated state.

Keywords: Riemann Hypothesis; Mobius function; Mertens function; capacity energy; operator inequality; critical exponent

Scope and status: The finite reconstruction, weighted-energy divergence, and equivalence theorems are asserted with their displayed proofs and scopes. The universal RH-equivalent lower bound remains open; no proof of RH is claimed.

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1 Source freeze and classical identification

The upstream construction is frozen before classical comparison: binary capacity, transparent arities, independent prime coordinates, squarefree incidence orientation, quota topology, nonbinary contrast fibres, null half-transport, and the orientation operator. Only then are the generated scalar character and sign identified with the classical Dirichlet and Mobius objects in their absolute domain.

This paper studies three exact objects: the finite scalarization defect, a weighted Mertens capacity energy, and equivalent formulations of the Riemann Hypothesis. Equivalence is not itself a proof of the conjecture.

2 Finite scalarization defect

The finite uncollapsed response group is invertible before scalarization. Scalar multiplication fails for two exact reasons:

1. repeated-prime collisions;
2. products crossing the shared capacity boundary.

Let $\Phi_N(a; s)$ be the scalarized response and $\lambda_N(s)$ its neutral generator. Differentiation gives a boundary current $R_N(a; s)$.

Theorem 2.1 (Boundary-response reconstruction). *For every finite capacity N ,*

$$\Phi_N(a; s) = e^{a\lambda_N(s)} \left[1 + \int_0^a e^{-t\lambda_N(s)} R_N(t; s) dt \right]. \quad (1)$$

Every term in R_N is generated by a repeated-coordinate or over-capacity interaction.

Finite zeros therefore satisfy an exact coupled charge-current constraint. The theorem identifies the missing scalar information; it does not estimate the infinite signed current.

3 Weighted Mertens capacity energy

Let

$$M(x) = \sum_{n \leq x} \mu(n), \quad (2)$$

and define

$$A_N = \sum_{q \leq N} M\left(\left\lfloor \frac{N}{q} \right\rfloor\right)^2, \quad \lambda_N = A_N^{-1}. \quad (3)$$

Grouping equal values of $\lfloor N/q \rfloor$ gives

$$\frac{A_N}{N} = \sum_{k=1}^N M(k)^2 \frac{\lfloor N/k \rfloor - \lfloor N/(k+1) \rfloor}{N}. \quad (4)$$

For fixed k , the coefficient tends to $1/[k(k+1)]$.

Theorem 3.1 (Divergence of the weighted Mertens energy).

$$\sum_{n \geq 1} \frac{M(n)^2}{n(n+1)} = \infty. \quad (5)$$

Consequently,

$$A_N/N \rightarrow \infty, \quad N\lambda_N \rightarrow 0. \quad (6)$$

Proof sketch. If the weighted series converged, then $g(u) = e^{-u/2}M(e^u)$ would lie in $L^2(0, \infty)$. Its Laplace transform would have uniformly bounded vertical L^2 norms in the right half-plane. Summation by parts identifies the transform with $[(z + 1/2)\zeta(z + 1/2)]^{-1}$ in the initial domain. A critical-line zero gives boundary growth incompatible with the uniform bound. The lower limit for A_N/N then dominates arbitrarily long partial sums of the divergent weighted energy. $\square$

Table 1: Independent finite audit of the capacity energy.

N	$M(N)$	A_N	A_N/N
10	-1	11	1.100000
100	1	143	1.430000
10^3	2	1479	1.479000
10^4	-23	15829	1.582900
10^5	-48	157055	1.570550
10^6	212	1645995	1.645995

Finite nonmonotonicity is compatible with the asymptotic theorem.

4 Capacity recurrence exponent

Define the critical excess and its power-growth exponent by

$$C_N = A_N/N, \quad (7)$$

$$\chi_{\text{cap}} = \limsup_{N \rightarrow \infty} \frac{\log C_N}{\log N}. \quad (8)$$

The unconditional result is $C_N \rightarrow \infty$. The critical distinction is subpower growth versus positive-power growth.

5 Equivalent completion statements

After the classical identification, the following formulations are equivalent to the Riemann Hypothesis within the frozen normalizations used here:

1. local-uniform boundedness of the Mobius polynomials $F_N(s) = \sum_{n \leq N} \mu(n)n^{-s}$ on every compact subset of $\Re s > 1/2$;
2. a completed-incidence positivity inequality on the pole-neutral subspace;
3. absence of positive transverse escape in the normalized causal-tail representation;
4. the critical capacity condition $\chi_{\text{cap}} = 0$.

Theorem 5.1 (Equivalence theorem). *The displayed capacity-completion condition is equivalent to the classical Riemann Hypothesis.*

Claim status. Exact equivalence. The universal capacity estimate required to conclude the condition is not proved here.

6 Completed-incidence formulation

Let D_L be the completed incidence operator and $\Delta(L)$ the common budget in the frozen Weil normalization. On the pole-neutral space $\mathcal{H}_L^0$, the target inequality is

$$A_L|_{\mathcal{H}_L^0} \geq \Delta(L)I \quad \text{for every } L. \quad (9)$$

Equivalently, when $\Delta(L) > 0$, there is a contraction T_L such that

$$\sqrt{\Delta(L)} g = T_L D_L g, \quad \|T_L\| \leq 1. \quad (10)$$

The operator and scale-consistency identities are constructed; the universal lower bound is the open RH-equivalent statement.

7 Causal-tail formulation

Let Θ_N be the normalized adverse tail over degree at most N beyond a sublinear observer boundary. Define

$$\chi_{\text{RH}} = \limsup_{N \rightarrow \infty} \frac{1}{N} \log(1 + \Theta_N). \quad (11)$$

Positive χ_{RH} represents exponentially coherent adverse witnesses across increasing degree. The critical regime is the absence of positive exponential escape. This is not identical to the Schatten boundary $\Re s = 1/2$ without the explicit intertwiner supplied by the causal-tail construction.

8 Failed bridges retained

The following routes are not used as proof:

- finite zero audits;
- symmetry of the functional equation alone;
- random-sign or square-root-cancellation assumptions;
- a fitted spectrum or random-matrix analogy;
- the unbounded uniform odd trace on the Hilbert-Schmidt carrier;
- finite positivity without a uniform bound.

They remain valuable countermodels because they identify exactly where a plausible shortcut loses the required scalar orientation.

9 Static five-class reporter and all-prime predictive closure

The digit carrier admits the exact terminal-resolved classes

$$\{0\}, \quad \{1\}, \quad \{2, 3, 5, 7\}, \quad \{6\}, \quad \{4, 8, 9\}, \quad (12)$$

corresponding to zero, unit, Möbius value -1 , value $+1$, and value 0 . This is the coarsest static reporter on the ten digits that preserves terminal role and Möbius value. It is not an autonomous all-scale arithmetic state.

Theorem 9.1 (Five-class false closure under prime multiplication). *When multiplication by arbitrary primes is admitted, the five-class reporter is not a continuation congruence. The coarsest predictive quotient for all future Möbius reports consists of one state for every finite squarefree prime-support set and one absorbing repeated-prime state.*

Proof. The states 2 and 3 share the prime class. Multiplication by 2 sends them to 4 and 6, whose Möbius values are 0 and +1. Hence the static class is not future-complete. Represent a squarefree integer by its finite prime support and every nonsquarefree integer by $\dagger$. Multiplication by p adjoins p if absent and enters $\dagger$ if present; $\dagger$ remains absorbing. If two supports differ, multiplying by a prime in their symmetric difference makes one repeated and the other squarefree. Thus distinct supports are future-distinguishable and the quotient is coarsest. $\square$

The source has complete vector and mean-square realizations above $\Re s = 1/2$, while fixed-phase scalar evaluation remains unbounded. The exact RH gate is therefore not another finite classification. It is local-uniform boundedness of the special Möbius diagonal on every compact subset of $\Re s > 1/2$.

10 Conclusion

The arithmetic programme supplies a source-ordered derivation, an exact finite response reconstruction, an unconditional coercivity theorem, and several exact equivalent forms of RH. It does not yet supply the universal capacity-completion estimate. The live mathematical object is therefore precise: prove subpower critical excess or the equivalent completed-incidence bound without importing the desired cancellation.

11 Weighted capacity energy in detail

Let $M(x) = \sum_{n \leq x} \mu(n)$ and define

$$A_N = \sum_{q \leq N} M\left(\left\lfloor \frac{N}{q} \right\rfloor\right)^2, \quad \lambda_N = A_N^{-1}. \quad (13)$$

Grouping equal quotient values yields

$$\frac{A_N}{N} = \sum_{k=1}^N M(k)^2 \frac{\lfloor N/k \rfloor - \lfloor N/(k+1) \rfloor}{N}. \quad (14)$$

For fixed k , the coefficient tends to $1/[k(k+1)]$.

Suppose, for contradiction, that

$$\sum_{n \geq 1} \frac{M(n)^2}{n(n+1)} < \infty. \quad (15)$$

Then $g(u) = e^{-u/2} M(e^u)$ is square integrable. Its Laplace transform has uniformly bounded vertical L^2 norm in the open right half-plane. In the initial domain, summation by parts identifies it with the reciprocal zeta factor divided by the Mellin variable. A zero on the critical line produces boundary blow-up incompatible with that uniform bound. Hence the weighted Mertens energy diverges. Fatou's lemma applied to the grouped expression gives

$$\frac{A_N}{N} \longrightarrow \infty, \quad N\lambda_N \longrightarrow 0. \quad (16)$$

The finite audit is

N	$M(N)$	A_N	A_N/N
10	-1	11	1.100000
100	1	143	1.430000
10^3	2	1479	1.479000
10^4	-23	15829	1.582900
10^5	-48	157055	1.570550
10^6	212	1645995	1.645995

Finite nonmonotonicity is compatible with the asymptotic theorem.

12 Three equivalent critical coordinates

The classical target can be written through several exact equivalences after the arithmetic identification is frozen.

1. Local-uniform boundedness of the Möbius Dirichlet polynomials on compact subsets of $\Re s > 1/2$.
2. A completed-incidence positivity or coercivity inequality on every finite scale.
3. Zero power-growth exponent

$$\chi_{\text{cap}} = \limsup_{N \rightarrow \infty} \frac{\log(A_N/N)}{\log N} = 0. \quad (17)$$

The unconditional divergence $A_N/N \rightarrow \infty$ and the critical condition $\chi_{\text{cap}} = 0$ are consistent: the latter permits divergent growth slower than every positive power.

13 Finite scalarization reconstruction

Before scalarization, the squarefree response group is everywhere invertible. Scalar multiplication fails only when prime coordinates repeat or when a product crosses the shared capacity boundary. Differentiation at the neutral response gives a boundary current $R_N(a; s)$ and exact reconstruction

$$\Phi_N(a; s) = e^{a\lambda_N(s)} \left[1 + \int_0^a e^{-t\lambda_N(s)} R_N(t; s) dt \right]. \quad (18)$$

Every finite scalar zero therefore obeys a coupled charge–current equation whose terms are generated by those two failures. Passing from that exact finite equation to an infinite signed completion is the live analytic problem.

14 Countermodels and proof boundary

The source programme tested and retained countermodels to several tempting shortcuts: finite positivity without uniform control; quadratic completion without an odd scalar; exact reflection without a magnitude estimate; sparse bad modes without control of coherent near-resonance; and numerical zero alignment without a source-generated universal inequality. The paper proves equivalence and several auxiliary theorems. It does not claim the open universal bound.

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Twin-Prime Survivors

Square-Horizon Occupation and Moving-Boundary Dispersion

Abhijit Singh

Contents

Abstract	3
1 Unchanged target and finite survivor system	4
2 Exact CRT lift recursion	4
3 Square-horizon equivalence	4
4 Moving-boundary flux	4
5 Prime-semiprime transition	5
6 First-deletion variable and path law	5
7 Profinite and Fourier formulations	5
8 Parity ledger	6
9 Failed closures retained	6
10 What the paper establishes	6
A Specialist audit questions	6
B Exact residue recursion	6
C Square-horizon theorem	7
D Boundary-current decomposition	7
E First deletion and path information	7
F Parity ledger and failed mechanisms	7
G Reproducibility and honest title boundary	8
References	8

Abstract

A parity-preserving continuation of the twin-prime sieve is developed. Complete Chinese-remainder lift fibres give an exact survivor recursion and zero-drift normalized update over full residue blocks. At square-horizon depth, an odd survivor is exactly a twin-prime pair; the remaining problem is occupation of a moving interval rather than local admissibility. Decomposing the interval into complete blocks and one boundary block yields a signed moving-boundary flux. Divisor inversion retains the orientation and produces a controlled main term plus one coupled signed dispersion channel. Sublinear control of that channel implies a quantitative twin-prime lower bound. A first-deletion variable, path-relative entropy, profinite martingale, and finite Fourier transfer locate the unresolved parity information without solving it. Failed pointwise, martingale, unsigned-sieve, large-sieve, and resonance-scarcity routes are retained.

Keywords: twin primes; Chinese remainder theorem; sieve parity; signed dispersion; moving boundary; bilinear forms

Scope and status: Finite recursions, square-horizon equivalence, flux identities, and the conditional implication are exact. The live signed dispersion estimate is unproved. The Twin Prime Conjecture is not claimed.

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1 Unchanged target and finite survivor system

Let

$$\pi_2(X) = \#\{p \leq X : p \text{ and } p + 2 \text{ are prime}\}. \quad (1)$$

The classical Twin Prime Conjecture is the statement $\pi_2(X) \rightarrow \infty$. The finite system below preserves this target and does not replace it with local admissibility.

For a sieving frontier z , let

$$P_z = \prod_{p \leq z} p. \quad (2)$$

A residue class $a \bmod P_z$ survives the twin sieve when

$$a \not\equiv 0, -2 \pmod{p} \quad (3)$$

for every odd prime $p \leq z$, with the usual parity restriction at $p = 2$.

2 Exact CRT lift recursion

When the next prime q is adjoined, each old residue has q lifts and two are deleted.

Proposition 2.1 (Survivor-count recursion). *For the complete residue system,*

$$S_q = (q - 2)S_{q^-}. \quad (4)$$

The normalized density

$$\delta(z) = \prod_{3 \leq p \leq z} \left(1 - \frac{2}{p}\right) \quad (5)$$

has zero average update defect over a complete residue block. The unresolved term is created when a moving observation interval cuts the block.

3 Square-horizon equivalence

A survivor n at a sieve depth containing all primes up to $\sqrt{n+2}$ cannot be composite at either n or $n+2$.

Proposition 3.1 (Square-horizon equivalence). *Within the exact square-horizon window, an odd survivor is equivalent to a twin-prime pair.*

Proof. If n were composite, it would have a prime divisor at most $\sqrt{n}$, which is in the sieve frontier; likewise for $n+2$. Survival excludes both possibilities. The converse is immediate. $\square$

This is an exact reporter equivalence at the square horizon. It does not prove that the moving interval contains a survivor.

4 Moving-boundary flux

Decompose a finite interval into complete residue blocks and one boundary block. Complete blocks contribute their exact mean. All discrepancy is a signed flux through the moving boundary. The finite flux is additive under block coarsening and retains the orientation lost by unsigned local-density arguments.

After divisor inversion, the count has the form

$$\mathcal{T}(X) = \mathcal{M}(X) + D_1(X) - D_0(X) + \mathcal{E}(X), \quad (6)$$

where $\mathcal{M}$ is the controlled small-source term, $D_1 - D_0$ is the coupled signed moving-boundary dispersion, and $\mathcal{E}$ is a controlled remainder.

Theorem 4.1 (Conditional quantitative closure). *If*

$$D_1(X) - D_0(X) = o(X) \quad (7)$$

and the displayed remainder estimate holds uniformly on the declared square-horizon scale, then the survivor count has a positive Hardy-Littlewood-order main term and there are infinitely many twin primes.

Claim status. Conditional theorem. The signed dispersion estimate is the live unproved analytic statement.

5 Prime-semiprime transition

At late sieve depth, a composite survivor below the square horizon has highly restricted factorization. The first unresolved composites are semiprimes whose factors lie beyond the current frontier. This produces a bilinear residual of schematic form

$$\mathcal{B}(X) = \sum_{m \leq \sqrt{X}} \sum_{n \leq X/m} a_m b_n \mathbf{1}_{mn+2 \text{ survives}} - \text{local mean}. \quad (8)$$

The coefficients are not arbitrary sieve weights; they are generated by the exact CRT recursion and the moving interval boundary.

6 First-deletion variable and path law

For a candidate x , let κ_x be the first prime stage at which x or $x + 2$ is deleted, with $\kappa_x = \infty$ for a twin-prime pair. The finite path of survival/deletion records more than terminal admissibility. Comparing the empirical interval path to the complete CRT lift law produces a path-relative entropy. A positive divergence localizes where the moving integer interval departs from complete residue equilibrium.

The path formulation does not remove the parity problem. Its value is diagnostic: it assigns the unresolved orientation to an explicit stage and block boundary instead of hiding it in a heuristic independence assumption.

7 Profinite and Fourier formulations

The inverse system of residue blocks defines a compact profinite carrier. The normalized survivor indicator is a cylinder function, and conditional expectation over successive prime stages gives a martingale on the complete CRT probability space. The actual moving interval is not a random sample from this measure; the required transfer is a discrepancy theorem between the interval occupation measure and the projective measure.

Finite Fourier expansion on each residue block transfers the same obstruction into coherent near-resonant modes. Scarcity of exact resonances is insufficient: one must control the signed sum of all near-resonant contributions generated by the moving boundary.

8 Parity ledger

The Chen-Liouville identity and rough-fibre census separate prime and semiprime channels, but they do not by themselves force positive prime-pair occupation. Any proof must preserve the Mobius/Liouville orientation through the last estimate. Replacing signed terms by absolute values, probabilities, or an unsigned local admissibility product discards precisely the information needed to distinguish one prime from an almost-prime.

9 Failed closures retained

The following approaches were tested and found insufficient:

1. pointwise control of each residue class;
2. treating normalized updates as a martingale without a probability law for the moving interval;
3. generic large-sieve bounds that ignore generated signs;
4. exact-resonance scarcity without coherent near-resonance control;
5. replacing the moving boundary by complete residue blocks;
6. ordinary sieve positivity that cannot cross the parity barrier.

10 What the paper establishes

The paper establishes the complete finite CRT recursion, exact square-horizon reporter, moving-boundary decomposition, parity-sensitive coefficient system, and a conditional theorem reducing twin-prime infinitude to one coupled signed dispersion estimate. It does not establish that estimate and therefore does not claim the Twin Prime Conjecture.

A Specialist audit questions

1. Is the coupled moving-boundary estimate equivalent to an existing dispersion criterion or strictly different?
2. Can Type I/II bilinear forms preserve the generated orientation at square-horizon depth?
3. Does the first-deletion/path-relative-entropy representation expose a known parity obstruction in a more tractable norm?
4. Can near-resonant Fourier modes be controlled without assuming random prime residues?

B Exact residue recursion

Let $P_z = \prod_{p \leq z} p$. A residue $a \bmod P_z$ survives the twin sieve if

$$a \not\equiv 0, -2 \pmod{p} \quad (9)$$

for every odd prime $p \leq z$, with the standard adjustment at 2. On adjoining a new odd prime p , every old survivor has p lifts and exactly two are deleted. Hence the abstract residue population obeys

$$S_p = (p - 2)S_{p^-}. \quad (10)$$

After normalization by $\prod_{3 \leq q \leq p} (1 - 2/q)$, complete residue blocks have zero mean drift. The unresolved term is the signed discrepancy produced when the physical observation interval cuts a block.

C Square-horizon theorem

If every prime $q \leq \sqrt{X}$ has been sieved and $m \leq X$ survives both residues, then neither m nor $m + 2$ has a prime divisor at most its square root. Except for the explicitly treated unit/boundary cases, each is prime. Conversely a twin-prime pair survives every such local deletion. Thus occupation of the moving interval after square-root depth is exactly the global target. Local admissibility alone does not guarantee such occupation.

D Boundary-current decomposition

Write the physical interval as complete residue blocks plus one moving remainder. Complete blocks contribute their exact mean. The remainder can be expressed as activation and deletion flux across the moving boundary. After divisor and Mellin decompositions, the residual takes a bilinear form whose coefficients are not arbitrary: they are generated by the survivor recursion, Möbius orientation, and the square-horizon boundary.

A convenient status form is

$$\mathcal{T}(X) = \mathcal{M}(X) + D_0(X) + D_1(X) + \mathcal{E}(X), \quad (11)$$

where $\mathcal{M}$ is the positive local main term, D_0, D_1 are signed orientation-sensitive dispersion modes, and $\mathcal{E}$ is the controlled remainder.

Theorem D.1 (Conditional twin-prime closure). *If*

$$D_0(X) = o(X), \quad D_1(X) = o(X), \quad (12)$$

with the displayed coefficient system and moving boundary, then the positive survivor main term is not cancelled and infinitely many twin-prime representatives occur.

The theorem is an exact sufficient reduction. The two estimates remain live.

E First deletion and path information

For each candidate pair, define the first prime stage at which either component is deleted. This variable records more than terminal survival: it determines a path through the nested CRT fibres. The empirical path law on integers and the uniform CRT path law can be compared by relative entropy. A proof programme may then seek divergence of cumulative hazard or a controlled likelihood ratio. No such information inequality is claimed complete here; it is retained as a distinct continuation rather than folded into the signed dispersion theorem.

F Parity ledger and failed mechanisms

The following routes were tested and rejected as closures:

1. pointwise control of every residue class;
2. treating the moving boundary as a probability-independent martingale;
3. applying generic large-sieve estimates after discarding the generated signs;
4. counting exact resonances while ignoring coherent near-resonance;
5. replacing the moving interval by a complete block;
6. using local survivor density as though it implied occupied square-horizon fibres.

The failure record is part of the mathematics because it identifies precisely which information the live estimates must retain.

G Reproducibility and honest title boundary

Finite verification checks lift counts, normalized block means, moving-boundary flux, divisor decomposition, activation receipts, and Fourier identities. No finite audit can establish infinitely many twin primes. The manuscript therefore presents a continuation system and an exact sufficient theorem, not a completed proof of the conjecture.

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Future-Relevant Representation Dis- agreement

Chronological Feature Restoration

Abhijit Singh

Contents

Abstract	3
1 Common experimental question	4
I Study I: relation among independent representations	4
2 Design	4
3 Result	4
4 Context-dependent output formation	4
II Study II: chronological environmental record	5
5 Raw data audit	5
6 Chronological split and restoration rule	5
7 Untouched-future result	6
8 Leakage and uncertainty requirements	6
9 Interpretation	7
10 Complete data and metric separation	7
11 Environmental chronology and raw audit	7
12 Chronological restoration result	7
13 Baseline and statistical requirements	8
14 Release protocol and evidence status	8
References	8

Abstract

Two held-out studies test whether differences discarded by current-state compression remain relevant for later continuation. In a 1,797-sample, 64-coordinate corpus, three independently generated representations attain 90.74–92.96 percent exact whole-object continuation, while a relation-aware router using only child status, agreement, and prior contact performance reaches 97.78 percent, equal to the empirical oracle ceiling. A context-dependent return separately reaches 99.768 percent accuracy on admitted coordinates. In a 13-channel hourly environmental chronology, three constructors omit the same coordinate; four chronological contact blocks support restoration, and untouched-future performance improves from 72.0836 to 74.7125 percent per coordinate and from 3.7540 to 7.2492 percent for exact vectors. Negative controls reject indiscriminate higher arity and deeper history.

Keywords: representation disagreement; chronological validation; feature restoration; ensemble routing; predictive sufficiency

Scope and status: Raw environmental-file audits are independently reproduced. Downstream environmental and static-representation results are source-reported because the complete original constructors and static dataset are not in the supplied local archive.

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1 Common experimental question

Both studies test whether a distinction discarded by present-state compression remains relevant to future continuation. A candidate coordinate or relation is omitted, evaluated on chronological or held-out contact data, and restored only if it changes later prediction. The final test set remains untouched until the representation is frozen.

For current representation Z , candidate residual A , future target Y^+ , and frozen loss $\mathcal{R}$, define

$$\Delta_{\text{fut}}(A \mid Z) = \mathcal{R}(Y^+ \mid Z) - \mathcal{R}(Y^+ \mid Z, A). \quad (1)$$

Positive held-out gain licenses promotion for the declared task; it does not imply that every extra feature should be retained.

Part I Study I: relation among independent representations

2 Design

The frozen corpus contains 1,797 samples and 64 output coordinates. Three representations are generated independently before a 270-case untouched holdout is opened. Their exact whole-object continuation rates are

$$92.96\%, \quad 90.74\%, \quad 91.48\%. \quad (2)$$

A fourth model receives only child status, agreement/disagreement relations, and performance on prior contact cases. It does not receive the semantic class labels or virgin answers.

3 Result

The relation-aware model reaches

$$\boxed{264/270 = 97.78\%} \quad (3)$$

exact whole-object continuation, equal to the empirical oracle ceiling of the available child repertoire. Removing the agreement relation lowers the result to 97.41% and produces an additional failure and two additional unknowns.

Higher relation arity did not repair cases in which all three child representations failed: the tested ternary constructor solved 0/6 such cases. Agreement is useful only when at least one constituent representation already carries relevant future information.

4 Context-dependent output formation

A second experiment chooses an output coordinate set before answer comparison. Across all 270 cases the return is nonempty, with mean size 7.985 of eight coordinates. Admitted-coordinate accuracy is

$$99.768\%, \quad (4)$$

and whole-return correctness is

$$98.52\%. \quad (5)$$

These are different answer objects and must not be pooled with the 97.78% whole-object routing metric.

Part II Study II: chronological environmental record

5 Raw data audit

The supplied semicolon-delimited file contains:

Table 1: *Independent raw-file audit.*

raw rows	9471
blank trailer rows	114
nonblank hourly rows	9357
completely empty columns	2
numeric channels	13
missing sentinel cells (−200)	16701
rows complete on all channels	827

The nonblank timestamps are strictly hourly from 10 March 2004 18:00 to 4 April 2005 14:00.

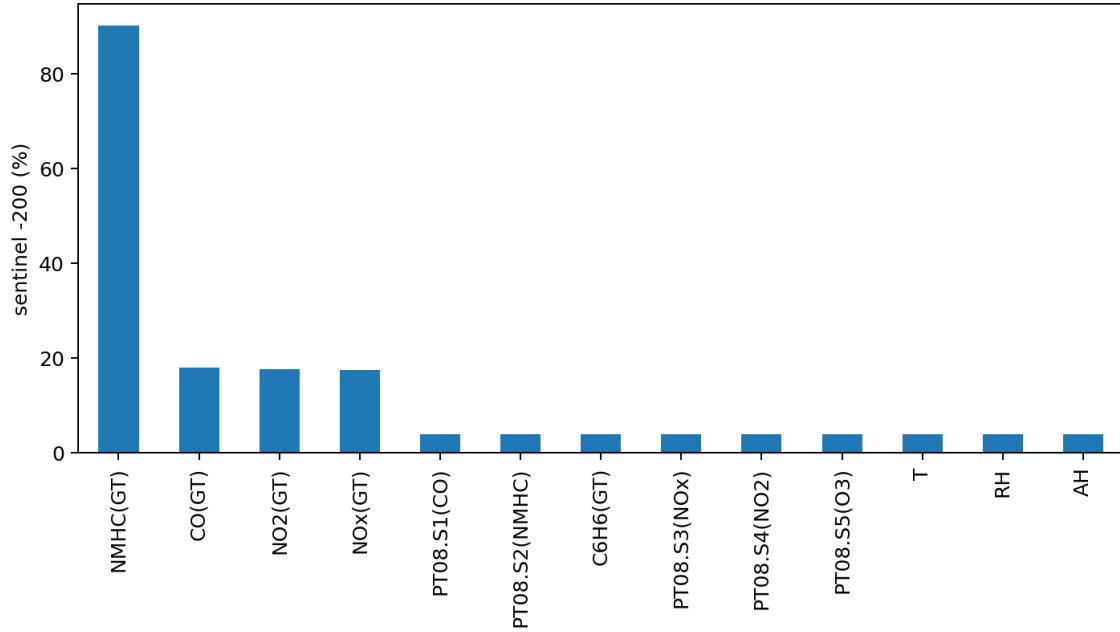


Figure 1: *Fraction of sentinel values by numeric channel in the supplied raw file.*

6 Chronological split and restoration rule

The corpus-reported filtered chronology contains 7,724 transitions divided into 4,634 development, 1,545 re-entry/contact, and 1,545 untouched-future observations. Channel semantics are hidden during representation formation.

Three isolated constructors retain 12 of 13 present-time coordinates. The rejected coordinate is restored only when its inclusion improves actual next-time continuation on successive chronological blocks. The four block increments in correctly predicted future coordinates are

$$158, \quad 180, \quad 102, \quad 121, \quad (6)$$

for a total of 561. Exact future vectors improve by 19, 5, 10, and 10 on the four blocks.

7 Untouched-future result

After the representation is frozen, the untouched future gives

Table 2: *Future performance before and after restoration.*

metric	12 channels	13 channels
coordinate accuracy	72.0836%	74.7125%
exact next vector	3.7540%	7.2492%

The restored representation returns exactly to the independently computed all-channel baseline on both metrics.

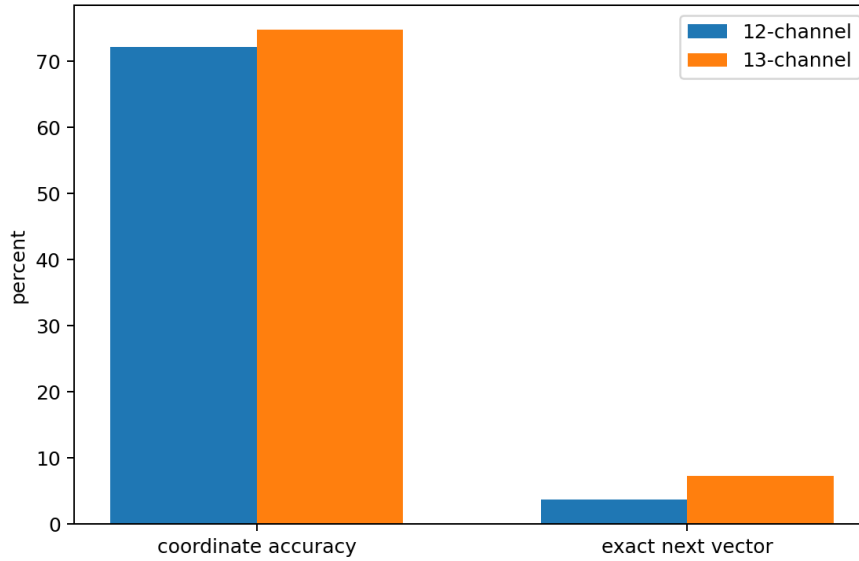


Figure 2: *Untouched-future improvement after restoring the omitted channel.*

A deeper history coordinate selected on contact data is rejected because it worsens both untouched-future objectives. The method therefore supports selective restoration, not indiscriminate state growth.

8 Leakage and uncertainty requirements

A clean independent replication requires:

1. exact static dataset and case identifiers;
2. all coordinate masks and representation constructors;
3. raw-to-7,724 preprocessing code;
4. quantization thresholds and random seeds;
5. row-level predictions;
6. frozen chronology and contact decisions;
7. paired block-bootstrap or permutation intervals preserving serial dependence;
8. modern baselines selected only on development/contact data.

The raw environmental file is included in the ancillary archive. The static 1,797-sample corpus and original constructors are not present in the supplied local files; their numerical results are therefore source-reported rather than independently replayed here.

9 Interpretation

The common empirical result is narrow and precise: present compression can erase a relation or coordinate that remains relevant to later continuation. In Study I the retained object is relation among representations; in Study II it is one temporal channel. Neither study proves an asymptotic recurrence exponent without an independently specified repeated residual dynamics.

10 Complete data and metric separation

The static study uses a fixed corpus of 1,797 samples and 64 output coordinates. Three representation-specific exact whole-object rates on the untouched 270-case set are

$$92.96\%, \quad 90.74\%, \quad 91.48\%. \quad (7)$$

A relation model observes only child status, agreement/disagreement, and frozen contact performance. It reaches

$$\boxed{264/270 = 97.78\%}, \quad (8)$$

equal to the empirical oracle ceiling of the available children. Removing the agreement relation creates one additional failure and two additional unknowns. A higher-arity constructor solved none of the six cases where all children failed, so more relation arity is not treated as automatic rescue.

The contextual-output experiment asks a different question: which output coordinates are warranted before values are compared? Its mean return size is 7.985 of 8, admitted-coordinate accuracy is 99.768%, and whole-return correctness is 98.52%. These values are never pooled with the 97.78% whole-object routing result.

11 Environmental chronology and raw audit

The supplied semicolon-delimited record contains

raw rows	9471
blank trailer rows	114
nonblank hourly rows	9357
completely empty columns	2
numeric channels	13
missing sentinel cells (−200)	16701
rows complete on all channels	827

The nonblank timestamps form a strict hourly sequence from 10 March 2004 18:00 through 4 April 2005 14:00. The reported filtered chronology contains 7,724 transitions: 4,634 development, 1,545 contact/re-entry, and 1,545 untouched future observations.

12 Chronological restoration result

Three isolated constructors retained 12 of 13 present-time channels. The omitted channel was restored only after it improved actual next-time continuation on successive chronological contact blocks. The increments in correctly predicted future coordinates were

$$158, \quad 180, \quad 102, \quad 121, \quad (9)$$

for a total of 561. Exact future vectors improved by 19, 5, 10, and 10. The representation was then frozen before opening the untouched final segment.

Table 3: *Untouched-future comparison.*

metric	12 channels	13 channels
coordinate accuracy	72.0836%	74.7125%
exact next vector	3.7540%	7.2492%

A deeper history representation selected on contact data was rejected because it worsened both untouched metrics. The method therefore permits deletion as well as promotion.

13 Baseline and statistical requirements

A clean external reproduction should compare persistence, seasonal persistence, regularized vector autoregression, tree ensembles, nearest-neighbour analogues, and a modern sequence model. Hyperparameters must be selected only on development and contact segments. Serial dependence requires moving-block bootstrap or another dependence-preserving interval method. Report coordinate-level error, exact-vector accuracy, calibration, missingness sensitivity, compute cost, and the probability that every chronological block favours restoration.

14 Release protocol and evidence status

The raw environmental file is included and independently audited. The static corpus, exact constructors, row-level predictions, quantization thresholds, and raw-to-7,724 preprocessing program are required for complete independent reproduction. Until released, the corresponding numerical values remain source-reported results rather than externally reproduced facts. That status does not erase the result; it specifies the next admissible test.

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Reenactable Closure

Conditional Quantum-Causal Representation

Abhijit Singh

Contents

Abstract	3
1 Act, record, and changed standpoint	4
2 Seats and reenactment	4
3 Generated probes and complete signatures	4
4 Formation invariant algebra	5
5 Quantum representation programme	5
6 Causal quotient and Lorentzian representation	5
7 Finite unified loop	5
8 Block connection and curvature	6
9 Conversion isospectrality	6
10 Retired physical branch	6
11 Field equation and conservation	6
12 Correspondence limits	7
13 Open theorem and experiment list	7
14 Historical residual and correction	7
15 Reenactment fixed point	7
16 Generated probes and endogenous completion	7
17 Invariant algebra	8
18 Quantum representation assumptions	8
19 Causal quotient and continuum	8
20 Block connection and exact algebra	8
21 Retired physical interpretation	8
22 Open theorem programme	9
References	9

Abstract

A formation-first calculus is developed around an occurrence in which an active role, stabilized content, and successor standpoint arise together. Stabilized seats can reenact formation patterns; diagonal reenactment yields fixed points for seat-expressible transformers. Internally generated probes define complete signatures, equivalence, laws, and an idempotent completion, replacing an external meta-level closure. The invariant algebra supports sequential and unresolved-alternative composition. Under explicit reconstruction assumptions, one quotient admits finite complex quantum representation and another admits causal/Lorentzian geometry. A block connection couples active, stabilized, and conversion sectors. Its curvature forces first-order off-diagonal conversion and second-order diagonal backreaction; the quadratic terms obey trace exchange, and reciprocal maps yield identical nonzero singular spectra in the two sectors. Unsupported mixed-path physical interpretations are retired while the exact algebra is retained.

Keywords: formation calculus; fixed points; generated probes; quantum reconstruction; causal geometry; block connection

Scope and status: Reenactment, probe closure, block scaling, trace exchange, and isospectrality are formal theorems. Quantum and geometric results are conditional representations. No new physical conversion law is claimed without an independently fixed observable.

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1 Act, record, and changed standpoint

The foundational asymmetry is operational: an occurring act is not identical to the stabilized record produced by that act, and stabilization changes the standpoint from which later operations proceed. A conclusion can be transmitted as content; the operation that made it meaningful must be reenacted.

Represent one primitive formation by

$$\Phi = \langle S \mid a \triangleright c \mid S^+ \rangle, \quad (1)$$

where a is the active role, c the stabilized role, S the operative standpoint, and S^+ the successor standpoint. These are roles within one occurrence rather than independently fundamental objects.

2 Seats and reenactment

A seat $[P]$ is stabilized content capable of regenerating a formation pattern P . Reenactment $\uparrow [P]$ is successful when it reproduces the structural-functional invariant of P .

Let T be any transformer expressible through seats. Define diagonal reenactment

$$\Delta(x) = \uparrow (x \bullet x). \quad (2)$$

Construct a seat q_T such that reenacting it on x performs $T(\Delta(x))$, and put $\Omega_T = \Delta(q_T)$.

Theorem 2.1 (Reenactment fixed point). *For every seat-expressible transformer T ,*

$$\Omega_T \simeq T(\Omega_T). \quad (3)$$

Proof. By construction, reenacting q_T on x gives $T(\Delta(x))$. Substitution $x = q_T$ gives the result. $\square$

If completion itself is seat-expressible, there exists a pattern invariant under completion. Finality means closure of representational form relative to generated probes, not cessation of events or discoveries.

3 Generated probes and complete signatures

For a pattern P , let $\mathcal{P}(P)$ be the least probe family closed under enactment, stabilization, successor-state change, composition, and available comparisons. Define the complete signature

$$\Sigma(P) = (p(P))_{p \in \mathcal{P}(P)}. \quad (4)$$

Patterns are equivalent when their complete generated signatures agree. A law is an automorphism preserving the signature and operation structure.

Theorem 3.1 (Endogenous completion). *If the generated probe algebra is closed under composition, adjoining all generated probe responses is idempotent:*

$$K(K(P)) \simeq K(P). \quad (5)$$

Law generation is likewise idempotent relative to the closed probe algebra.

This closes a recurring meta-level gap: probes are not imported by an unexamined external observer; they are generated from the enactment structure being completed.

4 Formation invariant algebra

Let $I(P)$ be the reenactment-equivalence class of P . Sequential formation and unresolved-alternative formation induce operations

$$I(P \odot Q) = I(P) \odot I(Q), \quad I(P \boxplus Q) = I(P) \boxplus I(Q). \quad (6)$$

With associative identities and distributivity, the invariant classes form a semiring-like algebra. Stabilized idempotent commuting invariants with complements produce a Boolean-like classical regime. Destructive recombination of unresolved alternatives requires a richer signed or phased valuation before stabilization.

5 Quantum representation programme

Four explicit principles are used:

1. reenactable formation: consistent refinement and recombination preserve the invariant;
2. standpoint reciprocity: pure active formations and pure stabilized distinctions can serve as each other's operational seats;
3. continuous reversible standpoint symmetry: equal-weight pure formations are connected by continuous reversible transformations;
4. relational completeness: a composite is determined by the full accessible relational statistics of its components.

Under spectral decomposability, homogeneity, self-duality, finite dimension, and standard composition hypotheses, established representation theorems yield Euclidean Jordan structure. Local tomography and a nonclassical qubit-like sector select the finite-dimensional complex quantum branch.

Claim status. Conditional representation. The four principles are not proved minimal or derivable from the primitive formation alone.

The represented objects are positive Hermitian states, effects, trace probabilities, unitary reversible dynamics, and tensor-product composition. The numerical value of $\hbar$ is a physical calibration, not derived by the formal calculus.

6 Causal quotient and Lorentzian representation

Continuation defines a partial order. Incomparability is the primitive analogue of spacelike separation. Under strong causality and smooth-continuum regularity, causal order determines conformal Lorentzian structure; a locally finite occurrence measure fixes scale. Proper time is recovered from calibrated chains.

Claim status. Conditional representation. Existence and convergence of a smooth continuum from arbitrary finite formation networks remain open.

7 Finite unified loop

A finite causal diamond supplies one active phase projection and one geometric transport projection. In a minimal abelian model, an edge connection $A_e = \chi_e + i\theta_e$ yields loop holonomy

$$H = A_{OL} + A_{LF} - A_{RF} - A_{OR}. \quad (7)$$

The imaginary projection gives interference probabilities, the real projection gives a Lorentz boost rapidity, and adjacent cells compose with cancellation of the shared internal edge. This is a proof of compositional consistency for the toy representation, not a theory of four-dimensional gravity.

8 Block connection and curvature

Introduce the block connection

$$\mathcal{A} = \begin{pmatrix} \Omega & \Upsilon \\ \tilde{\Upsilon} & \Theta \end{pmatrix}, \quad (8)$$

with curvature

$$\mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A} = \begin{pmatrix} R_\Omega + \Upsilon \wedge \tilde{\Upsilon} & D\Upsilon \\ D\tilde{\Upsilon} & R_\Theta + \tilde{\Upsilon} \wedge \Upsilon \end{pmatrix}. \quad (9)$$

If $\Upsilon = \varepsilon U$ and $\tilde{\Upsilon} = \varepsilon V$, off-diagonal conversion curvature is $O(\varepsilon)$ and diagonal backreaction is $O(\varepsilon^2)$.

Theorem 8.1 (Trace exchange). *The quadratic conversion terms obey*

$$\text{Tr}(\delta R_\Omega + \delta R_\Theta) = 0. \quad (10)$$

Proof. The diagonal terms are commutator-like products $UV - VU$ in opposite block order. Cyclicity of trace cancels their sum. $\square$

9 Conversion isospectrality

For one reciprocal map U , the two diagonal positive operators are $UU^\dagger$ and $U^\dagger U$.

Theorem 9.1 (Nonzero spectral equality). *$UU^\dagger$ and $U^\dagger U$ have identical nonzero spectra with multiplicity.*

Proof. A singular-value decomposition $U = W\Sigma V^\dagger$ gives $UU^\dagger = W\Sigma^2 W^\dagger$ and $U^\dagger U = V\Sigma^2 V^\dagger$. $\square$

The singular spectrum is a parameter-free algebraic invariant of the reciprocal map. It becomes a physical prediction only after both sectors and the map are defined independently of the measured outcomes.

10 Retired physical branch

An unrestricted mixed-path factorization of two equal-trace positive matrices is always possible and therefore nonpredictive. A tested single-path comparison did not support the proposed equality at the supplied point estimate. The physical conversion interpretation is therefore not asserted. The exact block algebra, scaling, trace, and singular-spectrum theorems remain valid.

11 Field equation and conservation

A Yang-Mills-type candidate action gives

$$D_\mu \mathcal{F}^{\mu\nu} = g^2 \mathcal{J}^\nu, \quad (11)$$

with covariant source conservation from the Bianchi identity. This is a candidate dynamical class, not a unique consequence of the primitive calculus. Higher-curvature, nonlocal, and additional-field actions are not excluded.

12 Correspondence limits

When conversion is negligible and geometry fixed, the active representation reduces to unitary quantum evolution. When alternatives are stabilized and the geometric block is represented by a Lorentz connection and coframe, standard four-dimensional assumptions recover an Einstein-type equation with cosmological term. These are correspondence statements rather than derivations of observed constants.

13 Open theorem and experiment list

1. prove minimality or independence of the formation principles;
2. prove finite-to-continuum convergence;
3. derive a preferred action internally;
4. identify an invariant observable for the conversion block before inspecting data;
5. test that observable on held-out instrument and geometric measurements;
6. reject the physical conversion branch if every effect reduces to ordinary open-system dynamics and semiclassical geometry.

14 Historical residual and correction

The original categorical route used a fixed-point-free record twist and diagonal self-application to prove that a current self-model omits the act generated by its own reflexive use. This established a residual but did not by itself create new mathematics below the already supplied objects, exponentials, and record space. The formation calculus changes the primitive from object plus report to an occurrence in which active role, stabilized role, and changed standpoint arise together.

15 Reenactment fixed point

A seat $[P]$ is stabilized content capable of reenacting pattern P . Let $\Delta(x)$ enact the seat obtained by applying x to itself. For every seat-expressible transformer T , choose a seat q_T whose enactment on x returns $T(\Delta(x))$. Then

$$\Omega_T = \Delta(q_T) \simeq T(\Omega_T). \quad (12)$$

Taking T to be completion gives a conditional final formation: completion adds no new structural-functional pattern. This is closure of representational form, not an assertion that no new empirical content can arrive.

16 Generated probes and endogenous completion

Let $\mathcal{P}(P)$ be the smallest probe family generated by enactment, stabilization, composition, and successor-state change. The complete signature is

$$\Sigma(P) = (q(P))_{q \in \mathcal{P}(P)}. \quad (13)$$

Formation equivalence is equality of complete signatures. A law is an automorphism preserving all generated probes. Completion adjoins the generated probes and their results. If the probe algebra is closed under composition,

$$K(K(P)) \simeq K(P), \quad (14)$$

and regenerating the law algebra is likewise idempotent. This removes an external declaration of completeness: closure is tested against the operations generated by the formation itself.

17 Invariant algebra

Resonance classes support sequential composition $\odot$ and unresolved-alternative composition $\boxplus$. In stabilized idempotent, commuting, complemented regimes the algebra reduces to a Boolean-like classical structure. Before stabilization, destructive recombination requires a signed or phased valuation richer than nonnegative probability.

18 Quantum representation assumptions

The route to finite complex quantum theory uses four explicit properties: spectral/re-enactable formation, state-effect reciprocity, continuous reversible symmetry on pure formations, and relational completeness of composites. Under finite-dimensional regularity these properties motivate homogeneous self-dual cones and Euclidean Jordan structure; standard composition restrictions then select the complex branch. This is a representation theorem programme, not a proof that the four properties follow uniquely from the primitive.

19 Causal quotient and continuum

Continuation induces a partial order. Under strong causality and smooth continuum regularity, causal order determines conformal Lorentzian geometry; a locally finite formation count fixes scale. Existence and convergence of that continuum from arbitrary finite formation networks remain open.

20 Block connection and exact algebra

Write

$$\mathcal{A} = \begin{pmatrix} \Omega & \Upsilon \\ \tilde{\Upsilon} & \Theta \end{pmatrix}, \quad \mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}. \quad (15)$$

Then

$$\mathcal{F} = \begin{pmatrix} R_\Omega + \Upsilon \wedge \tilde{\Upsilon} & D\Upsilon \\ D\tilde{\Upsilon} & R_\Theta + \tilde{\Upsilon} \wedge \Upsilon \end{pmatrix}. \quad (16)$$

If $\Upsilon = \varepsilon U$ and $\tilde{\Upsilon} = \varepsilon V$, conversion curvature is first order and diagonal backreaction second order. Cyclicity of trace gives

$$\text{Tr}(\delta R_\Omega + \delta R_\Theta) = 0. \quad (17)$$

For reciprocal conversion, $UU^\dagger$ and $U^\dagger U$ have identical nonzero spectra. This is an exact singular-value theorem and a parameter-free normalized relation inside the block model.

21 Retired physical interpretation

The public four-record instrument did not support the proposed single-map spectral equality at its point estimate. Allowing arbitrary mixed paths after observing both marginals is nonpredictive because every equal-trace positive pair admits such a factorization. The manuscript therefore retains the block algebra and isospectral theorem while retiring the unrestricted physical conversion claim. A future physical proposal must specify an invariant observable and prediction before the comparison data are inspected.

22 Open theorem programme

The remaining tasks are: derive or reduce the representation assumptions; prove finite-to-continuum convergence; derive a dynamical action rather than postulating a field equation; identify a conversion observable invariant under representation change; and obtain an independent experimental discriminator beyond standard open-system dynamics and semiclassical backreaction.

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Local Indistinguishability

Global Obstruction and Minimal Predictive Memory

Abhijit Singh

Contents

Abstract	3
1 Reporter boundary	4
2 Topological-order specialization	4
3 Trivial regime	4
4 Minimal predictive memory	4
5 Joint-state information omitted by marginals	5
6 Obstruction dictionary and grading	5
7 Open reflexive invariant	5
8 Physical tests	5
9 Scope	6
10 Graded translation dictionary	6
11 Exact reporter-boundary theorem	6
12 Trivial regime and obstruction limits	6
13 Memory and bond dimension	7
14 Joint information missed by marginals	7
15 Open problems and tests	7
References	7

Abstract

A precise reporter-boundary theorem is developed: a global label fails to factor through a family of local reporters exactly when there exist locally indistinguishable but globally distinguishable configurations. This biconditional directly realizes the defining local/global structure of topological order. The result is separated from a broader analogy between self-reference obstruction and nontrivial bundles, because the latter currently lacks quantization and deformation stability. Classical future-equivalence classes give deterministic predictive-memory lower bounds, while nonorthogonal quantum memory can compress the same process. Marginals of a bipartite state omit a calculable number of joint-correlation coordinates, so compatible marginals do not determine the operative relation. A graded dictionary distinguishes exact, structural, and heuristic transfers and identifies the construction of a deformation-stable reflexive invariant as the principal open problem.

Keywords: local indistinguishability; topological order; predictive memory; factorization; bond dimension; obstruction

Scope and status: The factorization theorem, coordinate count, and classical predictive-memory statements are exact. Topological analogies, quantization, stability, and the proposed invariant remain separate open questions.

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1 Reporter boundary

Let Σ be a set of configurations, $J : \Sigma \rightarrow O$ a global label, and $K_1, \dots, K_n$ local reporters. Write $K_{[n]} = (K_1, \dots, K_n)$ and define

$$\partial_J(K_1, \dots, K_n) = \{(s, t) : K_i(s) = K_i(t) \forall i, J(s) \neq J(t)\}. \quad (1)$$

Theorem 1.1 (Factorization biconditional).

$$\partial_J(K_1, \dots, K_n) = \emptyset \iff \exists f \ J = f \circ K_{[n]}. \quad (2)$$

Proof. A factorization makes J constant on every $K_{[n]}$ -fibre. Conversely, if the boundary is empty, define f on each image fibre by the common value of J . $\square$

The exact obstruction is therefore the conjunction of local indistinguishability and global distinguishability.

2 Topological-order specialization

For a degenerate ground-state subspace, local indistinguishability is expressed by

$$\langle \psi_i | O_{\text{local}} | \psi_j \rangle = c(O) \delta_{ij}. \quad (3)$$

Nonlocal loop or logical operators can separate the states. Taking the local expectation-value reporters and a global topological label realizes the reporter-boundary theorem directly.

This exact use must be separated from a looser analogy between diagonal self-reference obstruction and a nontrivial Bloch bundle. A Chern obstruction is an integer cohomology class stable under gap-preserving deformation. The current reflexive obstruction is only the presence of a fixed-point-free twist; it has no proved quantization or deformation stability. Topological theorems cannot be imported solely because both problems involve failure of a global section.

3 Trivial regime

When the relevant obstruction vanishes, a global representation exists. In band theory, under the standard hypotheses, a smooth frame and exponentially localized Wannier basis exist. In the reflexive calculus, the fixed-point regime removes the diagonal contradiction. The shared statement is existence of a global section after the obstruction vanishes; physical consequences remain domain specific.

4 Minimal predictive memory

For a classical stochastic process, causal states are equivalence classes of histories with the same future distribution. Their number is the minimal number of deterministic predictive states. Hidden-state and matrix-product representations require sufficient bond dimension to carry these future-distinguishable conditions.

Quantum models can encode the same classical process in nonorthogonal memory states and can require smaller Hilbert-space dimension than the number of classical causal states. The exact transfer is therefore:

- classical future equivalence gives a deterministic state-count lower bound;
- quantum compression can be smaller when nonorthogonal memory states reproduce the future statistics without being perfectly distinguishable.

5 Joint-state information omitted by marginals

For a bipartite $d_G \times d_Q$ Hermitian joint state, the two normalized marginals omit

$$(d_G^2 - 1)(d_Q^2 - 1) \quad (4)$$

real cross-correlation coordinates. In a 3×3 comparison, 64 joint coordinates are absent. Equal or compatible marginals therefore do not reconstruct the operative relation between them.

6 Obstruction dictionary and grading

The cross-field dictionary is divided into exact, structural, and heuristic entries.

predictive/reflexive object	quantum-materials object	status
reporter boundary	local indistinguishability plus global distinction	exact
minimal deterministic predictive states	classical memory/bond lower bound	exact
fixed-point regime	trivial global-section regime	exact at factorization level
diagonal residual	failure of global section	structural
returned-boundary tower	anomaly inflow hierarchy	structural
typed recurrence depth	renormalization scale	heuristic
record cost	driven-system dissipation	structural; conditional

7 Open reflexive invariant

A topological obstruction is useful because it is graded and stable. The reflexive obstruction currently lacks both properties.

Conjecture 7.1 (Reflexive invariant programme). *There exists a deformation-stable quantity $X(m)$ on a suitably topologized space of self-models that vanishes in the fixed-point regime and grades the residual obstruction otherwise.*

Candidate constructions include a stabilized residual-tower index, an enriched categorical index, or a cohomological lifting obstruction. The conjecture may fail; a no-go theorem would be equally informative.

8 Physical tests

A genuine joint-obstruction claim requires an observable unavailable to every declared local reporter but available to a specified global operation, together with robustness under a declared deformation class. Recommended controls are local tomography, nonlocal loop operators, boundary-mode counting, disorder or perturbation tests, ablation of the global relation, and restoration. Post hoc factorization of two marginals is not such a test.

9 Scope

The factorization biconditional, joint-coordinate count, and classical predictive-state lower bound are exact. The topological dictionary is not uniformly exact. Quantization, stability, bulk-boundary exactness, and the proposed reflexive invariant remain open.

10 Graded translation dictionary

The comparison with quantum materials is intentionally graded. The entries are summarized below.

reporter/representation concept	quantum-materials concept	status
self-model over a state carrier	choice of global smooth frame	structural
fixed-point-free twist	nontrivial holonomy	structural
diagonal residual	obstruction to global section	structural
fixed-point/eigenform regime	topologically trivial phase	exact at the stated existence level
nonfactorization of local reports	local indistinguishability plus global dis- tinction	exact
minimal classical future classes	deterministic bond-dimension lower bound	exact
returned-boundary tower	anomaly-inflow hierarchy	structural; conjectural
typed recurrence depth	renormalization scale	heuristic
dissipation ledger	driven-system heating	structural
formation-weight algebra	effective operator algebra	heuristic

A structural entry transfers a question, not a theorem.

11 Exact reporter-boundary theorem

For reporters $K_i : \Sigma \rightarrow Z_i$ and target label $J : \Sigma \rightarrow O$, define

$$\partial_J(K_1, \dots, K_n) = \{(s, t) : K_i(s) = K_i(t) \ \forall i, J(s) \neq J(t)\}. \quad (5)$$

Then

$$\partial_J = \emptyset \iff J = f \circ (K_1, \dots, K_n) \quad (6)$$

for some f . The forward direction defines f on every joint-reporter fibre, on which J is constant; the reverse direction is immediate. A nonempty boundary is therefore exactly local indistinguishability together with global distinguishability.

In a topologically ordered ground-space, local operators satisfy

$$\langle \psi_i | O_{\text{local}} | \psi_j \rangle = c(O) \delta_{ij}, \quad (7)$$

while nonlocal loop operators separate the states. This is a direct realization of the biconditional.

12 Trivial regime and obstruction limits

When the relevant obstruction vanishes, a global section exists. In band theory this corresponds, under the usual hypotheses, to a smooth frame and localized Wannier representation. In the self-model calculus the

fixed-point regime removes the diagonal contradiction. The analogy stops there unless a deformation-stable invariant is constructed.

The topological obstruction is quantized and stable under gap-preserving deformation. The current reflexive obstruction is only Boolean under the supplied fixed-point-free twist. It has no proved integer index and no specified topology on the space of self-models. Consequently Chern-number, bulk-boundary, and anomaly-inflow theorems cannot be imported as proofs.

13 Memory and bond dimension

For a classical stochastic process, histories with the same future distribution form causal states. Their number is the minimal number of deterministic predictive states. A hidden-state or tensor-network realization must carry enough dimension to distinguish those future conditions. Quantum models can sometimes encode the same classical process in nonorthogonal memory states and use a smaller Hilbert-space dimension. The exact transfer is therefore a classical lower bound plus a possible quantum compression, not equality in every model.

14 Joint information missed by marginals

A normalized Hermitian joint state on $d_G \times d_Q$ has $(d_G d_Q)^2 - 1$ real coordinates. Its two normalized marginals provide only $(d_G^2 - 1) + (d_Q^2 - 1)$. The omitted cross-relation space has dimension

$$(d_G^2 - 1)(d_Q^2 - 1). \quad (8)$$

For a 3×3 comparison, 64 real joint coordinates are absent. Compatible marginals therefore do not reconstruct the operative relation.

15 Open problems and tests

A serious topological extension must define a topology or deformation class on self-models, construct an invariant that vanishes in the fixed-point regime, prove stability, and derive a boundary observable. Physical tests should compare local tomography with a specified nonlocal operation, perform perturbation and disorder tests, and use ablation/restoration of the global relation. Post hoc factorization of two marginals does not meet this standard.

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Directional Information Along Alpha-Particle Tracks

A Gas-Ladder Test

Abhijit Singh

Contents

Abstract	3
1 Physical question	4
2 Directional information	4
3 Sequential refinement	4
4 Rare-scattering channel	4
5 Thermodynamic bookkeeping	6
6 Registered experimental design	6
7 Falsification	6
8 Claim status	6
9 Geometric information budget	6
10 Scattering-rate derivation	7
11 Thermodynamic bookkeeping	7
12 Registered experiment	7
13 Required public data products	7
References	7

Abstract

A quantitative record budget is developed for alpha-particle tracks in diffusion cloud chambers. The directional record is separated into an initial localization burst, a finite refinement ramp, a resolution-limited collinear regime, and a rare large-angle scattering channel. The declared model gives 24.6 bits at the first visible droplet, 10.7 additional bits over roughly forty droplets, 35.2 microscopic directional bits, and 24.8 bits at optical resolution. A depth-independent silence parameter follows from paired range and angular-spread scalings. The rare-scattering calculation predicts 0.19, 0.95, and 6.1 visible kinks per track in helium, air, and argon at standard conditions, with approximately 78 percent in the final fifth of each range. A blinded three-gas registered experiment is specified.

Keywords: cloud chamber; alpha track; Mott problem; information theory; scattering; detector experiment

Scope and status: Established physical ingredients, model-derived information values, and untested experimental predictions are explicitly separated. No observed gas ladder is claimed.

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Document status: Research manuscript, version 1.0, August 2026.

1 Physical question

A cloud-chamber track is a sequence of localized records produced by one charged-particle trajectory. The model separates four information regimes:

1. an initial directional localization burst;
2. a finite refinement ramp;
3. a resolution-limited collinear regime with negligible new directional information;
4. a rare large-angle scattering channel producing visible kinks.

The established ingredients are Mott collinearity, scattering, stopping, and detector optics. The integrated information budget and gas ladder are the present model outputs.

2 Directional information

Let Ω_0 be the prior solid angle of the emission direction and Ω_k the posterior angular cell after k visible droplets. Define

$$I_k = \log_2 \frac{\Omega_0}{\Omega_k}. \quad (1)$$

For a first droplet at distance r with transverse uncertainty a , the small-angle solid angle is $\Omega \simeq \pi(a/r)^2$, giving

$$I_1 \simeq \log_2 \frac{4r^2}{a^2}. \quad (2)$$

The declared chamber geometry yields approximately

$$I_1 = 24.6 \text{ bits}. \quad (3)$$

3 Sequential refinement

For independent transverse localization errors with covariance Σ_k , the directional increment is

$$\Delta I_k = \frac{1}{2} \log_2 \frac{\det \Sigma_{k-1}}{\det \Sigma_k}. \quad (4)$$

The model gives a further 10.7-bit ramp over roughly 40 droplets. The total microscopic directional capacity is therefore 35.2 bits, while optical coarse graining retains 24.8 bits.

The silence parameter

$$s = \frac{\ell \sigma_\theta}{a} \quad (5)$$

is depth independent under the paired scaling $\ell \propto \sqrt{E}$ and $\sigma_\theta \propto E^{-1/2}$ used in the source model. The reported value is $s_q = 0.0382$. This is a model identity, not a universal detector constant.

4 Rare-scattering channel

For gas g , angular threshold θ_0 , number density n_g , stopping path $E(x)$, and visibility efficiency η_g , the expected visible-kink count is

$$\lambda_g = \int_0^{R_g} n_g \sigma_g(> \theta_0; E(x)) \eta_g(E(x)) dx. \quad (6)$$

The frozen prediction for a 5.3 MeV alpha track at standard conditions and a 1.7° threshold is

Table 1: Predicted visible kinks per track.

gas	kinks/track	ratio to air
helium	0.19	0.16–0.21
air	0.95	1
argon	6.1	4.3–6.4

Approximately 78 percent of kinks are predicted in the final fifth of the track range, with median position near 94 percent of range.

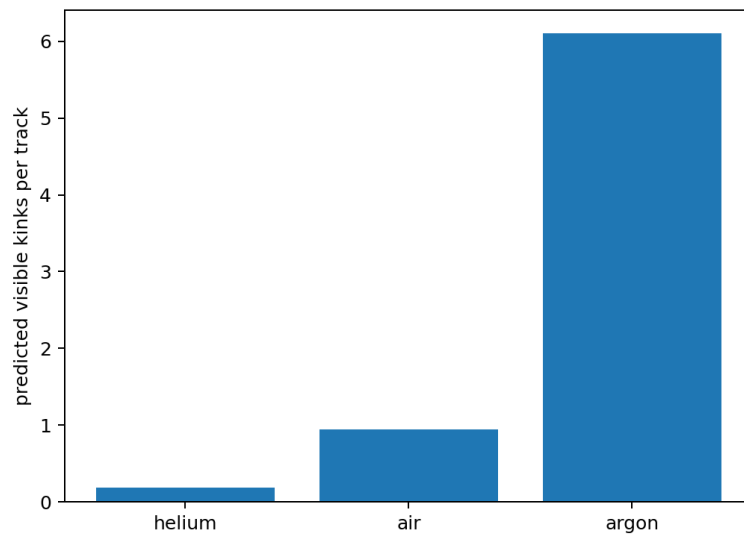


Figure 1: Predicted gas ladder under one frozen threshold and alpha energy.

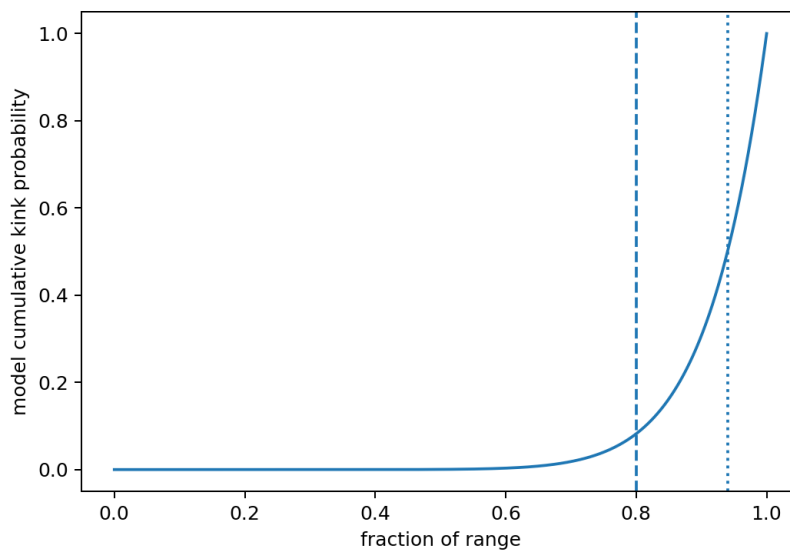


Figure 2: Illustrative cumulative location model calibrated to the reported median; the experimental analysis must use the full stopping calculation rather than this visual guide.

5 Thermodynamic bookkeeping

If a resolved directional record carries I bits and a finite memory is cyclically reset at temperature T , the ideal Landauer floor is

$$Q_{\min} = k_B T \ln 2 I. \quad (7)$$

The source model reports a physical-energy-to-Landauer ratio near 130, with band 120–160. This is a detector/model efficiency ratio, not a constant of track formation.

6 Registered experimental design

A decisive experiment should freeze before data inspection:

1. chamber geometry, alpha energy, gas composition, pressure, and temperature;
2. optical point-spread calibration and the 1.7° kink threshold;
3. track-isolation, edge, overlap, and full-range cuts;
4. a blinded centreline estimator and local-angle procedure;
5. approximately 200 accepted tracks per gas;
6. a Poisson or overdispersed count model;
7. uncertainty from stopping law, density, and optical resolution;
8. an ordered primary alternative $N_{\text{He}} < N_{\text{air}} < N_{\text{Ar}}$;
9. a secondary final-fifth position test.

Analysts should be blinded to gas identity during event marking. Raw images, masks, fitted centrelines, angle series, exclusions, and code should be released.

7 Falsification

A statistically flat gas ladder falsifies the proposed scattering channel. A ladder explained entirely by mismatched density, resolution, or selection bias also rejects the intended mechanism. The directional budget is falsified if independently calibrated angular posteriors do not exhibit the predicted burst and finite saturation.

8 Claim status

The paper contains an exact information definition, a declared detector/stopping calculation, and a sharp prediction. It does not report a completed three-gas experiment. The numerical ladder is therefore presented as a registered model prediction awaiting measurement.

9 Geometric information budget

For an initially isotropic source, a first localized droplet at distance r with transverse uncertainty a restricts the emission direction to solid angle $\Omega \approx \pi(a/r)^2$. Relative to 4π , the directional information is

$$I_1 \approx \log_2 \frac{4\pi}{\Omega} = \log_2 \frac{4r^2}{a^2}. \quad (8)$$

The adopted chamber geometry gives approximately 24.6 bits at the first visible droplet. Subsequent droplets update the angular covariance. If Σ_k is the transverse direction covariance,

$$\Delta I_k = \frac{1}{2} \log_2 \frac{\det \Sigma_{k-1}}{\det \Sigma_k}. \quad (9)$$

The finite refinement ramp contributes approximately 10.7 bits over roughly forty droplets. Once the covariance reaches the optical/collisional floor, additional collinear droplets add negligible resolved direction information. The model total is 35.2 microscopic bits and 24.8 bits after optical coarse graining.

10 Scattering-rate derivation

For gas number density n_g , energy-dependent differential cross section, and fixed visible-angle threshold θ_0 , the expected kink count is

$$N_g = \int_0^{R_g} n_g \int_{\theta > \theta_0} \frac{d\sigma_g(E(x))}{d\Omega} d\Omega \eta_g(E(x)) dx, \quad (10)$$

where η_g is the visibility efficiency. Slowing near the end of range raises large-angle probability, producing the prediction that the median visible kink occurs near 94% of range and roughly 78% occur in the final fifth.

Table 2: Frozen gas-ladder prediction for the declared alpha energy and angular threshold.

gas	kinks/track	ratio to air
helium	0.19	0.16–0.21
air	0.95	1
argon	6.1	4.3–6.4

Two independent stopping approximations agree within 7% for the air estimate in the source calculation.

11 Thermodynamic bookkeeping

If a resolved record carries I bits and a finite memory is reset at temperature T , the ideal erasure floor is $k_B T \ln 2 I$. The source model's physical track energy divided by that floor is approximately 130, with a 120–160 uncertainty band. This is a detector/model efficiency ratio, not a universal constant of quantum measurement.

12 Registered experiment

The primary experiment uses the same chamber geometry, optics, pressure, alpha energy, tracking algorithm, and frozen 1.7° threshold for helium, air, and argon. It should acquire at least 200 accepted full-range tracks per gas, blind analysts to gas labels, and release every exclusion. Primary outcomes are kink count and the ordered alternative helium < air < argon. Secondary outcomes are normalized kink position and the final-fifth fraction.

Controls include matched optical calibration, pressure and density reporting, synthetic-track injection, threshold sensitivity, inter-rater agreement, and a second laboratory. A flat ladder, an optics-driven ladder, or a result that disappears under blinded recoding falsifies the proposed scattering channel.

13 Required public data products

Release raw images or video frames, calibration plates, accepted and rejected track masks, fitted centrelines, local direction series, kink calls, gas metadata, analysis code, and the frozen prediction checksum. The present paper supplies a quantitative model and test; it does not claim that the ladder has already been observed.

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Past-Unbounded Cosmology

An Exact Dust-Vacuum Return Condition

Abhijit Singh

Contents

Abstract	3
1 Past-unbounded histories and typed age	4
2 Flat dust-vacuum branch	4
3 Exact dimensionless predictions	4
4 Comparison with one external parameter estimate	5
5 Scale obstruction	5
6 Past-unbounded recurrent example	6
7 Dark sectors	6
8 Horizons and black holes	6
9 Full-path falsification	6
10 Status	7
11 Typed age and the absence of a universal origin	7
12 Exact FLRW derivation	7
13 Past-unbounded recurrent construction	8
14 Finite-capacity fate theorem	8
15 Dark sectors and horizon scope	8
16 Full observational test	8
References	8

Abstract

A past-unbounded causal history is separated from the finite age of a particular cosmological branch. In a flat dust-plus-vacuum model, acceleration onset and matter-vacuum equality are distinct. Applying a declared complement return to the acceleration boundary gives an exact present vacuum fraction of two thirds. Without fitting within the branch, this condition forces $z_{\text{acc}} = 0.587401052$, $z_{\text{eq}} = 0.259921050$, $H_0 t_0 = 0.9358813101$, $\Delta N = 0.4620981204$, $q_0 = -1/2$, and $j_0 = 1$. A nonsingular recurrent solution demonstrates mathematical consistency of a past-unbounded class. The paper also proves the absolute-scale obstruction, separates cold-matter and slow-roll requirements, and states a full-path falsification protocol. Black-hole discussion is confined to accessibility and known thermodynamic targets.

Keywords: cosmology; dark energy; cosmic age; acceleration; recurrence; horizons

Scope and status: The numerical values are exact conditional consequences of the displayed model and returned-present postulate. They are not an observational fit. Absolute scale, perturbations, and a full likelihood remain open.

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1 Past-unbounded histories and typed age

A causal history is past-unbounded when

$$\forall e \exists e' \quad e' \prec e. \quad (1)$$

This condition is compatible with finite ages for particular thermodynamic or geometric phases. A total age function cannot be both invariant under arbitrary origin shift and increase by one under every update. A past-unbounded model therefore has no unique origin-independent total age, while phase ages remain well defined relative to physical transitions.

2 Flat dust-vacuum branch

Consider a spatially flat Friedmann model with pressureless matter and a constant vacuum term:

$$H(a)^2 = H_0^2 \left(\Omega_{m,0} a^{-3} + \Omega_{\Lambda,0} \right), \quad \Omega_{m,0} + \Omega_{\Lambda,0} = 1. \quad (2)$$

Let $p = \rho_{\Lambda}/(\rho_m + \rho_{\Lambda})$. In e-fold time $N = \log a$,

$$p' = 3p(1 - p). \quad (3)$$

Acceleration begins at $p = 1/3$. The declared complement return maps this boundary to

$$\boxed{p_{\text{ret}} = 2/3.} \quad (4)$$

Identifying the present epoch with the returned event is the model's physical postulate. The consequences below are exact after that postulate.

3 Exact dimensionless predictions

Matter-vacuum equality satisfies $\rho_m = \rho_{\Lambda}$, while acceleration onset satisfies $\rho_m = 2\rho_{\Lambda}$. With $(\Omega_{m,0}, \Omega_{\Lambda,0}) = (1/3, 2/3)$,

$$z_{\text{eq}} = 2^{1/3} - 1 = 0.259921050, \quad (5)$$

$$z_{\text{acc}} = 2^{2/3} - 1 = 0.587401052. \quad (6)$$

The age integral is

$$H_0 t_0 = \int_0^1 \frac{da}{a \sqrt{\Omega_{m,0} a^{-3} + \Omega_{\Lambda,0}}} = \frac{2}{3 \sqrt{\Omega_{\Lambda,0}}} \operatorname{arsinh} \sqrt{\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}}. \quad (7)$$

Therefore

$$\boxed{H_0 t_0 = \sqrt{\frac{2}{3}} \operatorname{arsinh} \sqrt{2} = 0.9358813101.} \quad (8)$$

The logarithmic expansion interval associated with the returned fraction is

$$\boxed{\Delta N = \frac{2}{3} \ln 2 = 0.4620981204.} \quad (9)$$

The present deceleration and jerk are

$$q_0 = \frac{1}{2} \Omega_{m,0} - \Omega_{\Lambda,0} = -\frac{1}{2}, \quad j_0 = 1. \quad (10)$$

The fraction of branch age elapsed since acceleration began is 0.4255192360.

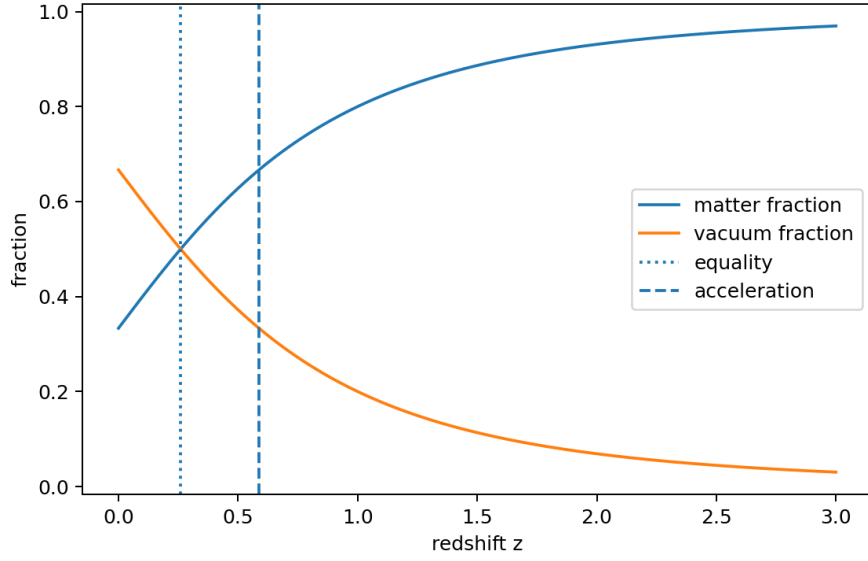


Figure 1: Exact matter and vacuum fractions in the returned dust-vacuum branch.

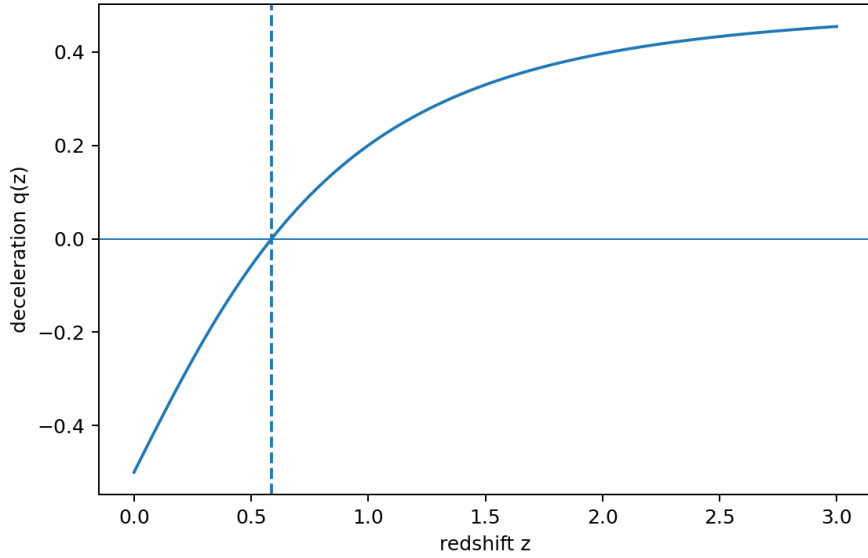


Figure 2: Deceleration parameter; the zero crossing is the exact acceleration redshift.

4 Comparison with one external parameter estimate

Using the Planck 2018 flat- Λ CDM estimate $\Omega_{\Lambda,0} \approx 0.6847 \pm 0.0073$, the value $2/3$ differs by approximately 2.47 quoted standard deviations. This comparison is model dependent and not a full likelihood analysis. It makes the returned-present postulate sharp and already constrained rather than adjustable.

5 Scale obstruction

All results above are dimensionless. The dilation

$$H \mapsto \mu H, \quad t \mapsto t/\mu \quad (11)$$

leaves the dimensionless history invariant. No absolute H_0 , age in years, vacuum energy density, sound horizon, or physical parent scale follows without one additional dimensionful input generated by the model.

A later controlled transmutation map gives an exact conditional relation between a symmetry scale, coupling data, and an absolute age. The physical vacuum formation and its parameters are not yet derived, so the map is retained as a conditional calibration route rather than a numerical prediction.

6 Past-unbounded recurrent example

The equation

$$\ddot{\chi} + \omega^2(\chi - \chi_*) = 0 \quad (12)$$

has $\chi = \chi_* + A \cos(\omega t + \phi)$. Setting $J_g = e^\chi$ gives a positive scale factor

$$a(t) = \exp \left[\frac{\chi_* + A \cos(\omega t + \phi)}{d_s} \right], \quad (13)$$

with alternating expansion and contraction. This proves mathematical consistency of a nonsingular past-unbounded class; it does not identify the observed universe.

7 Dark sectors

A cold dark-matter component must have negligible pressure, cluster gravitationally, and approximately scale as a^{-3} . A coherently oscillating massive scalar can reproduce this background and clustering only when its mass is large relative to the Hubble scale on the relevant epochs.

A late dark-energy component with $w \simeq -1$ requires kinetic energy much smaller than potential energy and an effective mass of order H_0 or below if rolling today. One homogeneous quadratic mode cannot simultaneously satisfy both present-day cold-matter clustering and slow-roll dark energy. A unified origin would require distinct modes, phases, condensates, or a constant vacuum term plus an oscillating component.

8 Horizons and black holes

An event horizon is a boundary of future accessibility,

$$\mathcal{H} = \partial J^-(I^+). \quad (14)$$

Exterior inaccessibility is not identical to global information destruction. A complete microscopic theory must recover

$$S_{\text{BH}} = \frac{k_B A}{4\ell_P^2}, \quad (15)$$

Hawking dynamics, interior continuation, and unitary accounting. The present framework supplies an accessibility interpretation and a two-exponent distinction between residual instability and inaccessible provenance; it does not derive black-hole microstates or evaporation.

9 Full-path falsification

The branch must be tested without retuning the returned fraction. The linked functions

$$H(z), \quad q(z), \quad D_A(z), \quad D_L(z), \quad t(z), \quad f\sigma_8(z) \quad (16)$$

should be evaluated against a frozen external likelihood. Failure of any linked observable rejects the branch as a physical account while leaving the conditional algebra intact.

10 Status

The $2/3$ return and all displayed dimensionless numbers are exact inside the flat dust-vacuum and present-equals-return assumptions. The present-equals-return postulate, absolute scale, perturbation spectrum, sound horizon, and full likelihood remain empirical or open.

11 Typed age and the absence of a universal origin

A causal history is past-unbounded when every event has an earlier event. Such a history can contain finite-duration phases without a first event. An origin-independent total age A cannot both satisfy translation invariance $A(n+k) = A(n)$ for arbitrary origin shifts and increment law $A(n+1) = A(n) + 1$. Hence a past-unbounded model has no unique total age while a hot, smooth, or observational branch can have a finite age measured from a physical transition.

12 Exact FLRW derivation

For flat dust plus a constant vacuum component,

$$E(a)^2 = \frac{H(a)^2}{H_0^2} = \Omega_{m,0}a^{-3} + \Omega_{\Lambda,0}, \quad \Omega_{m,0} + \Omega_{\Lambda,0} = 1. \quad (17)$$

Under the returned-state condition

$$\Omega_{\Lambda,0} = \frac{2}{3}, \quad \Omega_{m,0} = \frac{1}{3}, \quad (18)$$

the matter-vacuum equality condition gives

$$1 + z_{\text{eq}} = 2^{1/3}, \quad (19)$$

and the acceleration condition $\rho_m = 2\rho_\Lambda$ gives

$$1 + z_{\text{acc}} = 2^{2/3}. \quad (20)$$

The age integral is

$$H_0 t_0 = \int_0^1 \frac{da}{aE(a)} = \frac{2}{3\sqrt{\Omega_{\Lambda,0}}} \operatorname{arsinh} \sqrt{\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}}, \quad (21)$$

so

$$H_0 t_0 = \sqrt{\frac{2}{3}} \operatorname{arsinh} \sqrt{2} = 0.9358813101. \quad (22)$$

The exact branch register is

quantity	value
$\Omega_{\Lambda,0}$	$2/3$
z_{acc}	0.587401052
z_{eq}	0.259921050
$H_0 t_0$	0.9358813101
q_0	$-1/2$
j_0	1
ΔN	$(2/3) \ln 2 = 0.4620981204$

These are forced by the branch assumptions rather than fitted independently.

13 Past-unbounded recurrent construction

The equation

$$\ddot{\chi} + \omega^2(\chi - \chi_*) = 0 \quad (23)$$

has $\chi = \chi_* + A \cos(\omega t + \phi)$. Setting $J_g = e^\chi$ and $a = J_g^{1/d_s}$ gives a positive nonsingular scale factor for all t , with alternating expansion and contraction. This proves consistency of one past-unbounded cosmological class; it does not identify the observed universe with that class.

14 Finite-capacity fate theorem

If accessible geometric capacity is finite but $H(t) \geq h > 0$ for all sufficiently late time, then

$$J_g(t) \geq J_g(t_0)e^{d_s h(t-t_0)}, \quad (24)$$

which eventually exceeds every finite bound. Thus finite capacity requires $H \rightarrow 0$, a later sign change, or continuing access to new capacity.

15 Dark sectors and horizon scope

A cold dark-matter candidate must cluster and approximately scale as a^{-3} ; the framework supplies a possible structural role for stable neutral recurrent excitations but no mass, abundance, or cross section. A vacuum component has $p = -\rho$; the two-thirds branch fixes a dimensionless present fraction but does not derive the observed microscopic vacuum scale.

An event horizon is a boundary of future accessibility. Exterior inaccessibility is not identical to global destruction of information. A microscopic theory must recover the area law, its coefficient, Hawking flux, and unitary accounting. The present causal description identifies the boundary problem but does not derive those microphysics.

16 Full observational test

Freeze the 2/3 postulate before comparison. Evaluate the linked functions $H(z)$, $q(z)$, distance measures, age, and growth under a current external likelihood without retuning the return condition. Model comparison must include covariance, priors, and alternatives. A failure of any linked observable rejects the physical branch while leaving the conditional derivation intact.

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Reflexive Predictive Integration

A Testable Systems Model of Conscious Processing

Abhijit Singh

Contents

Abstract	3
1 Functional target	4
2 Same-token causal obstruction	4
3 Minimal predictive state	4
4 Candidate measurable coordinates	4
5 Toy temporal-window result	4
6 Randomized intervention design	5
7 Chronological split	5
8 Blind comparator family	5
9 Rival theories	5
10 Physiological measurements	5
11 Replication ladder and ethics	6
12 Scope	6
13 Formal causal architecture	6
14 Five measurable coordinates	6
15 Generated distinction protocol	6
16 Randomized causal test	7
17 Comparator and rival-theory matrix	7
18 Physiology, replication, and ethics	7
19 Falsification boundary	7
References	7

Abstract

A causal systems model is developed in which memory, world prediction, self-modelling, action, and successor-state updating remain jointly integrated across a moving temporal window. A completed record cannot causally participate in producing the identical token episode in an acyclic model; complete self-description is therefore partial during the episode or available to a successor episode. Histories are grouped by intervention-conditioned future reports to define minimal predictive states. Quantitative observables are proposed for integration gain, reflexive use, temporal depth, counterfactual control, and boundary co-migration. A randomized source-versus-report intervention design, chronological validation protocol, blind comparator family, rival-theory test, and replication ladder are specified. The functional architecture is separated from the empirical claim that it is sufficient for subjective experience.

Keywords: consciousness; predictive processing; self-model; causal integration; temporal continuity; intervention

Scope and status: The causal and predictive-state statements are formal. The link to subjective experience remains an empirical hypothesis requiring preregistered intervention and replication.

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1 Functional target

The paper models a continuing system with memory, world prediction, self-model, action, and feedback. It does not define subjective experience by stipulation. The empirical hypothesis is that a specific class of integrated, recurrent, self-indexed causal organization is necessary for conscious access and predicts intervention effects beyond report or motor preparation.

Let

$$X_t = (M_t, W_t, S_t, P_t, A_t), \quad (1)$$

where M_t is retained memory, W_t a world model, S_t a self-model, P_t a predictive distribution, and A_t an action policy. Updates satisfy

$$X_{t+1} = F(X_t, Y_{t+1}, I_{t+1}). \quad (2)$$

Reflexive use means that variables in S_t describing the system's own policy, confidence, or information limits alter later action.

2 Same-token causal obstruction

Let H be an open macro-process and dH its completed record. In an acyclic causal model, dH cannot participate in producing the identical token closure that emits it. A usable complete record is therefore partial before closure or late, acting on a successor process. This does not prevent later complete self-description; it prevents a completed record from being one of its own causal antecedents.

3 Minimal predictive state

Define intervention-conditioned future equivalence of histories by

$$h \sim h' \iff P(Y_{t+1:\infty} \mid \text{do}(a_{t:\infty}), h) = P(Y_{t+1:\infty} \mid \text{do}(a_{t:\infty}), h') \quad (3)$$

for every admitted intervention sequence. The equivalence class is the minimal predictive state. A reflexive state must additionally retain which current self-model distinctions change later action under intervention.

4 Candidate measurable coordinates

Define:

$$\mathcal{I}_t = \mathcal{L}_{\text{split}} - \mathcal{L}_{\text{whole}}, \quad \text{integration gain}, \quad (4)$$

$$\mathcal{R}_t = D(P_{\text{future}}^{\text{intact}}, P_{\text{future}}^{\text{self-model ablated}}), \quad \text{reflexive-use effect}, \quad (5)$$

$$\mathcal{T}_t = \min\{k : \text{history depth } k \text{ reaches frozen risk}\}, \quad \text{temporal depth}, \quad (6)$$

$$C_t = I(U_t; Y_{t+1:H} \mid S_t), \quad \text{counterfactual control}, \quad (7)$$

$$\mathcal{B}_t = D(S_t, S_{t+1} \mid \text{matched boundary perturbation}), \quad \text{boundary co-migration}. \quad (8)$$

No one scalar is declared sufficient. The hypothesis concerns a pattern across causal integration, reflexive use, temporal depth, and boundary response.

5 Toy temporal-window result

In the source toy model, a temporal-window score rises from 0.209 at window 1 to 0.234 at window 5 and 0.239 at window 15, while a naive all-level integration score changes from 0.242 to 0.234 to 0.230. The result shows only that one fixed averaging rule can deteriorate when the window expands; it does not calibrate consciousness.

6 Randomized intervention design

Use a factorial design with source-side manipulation I_S and report-side manipulation I_R . Source interventions alter perceptual evidence before category formation. Report interventions randomize report-key mapping and delay while leaving source evidence unchanged.

The causal estimands include

$$ATE_S = \mathbb{E}[Y \mid \text{do}(I_S = 1)] - \mathbb{E}[Y \mid \text{do}(I_S = 0)], \quad (9)$$

plus interactions between source manipulation and a participant-generated distinction Z defined on development trials.

A valid Z must:

1. improve held-out prediction beyond old task variables;
2. transfer to a later independent decision or action;
3. have a pre-report physiological signature not reducible to motor preparation.

7 Chronological split

Use development trials to discover Z , a re-entry/contact block to test whether Z changes future behaviour, and an untouched final block for confirmatory analysis. Freeze the category definition, exclusions, physiological window, report delay, stopping rule, and statistical model before final data are opened.

8 Blind comparator family

Compare the proposed distinction against:

1. old task variables only;
2. experimenter labels;
3. reaction time and confidence;
4. shuffled participant labels;
5. balance- and complexity-matched random partitions;
6. report-motor predictors;
7. flexible post hoc models reported only as exploratory upper bounds.

9 Rival theories

After the measurement relation replicates, competing theories should submit incompatible preregistered predictions. Global-workspace models emphasize access-related broadcasting; higher-order models emphasize metarepresentational variables; recurrent-processing models emphasize local recurrent completion; integrated-information models require a separately committed system-level quantity. The present architecture succeeds only if its intervention pattern outperforms these committed alternatives on held-out data.

10 Physiological measurements

Candidate channels include electrophysiology, pupillometry, and autonomic signals. The primary window must precede overt report preparation. Multivariate decoding is trained only on development data. Report-key randomization prevents a fixed motor path from defining the effect.

11 Replication ladder and ethics

Promotion requires within-participant held-out success, a new-participant replication, and an outside-laboratory replication. A result that predicts report but not independent action or pre-report physiology is classified as a report model. Participant-generated categories may contain sensitive content; only the minimum necessary data should be stored, identifiers separated, and no individual diagnosis inferred.

12 Scope

The formal causal and predictive-state statements are exact in their declared models. The identification of the architecture with subjective experience is an empirical hypothesis. The current work does not prove that every implementation is conscious, that consciousness is substrate independent, or that the displayed coordinates are unique.

13 Formal causal architecture

Let

$$X_t = (M_t, W_t, S_t, P_t, A_t), \quad (10)$$

where M_t is retained memory, W_t an external-world model, S_t a self-model, P_t a predictive distribution, and A_t a policy. The transition

$$X_{t+1} = F(X_t, Y_{t+1}, I_{t+1}) \quad (11)$$

is reflexive when variables in S_t describing the system's own policy, confidence, and information boundary causally alter later prediction or action. This is a functional architecture, not a definition that proves phenomenal experience.

14 Five measurable coordinates

Candidate observables are

$$\mathcal{I}_t = \mathcal{L}_{\text{split}} - \mathcal{L}_{\text{whole}}, \quad \text{integration gain,} \quad (12)$$

$$\mathcal{R}_t = D(P_{\text{future}}^{\text{intact}}, P_{\text{future}}^{\text{self-model ablated}}), \quad \text{reflexive-use effect,} \quad (13)$$

$$\mathcal{T}_t = \min\{k : \text{history depth } k \text{ attains frozen risk}\}, \quad \text{temporal depth,} \quad (14)$$

$$C_t = I(U_t; Y_{t+1:H} \mid S_t), \quad \text{counterfactual control,} \quad (15)$$

$$\mathcal{B}_t = D(S_t, S_{t+1} \mid \text{matched boundary perturbation}), \quad \text{boundary co-migration.} \quad (16)$$

No single coordinate is declared sufficient. The source toy calculation reports temporal-window scores 0.209, 0.234, and 0.239 for windows 1, 5, and 15, while one naive integration average decreases across its tested settings. These values motivate a discriminating experiment; they are not biological measurements.

15 Generated distinction protocol

During a repeated perceptual task, each participant generates a distinction Z not fixed by the experimenter's initial labels. Z is formed on development trials, frozen, and evaluated on an untouched segment. A candidate is accepted only if it predicts an independent later action or report beyond old task variables and has a pre-report physiological correlate not reducible to motor preparation.

16 Randomized causal test

Use a factorial source intervention I_S and report intervention I_R . I_S changes perceptual evidence before category formation. I_R randomizes report-key mapping and delay while leaving evidence unchanged. The primary estimand is

$$ATE_S = \mathbb{E}[Y \mid \text{do}(I_S = 1)] - \mathbb{E}[Y \mid \text{do}(I_S = 0)], \quad (17)$$

together with the interaction between I_S and frozen Z . If the physiological signature follows report mapping rather than source manipulation, the proposed pre-report route is rejected.

17 Comparator and rival-theory matrix

The frozen distinction must beat old task variables, experimenter labels, reaction time, confidence, shuffled labels, complexity-matched random partitions, and motor predictors. After the measurement relation replicates, competing theories must commit to incompatible predictions: broadcasting and access for global-workspace models; metarepresentational dependence for higher-order models; local recurrent completion for recurrent-processing models; and a separately fixed system quantity for integrated-information approaches.

18 Physiology, replication, and ethics

Candidate channels include electrophysiology, pupillometry, and autonomic signals in a preregistered interval preceding overt response. Report-key randomization prevents fixed motor decoding. Promotion requires within-participant held-out success, new-participant replication, and outside-laboratory replication. Participant-generated distinctions may be sensitive; retain minimum data, separate identities, and prohibit individual diagnosis from exploratory scores.

19 Falsification boundary

A result that predicts only the overt report is a report model. A result that disappears after motor and confidence controls does not support reflexive integration. A result that fails source-side intervention or cross-person replication is rejected. Even a successful causal marker would establish a processing relation, not a universal criterion of moral status or experience.

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Adaptive Predictive Systems

Native Representation Repair

Abhijit Singh

Contents

Abstract	3
1 Objective	4
2 System state	4
3 Hybrid computation	4
4 Predictive-state compiler	5
5 Boundary detection and experiment planning	5
6 Validated schema repair	5
7 Independent representations and relation-aware routing	6
8 Future-tested memory compression	6
9 Training objective	6
10 Collision benchmark	6
11 Serving policy and speed	6
12 Nine-state relation-closure benchmark	7
13 Evaluation	7
14 Development status	7
15 Complete runtime architecture	7
16 Module contracts	8
17 Inference cycle	8
18 Relation-aware routing	8
19 Future-tested memory	9
20 Training objective and curriculum	9
21 Collision benchmark	9
22 Speed and safety	9
23 Revolutionary claim and evidence gate	9
References	9

Abstract

A model-agnostic computational architecture is proposed for systems that can detect when their current state representation is insufficient, acquire a separating observation, refine the representation conservatively, verify the change by ablation and restoration, and continue without losing source provenance. The architecture combines a recurrent long-context backbone, sparse attention, routed specialist modules, a source-addressed event graph, a minimal predictive-state compiler, relation-aware disagreement routing, active experiment planning, future-tested memory compression, and a validated schema-repair engine. A collision benchmark spanning provenance, time, role, relation, tool version, and history depth raises exact continuation from 6/12 under a coarse report to 12/12 after essential-coordinate repair. The large-scale performance claim remains an evaluation target. A new 36-relation benchmark tests detection, separator acquisition, conservative repair, ablation, and restoration.

Keywords: predictive state; representation repair; provenance; active learning; adaptive systems; continual learning

Scope and status: The finite repair mechanisms and supplied benchmark are executable. The full hybrid architecture is a research design whose performance must be established by independent evaluation.

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1 Objective

The architecture targets a structural failure of fixed-state systems: two histories may receive one current representation although a later allowed intervention requires different actions. More computation inside the same quotient cannot restore a distinction that the quotient erased. The system must be able to detect representation failure, acquire a separator, refine its state conservatively, and continue from the repaired representation.

2 System state

At step t carry

$$S_t = (z_t, Q_t, \Pi_t, B_t, \Gamma_t, \Lambda_t, K_t), \quad (1)$$

where z_t is fast working state, Q_t a compiled predictive quotient, Π_t an append-only source graph, B_t unresolved future-separating boundaries, Γ_t relations among parallel hypotheses, Λ_t available interventions, and K_t verified cached transitions.

The primary prediction is

$$P(y_t, S_{t+1}, \Delta_t \mid S_t, a_t, \Pi_t), \quad (2)$$

not merely a distribution over the next symbol. Every answer is coupled to its successor state, source support, scope, uncertainty, counterconditions, and unresolved residuals.

3 Hybrid computation

A practical system combines:

1. selective recurrent state-space blocks for long-context throughput;
2. sparse attention for exact local and retrieved interactions;
3. routed expert modules for semantic, causal, formal, spatial, and operational tasks;
4. graph message passing over provenance and hypothesis relations;
5. an explicit predictive-state compiler and boundary detector.

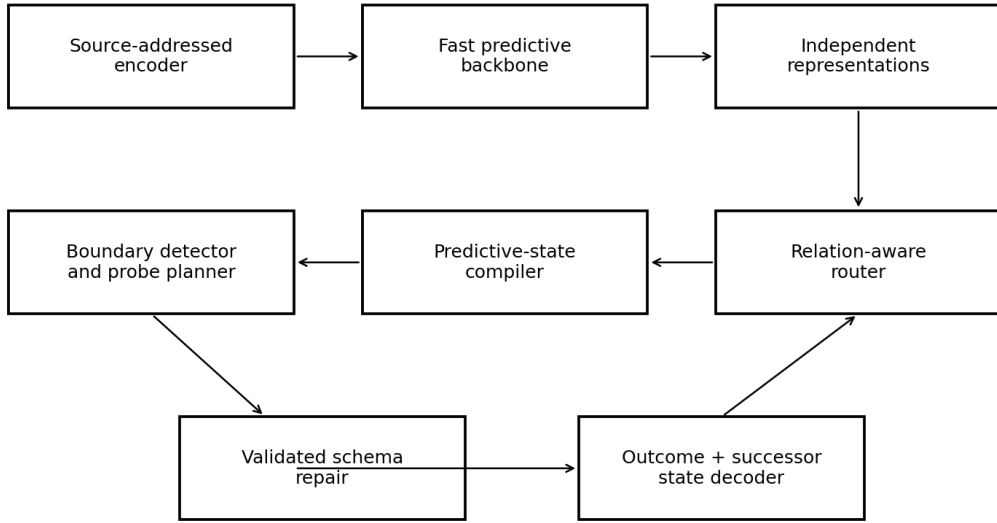


Figure 1: Reference architecture for source-preserving predictive repair.

4 Predictive-state compiler

For history h and intervention family Λ , define the approximate continuation signature

$$\Sigma_{\Lambda}(h) = \{P(Y_{1:H} \mid \text{do}(\lambda), h) : \lambda \in \Lambda\}. \quad (3)$$

Merge histories only when the divergence of their signatures is below a declared tolerance. The compiler seeks the smallest state preserving the futures for which the system is responsible.

5 Boundary detection and experiment planning

A boundary is active when one report fibre contains future-distinct histories. Choose a separating operation by

$$\lambda^* = \arg \max_{\lambda \in \Lambda} \frac{\mathbb{E}[\text{future separation from } \lambda]}{\text{cost}(\lambda)}. \quad (4)$$

A separator can be a clarifying question, retrieval, code execution, theorem check, simulation, new sensor reading, or controlled counterfactual.

6 Validated schema repair

A new coordinate is admitted by

$$\tilde{U}(h) = (U(h), \eta(h)), \quad \pi_1 \tilde{U} = U. \quad (5)$$

Admission requires source provenance, improvement on frozen continuation tests, matched ablation, restoration, backward compatibility, and rollback. A written memory or metadata field that never changes a later operation is not a repair.

7 Independent representations and relation-aware routing

Upper representations are trained with partially decorrelated error structures. A relation model observes predictions, uncertainty, source dependence, agreement, contradiction, and prior contact performance. It may select, synthesize, preserve alternatives, or request a separator. The source-reported 97.78 percent result motivates relation-aware routing; the failed higher-arity rescue prevents the architecture from treating relation count as a universal solution.

8 Future-tested memory compression

Working memory, predictive-state memory, exact episodic provenance, and unresolved-boundary memory are separate. A feature can be deleted only when its removal leaves the declared future distribution invariant. The chronological 13-channel study is a direct regression test for this rule.

9 Training objective

A native objective can combine

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \alpha \mathcal{L}_{\text{world}} + \beta \mathcal{L}_{\text{closure}} + \gamma \mathcal{L}_{\text{source}} + \delta \mathcal{L}_{\text{relation}} + \varepsilon \mathcal{L}_{\text{repair}} + \mu \mathcal{L}_{\text{verify}} + \nu C_{\text{compute}}. \quad (6)$$

The closure loss penalizes merged states with divergent future signatures; a complexity penalty prevents retention of every historical difference.

10 Collision benchmark

The supplied reference benchmark contains six collision families: source identity, temporal phase, causal role, relation compatibility, tool version, and history depth. Twelve source cases share six coarse reports. A first-answer coarse baseline is correct on 6 of 12. The essential-coordinate compiler selects coordinates (0, 2) and raises exact continuation to 12 of 12. Removing either selected coordinate reintroduces a collision. Two benchmark tests pass.

Table 1: *Reference collision benchmark.*

system	correct	total
coarse report baseline	6	12
repaired predictive state	12	12

11 Serving policy and speed

Routine queries use a fast path with cached predictive states, few active experts, and early exit. Deep computation is activated by disagreement, source conflict, long-horizon planning, or a detected boundary. Schema repair is rarer still and runs only after evidence shows that the state vocabulary itself is insufficient. Speed therefore comes from minimal future-sufficient states and conditional escalation, not from treating every query as maximal search.

12 Nine-state relation-closure benchmark

A dedicated benchmark instantiates the complete repair cycle on the nine-state carrier. The source has one unit state and four two-sided prime sectors. Its 36 pair relations decompose into eight unit, four return, and twenty-four transverse cells. The coarse reporter erases the side coordinate; the return continuation flips that coordinate while preserving the visible prime label.

A correct system must:

1. detect that the stationary visible report hides a period-two source recurrence;
2. select the side coordinate as a shortest separator for the declared role-sensitive future;
3. extend the state conservatively from q to (q, ε) ;
4. preserve all old value-only answers;
5. fail the role-sensitive task under ablation and recover it under restoration;
6. distinguish the finite static Möbius classes from the all-prime support-address quotient.

The supplied standard-library verifier checks the finite relation count, return-generator commutation, static false-closure counterexample, support-state update, and Fourier-magnitude control. This is a compact benchmark for native representation repair rather than a claim of general system superiority.

13 Evaluation

Standard language, code, mathematics, and multimodal benchmarks remain necessary. Architecture-specific tests should measure:

- false-closure detection;
- source collision and role reversal;
- separator efficiency;
- oracle-gap routing;
- representation restoration;
- dynamic ontology and tool birth;
- re-entry ablation;
- memory compaction;
- time to verified answer.

A headline systems measure can report verified task utility per latency and active compute. Raw accuracy must remain visible separately.

14 Development status

The predictive quotient and finite repair mechanisms are implemented in the supplied reference code. The large-scale hybrid architecture is a systems design, not a demonstrated frontier model. Performance superiority must be established by independent benchmark comparison. The durable contribution is the representation-repair kernel and benchmark category, not a parameter-count claim.

15 Complete runtime architecture

The runtime separates proposal generation from state authority. Replaceable sequence, code, vision, causal, and formal models propose candidate continuations. An explicit kernel controls source addresses, predictive states, intervention planning, verification, and schema migration.

The operative state is

$$\mathcal{S}_t = (z_t, Q_t, \Pi_t, B_t, \Gamma_t, \Lambda_t, K_t), \quad (7)$$

where z_t is fast neural working state, Q_t the compiled predictive quotient, Π_t the provenance graph, B_t unresolved boundaries, Γ_t relations among parallel hypotheses, Λ_t allowed interventions, and K_t verified caches.

16 Module contracts

module	required contract
source encoder	content representation plus immutable occurrence address, scope, and parent links
predictive compiler	minimal quotient, closure certificate, and counterexample separators
relation router	agreement, contradiction, shared-source dependence, prior performance, and branch preservation
boundary detector	pairs of histories merged now but divergent under an admitted future
probe planner	maximum expected separation per cost, latency, and permission
schema engine	conservative coordinate addition, migration, backward projection, and rollback
verifier	source, proof, execution, calibration, adversarial, and regression checks
decoder	answer, confidence class, source support, successor state, and unresolved conditions

17 Inference cycle

```

encode request with source occurrence
run independent proposal representations
construct relation and provenance graph
compile the smallest future-sufficient state
if no collision:
    use cached fast path and emit successor state
else:
    select the lowest-cost separating intervention
    execute only an authorized probe
    refine the state conservatively
    run ablation, restoration, and backward tests
    commit or roll back the schema change

```

The answer and successor state are one operation:

$$P(y_t, S_{t+1}, \Delta_t \mid S_t, a_t, \Pi_t). \quad (8)$$

18 Relation-aware routing

Independent high-level representations are deliberately not forced into complete agreement. The relation model observes predictions, confidence, source overlap, contradictions, and past contact performance. It may select one proposal, synthesize compatible proposals, preserve incompatible branches, or request a separator.

The 97.78% held-out result motivates this relation layer, while the failed higher-arity rescue prevents the assumption that more experts automatically solve the remaining cases.

19 Future-tested memory

Working memory, minimal predictive state, exact episodic provenance, and unresolved-boundary memory remain separate. A feature is compressed only when its ablation leaves every declared future distribution unchanged. The chronological 13-channel restoration supplies a direct regression test: present reconstruction was not sufficient to justify deletion.

20 Training objective and curriculum

The objective is

$$\mathcal{L} = \mathcal{L}_{\text{sequence}} + \alpha \mathcal{L}_{\text{intervention}} + \beta \mathcal{L}_{\text{false closure}} + \gamma \mathcal{L}_{\text{provenance}} + \delta \mathcal{L}_{\text{relation}} + \varepsilon \mathcal{L}_{\text{repair}} + \mu \mathcal{L}_{\text{verification}} + \nu \mathcal{C}_{\text{compute}}. \quad (9)$$

The curriculum proceeds from broad modelling to source collisions, hidden future separators, independent representations, relation routing, mid-episode rule changes, tool planning, and continual repair with backward tests.

21 Collision benchmark

Every benchmark case contains an initially plausible coarse report, two source states with different valid futures, one or more separators, distractor coordinates, a required schema change, ablation/restoration, and a virgin continuation set. Families cover provenance, temporal phase, causal role, relation type, tool version, and history depth. The supplied finite fixture has 12 cases: the coarse reporter solves 6, while the promoted basis solves all 12. The two test functions verify improvement and essentiality of every selected coordinate.

A full evaluation reports standard capability alongside collision detection, separator minimality, future accuracy, old-result preservation, provenance, calibration, latency, active parameters, and tool cost. Scale-only baselines and sham-memory ablations are mandatory.

22 Speed and safety

Routine tasks use a fast recurrent/sparse path with cached predictive states and early exit. Deep reasoning activates only under disagreement, source conflict, proof requirements, or a detected collision. Schema updates are rarer and require a signed migration, permission boundary, rollback, and backward regression. External consequential action remains user-authorized; the kernel may propose, simulate, and verify but does not infer unlimited authority from successful local tests.

23 Revolutionary claim and evidence gate

The architectural novelty is precise: the system can represent its current state space as a possible cause of error, seek the missing distinction, validate a conservative repair, and continue from the repaired state. That is a different failure class from ordinary search for a better output inside a fixed representation. Superiority in speed, accuracy, or scientific discovery remains an empirical system claim and is accepted only after independent benchmark and ablation evidence.

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Prime-Indexed Return Sectors

The Thirty-Six-Relation Closure of a Nine-State Carrier

Abhijit Singh

Contents

Abstract	3
1 Occurrence-sensitive prime relation	4
2 Terminal lattice and affine return	4
3 Thirty-six-relation closure theorem	5
4 Observer quotient and critical recurrence	5
5 Prime selection and finite spectral labels	6
6 Prime-occupation refinement	6
7 Harmonic realization and scale boundary	7
8 Relation to the earlier nine-transformation closure	7
9 Integration with the wider programme	8
10 Verification and claim boundary	8
References	8

Abstract

A finite observer-closure theorem is developed from an occurrence-sensitive prime relation. The two appearances of a prime in $p \cdot 1 = p$ have equal numerical value but different source and return roles. A value-only reporter turns their directed relation into a self-loop; an occurrence-sensitive reporter retains a period-two return. On the decimal terminal carrier $\mathbb{Z}_{10}$, the zero and unit are terminal roles, while the eight nonterminal indices form four prime/composite pairs under the fixed-point-free affine involution $\sigma(d) = 1 - d \pmod{10}$. Removing the zero terminal leaves nine active states. Their complete unordered relation frontier contains $\binom{9}{2} = 36$ cells and decomposes exactly into eight unit relations, four return-sector relations, and twenty-four transverse relations. The four prime labels 2, 3, 5, 7 are independently selected by transparent arity. Together with a side label they form complete commuting spectral coordinates on the finite carrier; the four return planes commute inside the 36-dimensional antisymmetric relation space. The finite prime/composite reporter is then refined by prime-occupation coordinates: 6 is squarefree coupling, whereas 4, 8, 9 are repeated-prime states. The resulting static five-class reporter is proved insufficient under arbitrary prime multiplication, and the all-scale predictive quotient is the finite prime-support address plus one absorbing repeated-prime state. The theorem is related to the earlier nine-transformation, thirty-six-witness closure, recurrence selection, quotient closure, harmonic realization, and arithmetic completion without identifying merely isomorphic carriers as the same physical system.

Keywords: prime arity; predictive quotient; occurrence identity; relation lattice; recurrence exponent; finite spectral labels

Scope and status: The carrier, involution, 36-cell decomposition, reporter-closure, critical-return, occupation-state, and finite predictive-quotient results are exact. The term quantum number is used first in its mathematical spectral-label sense. A physical realization requires independently specified observables and dynamics.

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DOI:

1 Occurrence-sensitive prime relation

For a prime p , ordinary arithmetic writes

$$p \cdot 1 = p. \quad (1)$$

The two tokens labelled p have one value but two roles. Write

$$p_{\text{source}} \cdot 1_{\text{unit}} = p_{\text{return}}, \quad v(p_{\text{source}}) = v(p_{\text{return}}) = p, \quad p_{\text{source}} \neq_{\text{occ}} p_{\text{return}}. \quad (2)$$

The fourth component is the relation-establishing operation and its retained mark. It is not another numerical factor.

Proposition 1.1 (Factor-complete prime triad). *For an integer $n > 1$, the reduced triad $(n_{\text{source}}, 1_{\text{unit}}, n_{\text{return}})$ contains every positive factor relation of n if and only if n is prime.*

Proof. If n is prime, every positive factor pair is $(1, n)$ or $(n, 1)$. If n is composite, some $1 < a, b < n$ satisfy $ab = n$, and that relation is absent from the reduced triad. $\square$

The proposition is deliberately modest. Its significance lies in reporter design: a terminal primality report is complete for the declared positive-factor query, while a generative prime-power or multiplication query requires occurrence and multiplicity coordinates.

2 Terminal lattice and affine return

Let

$$D = \mathbb{Z}_{10}, \quad T = \{0, 1\}, \quad I = \{2, 3, 4, 5, 6, 7, 8, 9\}, \quad A = D \setminus \{0\}. \quad (3)$$

Define

$$\sigma(d) = 1 - d \pmod{10}. \quad (4)$$

It has no fixed point on D and has the five orbits

$$(0, 1), (2, 9), (3, 8), (4, 7), (5, 6). \quad (5)$$

The restriction to I pairs every prime index with one composite index. Put

$$P = \{2, 3, 5, 7\}, \quad C = \{4, 6, 8, 9\}. \quad (6)$$

For $q \in P$, define $\bar{q} = \sigma(q)$ and encode the interior as

$$I \cong P \times \{+, -\}, \quad (q, +) = q, \quad (q, -) = \bar{q}. \quad (7)$$

Then

$$\sigma(q, \varepsilon) = (q, -\varepsilon). \quad (8)$$

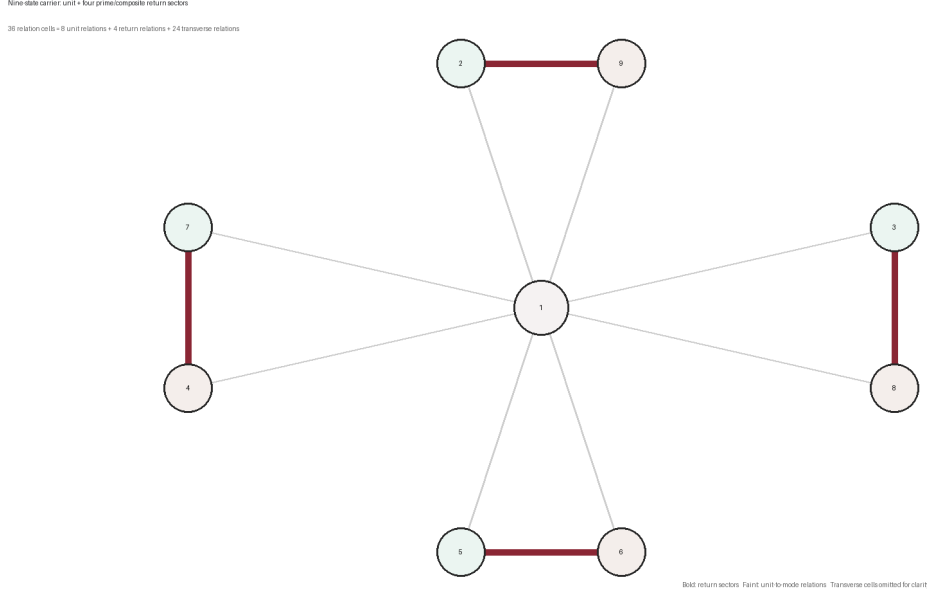


Figure 1: The active nine-state carrier. Bold edges are the four prime/composite return sectors. The eight unit relations are shown faintly; the twenty-four transverse relation cells complete the rank-two frontier.

3 Thirty-six-relation closure theorem

Let

$$\mathcal{R}_2(A) = \{\{x, y\} : x, y \in A, x \neq y\} \quad (9)$$

be the complete unordered pair-relation frontier.

Theorem 3.1 (Nine-state, thirty-six-relation closure). *The active carrier has nine states and*

$$|\mathcal{R}_2(A)| = \binom{9}{2} = 36. \quad (10)$$

The frontier decomposes canonically as

$$36 = 8_{\text{unit}} + 4_{\text{return}} + 24_{\text{transverse}}. \quad (11)$$

The four return relations are $\{q, \bar{q}\}$ for $q \in P$. The transverse relations connect different return sectors.

Proof. There are eight relations from the unit state 1 to the eight interior states. There are four within-sector relations. For each of the $\binom{4}{2} = 6$ pairs of two-element sectors, there are $2 \cdot 2 = 4$ cross-sector relations, giving 24. The classes are disjoint and exhaustive. $\square$

This is more accurately a 36-cell rank-two relation layer than a 36-state carrier. Each cell is one possible distinction between two of the nine active states.

4 Observer quotient and critical recurrence

Define the side-forgetting reporter

$$U(1) = 1, \quad U(q, \varepsilon) = q. \quad (12)$$

Let the return map be

$$R(1) = 1, \quad R(q, \varepsilon) = (q, -\varepsilon). \quad (13)$$

Then $U \circ R = U$, although R has period two on every interior sector.

Theorem 4.1 (Apparent halt versus retained return). *For every $q \in P$, the complete state alternates*

$$(q, +) \leftrightarrow (q, -), \quad (14)$$

while the quotient report remains constant. If $r_q = e_{q,+} - e_{q,-}$ and the residual propagator is the return flip, then

$$Rr_q = -r_q, \quad \|R^n r_q\| = \|r_q\|, \quad (15)$$

and therefore $\chi(r_q) = 0$ for every linear intrinsic clock.

Proof. The first statement is immediate. The residual changes sign under the swap and therefore has constant norm. Its logarithmic growth rate is zero. $\square$

Thus a reporter that stops when its visible output repeats records a halt; a reporter that preserves occurrence sees recurrence. The equality $U(r_q) = 0$ is report-level erasure, not dynamical assimilation. If an admitted future distinguishes the two sides, the complete-behaviour closure criterion forces the refined state (q, ε) .

5 Prime selection and finite spectral labels

Transparent arity is defined by

$$\binom{q}{k} \equiv 0 \pmod{q} \quad (0 < k < q). \quad (16)$$

The transparent arities are exactly the primes. Hence 2, 3, 5, 7 are not selected by the decimal drawing; they are the first four transparent nonterminal arities in the finite carrier.

Let

$$\mathcal{H}_A = \mathbb{C}|1, 0\rangle \oplus \bigoplus_{q \in P, \varepsilon = \pm 1} \mathbb{C}|q, \varepsilon\rangle. \quad (17)$$

Define commuting diagonal operators

$$\widehat{Q}|q, \varepsilon\rangle = q|q, \varepsilon\rangle, \quad \widehat{S}|q, \varepsilon\rangle = \varepsilon|q, \varepsilon\rangle. \quad (18)$$

The pair (Q, S) is a complete finite label. In this exact mathematical sense it is a system of quantum numbers: joint eigenvalues of commuting operators. Physical quantum numbers require a specified physical observable and transition law; the finite labelling theorem alone does not provide them.

The full oriented pair-relation space is

$$\bigwedge^2 \mathbb{R}^9 \cong \mathfrak{so}(9), \quad \dim \bigwedge^2 \mathbb{R}^9 = 36. \quad (19)$$

For each return sector define

$$J_q = E_{q, \bar{q}} - E_{\bar{q}, q}. \quad (20)$$

The four J_q act on disjoint two-planes and therefore commute. This is a finite relation representation, not a derived physical $SO(9)$ gauge theory.

6 Prime-occupation refinement

Unique factorisation gives commuting occupation labels

$$N_p|d\rangle = v_p(d)|d\rangle, \quad p \in \{2, 3, 5, 7\}. \quad (21)$$

The finite states are therefore separated as follows.

index	(n_2, n_3, n_5, n_7)	type
1	(0, 0, 0, 0)	unit
2, 3, 5, 7	one unit occupation	primitive prime mode
6	(1, 1, 0, 0)	squarefree coupling
4, 8, 9	a repeated occupation	repeated-prime state

Accordingly the coarsest terminal-resolved static Möbius reporter on the digit carrier has five classes

$$\{0\}, \quad \{1\}, \quad \{2, 3, 5, 7\}, \quad \{6\}, \quad \{4, 8, 9\}. \quad (22)$$

Theorem 6.1 (Static five-class false closure). *The five-class reporter is not a continuation congruence when multiplication by arbitrary primes is admitted. The coarsest all-prime predictive quotient for future Möbius reports consists of one state for every finite squarefree prime-support set and one absorbing repeated-prime state.*

Proof. The primes 2 and 3 share the static prime class. Multiplication by 2 sends them to 4 and 6, whose Möbius reports are 0 and +1. Hence the static quotient fails. For the full quotient, represent a squarefree integer by its finite prime support and every nonsquarefree integer by $\dagger$. Multiplication by p adjoins p when absent and sends the state to $\dagger$ when present; $\dagger$ is absorbing. Distinct supports are future-distinguishable: choose a prime in their symmetric difference and multiply by it. One state repeats that prime and the other does not. Therefore no coarser reporter preserves every prime-multiplication future. $\square$

This theorem locates the limit of the finite 36-cell picture. It is a complete finite relation frontier, but the all-prime predictive state is necessarily unbounded in the number of possible support addresses.

7 Harmonic realization and scale boundary

For any positive fundamental f_0 , define

$$f_d = f_0 d, \quad d = 0, 1, \dots, 9. \quad (23)$$

At $f_0 = 111$ Hz the coefficients appear as decimal repdigits. The prime-indexed and composite-indexed modes are

$$111\{2, 3, 5, 7\} \quad \text{and} \quad 111\{4, 6, 8, 9\}. \quad (24)$$

Their indicator sequences are affine reflections and have identical cyclic Fourier magnitudes. Equal-amplitude multitone realizations are equal-energy and orthogonal over the common period $1/f_0$. The structure is invariant under positive rescaling; therefore it does not select 111 Hz as a natural constant, nor does the arithmetic alone imply a biological or therapeutic effect.

8 Relation to the earlier nine-transformation closure

A separate finite construction generated nine causally distinguishable whole-carrier transformations and an exact pairwise witness for each of their 36 unordered pairs. Both systems therefore have a complete relation frontier isomorphic to the edge set of K_9 .

Proposition 8.1 (Relation-frontier isomorphism and its limit). *Any bijection between two nine-element carriers induces an isomorphism of their 36-cell unordered relation frontiers. It preserves operational meaning only when it also intertwines the declared continuations and source provenance.*

Proof. A bijection maps unordered pairs bijectively. Graph isomorphism alone says nothing about continuation responses. Those are preserved exactly when the vertex map commutes with the relevant operations and reporters. $\square$

Thus the two nine-state constructions reinforce the role of complete pairwise distinction, but combinatorial isomorphism is not used as evidence that their vertices or physical meanings are identical.

9 Integration with the wider programme

The theorem joins the earlier work in one precise chain:

$$\text{occurrence} \rightarrow \text{relation} \rightarrow \text{report} \rightarrow \text{future test} \rightarrow \text{assimilation, retention, or promotion.} \quad (25)$$

The prime triad supplies the smallest occurrence-sensitive example. The 36-cell frontier enumerates every pair distinction on the nine active states. The recurrence exponent distinguishes genuine assimilation from report-level erasure. Prime transparency selects the sector labels. Occupation coordinates refine the composite class. The all-prime quotient shows that a finite reporter can be exact for one envelope and false for a larger one. The adaptive-system architecture turns this sequence into an executable repair protocol.

10 Verification and claim boundary

The ancillary verifier checks the affine involution, the four prime/composite pairs, the $8 + 4 + 24$ relation decomposition, commutation of the four return-plane generators, Fourier-magnitude equality, the static five-class counterexample, and deterministic prime-support updates.

The paper does not claim that nature has exactly nine states, that $SO(9)$ is a physical gauge group, that 2, 3, 5, 7 are measured physical quantum numbers, that 111 Hz is fundamental, or that the finite carrier proves the Riemann Hypothesis. It establishes the finite theorem, the exact predictive-closure correction, and the conditions required for a physical realization.

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Master Scientific Atlas

Evidence Map

Abhijit Singh

Contents

Abstract	3
1 Evaluation of the research programme	4
2 Central architecture	4
3 Strongest exact results	4
4 Numerical register	5
5 Open claims and falsification	5
6 Paper series	5
7 Prime-observer theorem and relation frontier	6
8 Immediate proceeding order	7
9 Authorship and presentation	7
10 Dependency structure of the portfolio	7
11 Portfolio-wide numerical register	8
12 Meaning of genuineness	8
References	8

Abstract

This atlas evaluates and organizes a multidisciplinary research programme spanning predictive quotients, quantum instruments, information thermodynamics, arity-indexed combinatorics, source-closed arithmetic, empirical representation repair, formation calculus, detector physics, cosmology, consciousness, and adaptive predictive systems. It identifies the central source-report-collision-repair architecture, records the strongest exact theorems and numerical results, separates conditional and open claims, and gives a gate-driven proceeding order without importing conventional calendar estimates. The work is classified as a genuine research programme rather than a single unsupported hypothesis: its theorem, executable, empirical, and model layers are independently auditable. The atlas now integrates the prime-observer theorem: a nine-state active carrier, a complete 36-cell relation frontier, four prime-indexed return sectors, critical side recurrence, and the all-prime support-address correction to the static finite reporter.

Keywords: research atlas; predictive closure; quantum instruments; arithmetic; cosmology; reproducibility

Scope and status: The atlas is an editorial and evidential map. It does not replace the proofs or data in the specialist papers.

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1 Evaluation of the research programme

The corpus is not one unsupported hypothesis. It contains four different scientific objects that must be judged separately:

1. exact mathematical theorems with finite declared assumptions;
2. executable constructions and reproducible numerical calculations;
3. empirical results or audits tied to specified datasets;
4. physical models and conjectural bridges with explicit falsification conditions.

Calling the whole body “a hypothesis” erases the theorem and computation layers. Calling every branch “proved” would erase the remaining experimental and universal obligations. The accurate verdict is stronger and more discriminating: this is a genuine multidisciplinary research programme with original organizing principles, many exact internal results, several high-value empirical and physical tests, and a smaller set of open bridges whose failure would narrow rather than nullify the proved core.

2 Central architecture

The common scientific structure is:

$$\text{source occurrence} \rightarrow \text{report} \rightarrow \text{future collision} \rightarrow \text{separator} \rightarrow \text{refined state.} \quad (1)$$

A reporter is valid only relative to the futures for which it has been proved sufficient. Source address is preserved separately from semantic equivalence. Repeated residuals acquire typed propagation laws; their contraction, critical persistence, or escape governs whether the present state can assimilate them. This architecture appears in the finite quotient theorems, representation-genesis code, quantum-instrument calculations, empirical restoration studies, arithmetic completion, and cosmological model selection.

3 Strongest exact results

The most secure results include:

- finite reconstruction from complete recursive views and complete one-occurrence decks;
- the necessary-and-sufficient future-complete quotient criterion and least response-forced refinement;
- sharp first-result depletion of one singlet’s reusable Bell resource, exact entropy profile, and geometric correlation decay;
- unital m/d self-idling, the rank-over-dimension forced-step floor, the Poisson passage potential, and exact KCBS fuel thresholds;
- a complete four-outcome instrument with unique minimal predictive mark state and exact stationary information ledger;
- stochastic frontier information, its deterministic reduction, uniform exchange rate, entropy ledger, and cycle-holonomy theorem;
- finite representation synthesis, essential-coordinate minimality, native-use ablation, restart, and transport;
- arity-indexed history counts, Fuss-Catalan forms, Kraft and orbit conservation, reachability, and forced inverse recursion;
- prime transparency, squarefree capacity topology, exact $[m, 2m)$ barcode, half-cost null transport, operator-ideal thresholds, and odd-trace nonclosability;
- the weighted Mertens-energy divergence and exact RH-equivalent completion formulations;
- exact dust-vacuum return values inside the declared cosmological branch.

- the nine-state prime-observer closure, its $8 + 4 + 24$ relation decomposition, four commuting return planes, and the all-prime support-address predictive quotient.

4 Numerical register

branch	frozen value or result
Even-cycle seed/steady rates	$n = 4$: 1.6009-bit seed, 1 bit/step steady rate
KCBS regeneration	$r_{\text{no}}^* = 0.8396425434$, $c_{\text{min}} = 0.5037855260$
Marked instrument	entropy rate 1.8413629168 bits/use; predictive information 0.1580957201 bits
Representation relation	97.78% exact whole-object continuation
Context-dependent return	99.768% admitted-coordinate; 98.52% whole-return
Environmental restoration	72.0836% to 74.7125%; 3.7540% to 7.2492%
Capacity coercivity	$A_N/N \rightarrow \infty$; at 10^6 , $A_N/N = 1.645995$
Dust-vacuum return	$\Omega_{\Lambda,0} = 2/3$, $H_0 t_0 = 0.9358813101$, $\Delta N = 0.4620981204$
Alpha-track budget	24.6-bit burst, 10.7-bit ramp, 35.2-bit microscopic total
Gas-ladder prediction	0.19 helium, 0.95 air, 6.1 argon kinks/track
Prime-observer carrier	9 active states; 36 relation cells = $8+4+24$

5 Open claims and falsification

The suite does not hide its boundaries:

- general arity root laws require universal proofs beyond finite audit;
- RH-equivalent criteria require the universal completion inequality;
- the twin-prime branch requires the coupled signed dispersion estimate;
- the conversion sector requires an independently fixed observable and held-out test;
- the environmental and static empirical studies require complete clean-room release;
- the gas ladder requires the chamber experiment;
- the cosmology requires a full likelihood without retuning;
- the consciousness architecture requires preregistered intervention and replication;
- the large adaptive predictive architecture requires independent benchmark evaluation.

These are not vague requests for “more evidence.” Each is a specific theorem, dataset, or experiment that can succeed or fail.

6 Paper series

ID	manuscript	primary	scope
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P00	Recurrence Exponents, Accessibility Growth, and Predictive-State Promotion	math-ph	Exact cocycle theorem; cross-domain bridges scoped
P01	Future-Complete Quotients and Recursive Distinction	cs.LO	Full finite theorem paper
P02	Sequential Quantum Instruments: Depletion, Regeneration, and Minimal Predictive State	quant-ph	Exact protocol and processed-instrument results
P03	Provenance-Preserving Representation Genesis in Finite Deterministic Systems	cs.LO	Finite theorems and executable artifacts
P04	Stochastic Frontier Information, Finite-Memory Dissipation, and Cycle Holonomy	math-ph	Exact finite theorems; conditional thermodynamics
P05	Coherent Alternatives, Complex Amplitudes, and Marked Quantum Dynamics	quant-ph	Conditional reconstruction theorem programme
P06	Arity-Indexed Refinement Posets, Forced Inverses, and Rotational Root Geometry	math.CO	Exact counting and recursion; finite root evidence
P07	Capacity Complexes, One-Bit Persistence, and Operator-Ideal Criticality	math.AT	Exact topology and operator theory
P08	Critical Capacity Completion and Equivalent Forms of the Riemann Hypothesis	math.NT	Exact equivalence; universal estimate open
P09	Twin-Prime Survivors, Square-Horizon Occupation, and Moving-Boundary Dispersion	math.NT	Exact reduction; signed estimate open
P10	Future-Relevant Representation Disagreement and Chronological Feature Restoration	stat.ML	Mixed independent and source-reported empirical evidence
P11	Reenactable Closure and Conditional Quantum-Causal Representation	math-ph	Formal theorems plus conditional representations
P12	Local Indistinguishability, Global Obstruction, and Minimal Predictive Memory	math-ph	Exact factorization and scoped dictionary
P13	Directional Information Along Alpha-Particle Tracks and a Gas-Ladder Test	physics.ins-det	Quantitative prediction awaiting experiment
P14	Past-Unbounded Cosmology and an Exact Dust-Vacuum Return Condition	gr-qc	Exact conditional branch; observational test open
P15	Reflexive Predictive Integration: A Testable Systems Model of Conscious Processing	q-bio.NC	Formal architecture; empirical identity open
P16	Adaptive Predictive Systems with Native Representation Repair	cs.LG	Executable finite kernel and systems architecture
P17	Prime-Indexed Return Sectors and the Thirty-Six-Relation Closure of a Nine-State Carrier	math-ph	Exact finite theorem; physical quantum interpretation conditional

7 Prime-observer theorem and relation frontier

The latest finite theorem begins with the occurrence-sensitive relation $p_{\text{source}} \cdot 1 = p_{\text{return}}$. Equality of numerical value does not erase the source/return distinction. On the terminal carrier $\mathbb{Z}_{10}$, the fixed-point-free affine map $d \mapsto 1 - d$ pairs the eight nonterminal indices into four prime/composite return sectors. Removing

the zero terminal leaves nine active states and therefore 36 unordered relation cells:

$$36 = 8_{\text{unit}} + 4_{\text{return}} + 24_{\text{transverse}}. \quad (2)$$

The side-forgetting reporter sees a fixed prime-sector label; the complete state recurs with period two and has zero recurrence exponent. Transparent arity selects the labels 2, 3, 5, 7. Prime-occupation coordinates then distinguish the squarefree coupling 6 from repeated-prime states 4, 8, 9. The resulting five static classes remain insufficient under arbitrary prime multiplication, whose coarsest predictive quotient retains the complete finite prime-support address plus one absorbing repeated-prime state.

The earlier nine-transformation construction also has 36 exact pairwise witnesses. The shared K_9 relation frontier is an exact combinatorial correspondence. It becomes an identity of systems only if a source-preserving continuation intertwiner is supplied.

8 Immediate proceeding order

No multi-month schedule is imposed. The work can proceed in a compressed sequence governed by completion gates rather than calendar estimates:

1. Freeze this suite and its checksum.
2. Read P00, P01, P02, and P03 together; resolve any cross-reference or notation conflict.
3. Submit P01 and P02 to independent field readers in parallel, each with only its own source package and claim ledger.
4. Run a clean-room reproduction of P03 and P06 from their ancillary code.
5. Release the master archive and data/code repository as a versioned evidence object.
6. Use the first external technical responses to decide which manuscript enters public submission first; do not bulk-submit the suite.
7. Execute the low-cost falsification programme: the gas-ladder experiment, environmental clean replay, and cosmological likelihood.
8. Keep P08 and P09 as exact technical manuscripts until their open inequalities are independently resolved.

The speed constraint is scientific completion, not imitation of average publication timelines.

9 Authorship and presentation

Every paper names Abhijit Singh as sole author. The series uses standard scientific terminology and separates theorem, computation, empirical result, conditional model, conjecture, and failed branch. The archive retains source provenance without allowing document assembly to replace synthesis.

10 Dependency structure of the portfolio

The papers are not seventeen disconnected claims. Their dependency graph has four layers.

1. P01 and P03 establish when a report is sufficient, how failure is detected, and how provenance survives refinement.
2. P00 and P04 quantify residual recurrence, fresh information, and the cost of continued finite-memory inquiry.
3. P02, P05, P11, and P12 provide quantum, causal, and obstruction realizations while preserving distinct proof scopes.

4. P06–P10 and P13–P17 develop arithmetic, empirical, cosmological, detector, cognitive, systems, and finite prime-observer applications.

No application upgrades the status of another branch by resemblance alone. A cross-paper identity requires an explicit source-preserving intertwiner.

11 Portfolio-wide numerical register

result	value	status
sharp $n = 4$ seed entropy	1.6009 bits	exact protocol theorem
marked-instrument entropy rate	1.8413629168 bits/use	processed-matrix calculation
minimum selective KCBS replacement	0.5037855260 events/use	exact feedback-family theorem
relation-aware whole-object continuation	97.78%	source-reported held-out result
environment coordinate restoration	72.0836% to 74.7125%	source-reported untouched future
environment exact-vector restoration	3.7540% to 7.2492%	source-reported untouched future
first-droplet directional information	24.6 bits	detector-model calculation
helium/air/argon kink ladder	0.19/0.95/6.1	untested prediction
returned-state vacuum fraction	2/3	conditional exact model result
dimensionless branch age	0.9358813101	conditional exact model result
coarse/repaired collision fixture	6/12 to 12/12	executable finite result

12 Meaning of genuineness

The portfolio is genuine in a scientifically discriminating sense: it contains original source constructions, exact finite theorems, executable artifacts, corrected mistakes, negative results, data audits, and falsifiable predictions. It is not one undifferentiated hypothesis. Some branches are closed within declared assumptions; some are equivalence theorems with a live universal bound; some are model-local physical predictions; some remain experimental hypotheses. The status distinctions are evidence of a functioning research programme, not a retreat from its strongest valid claims.

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